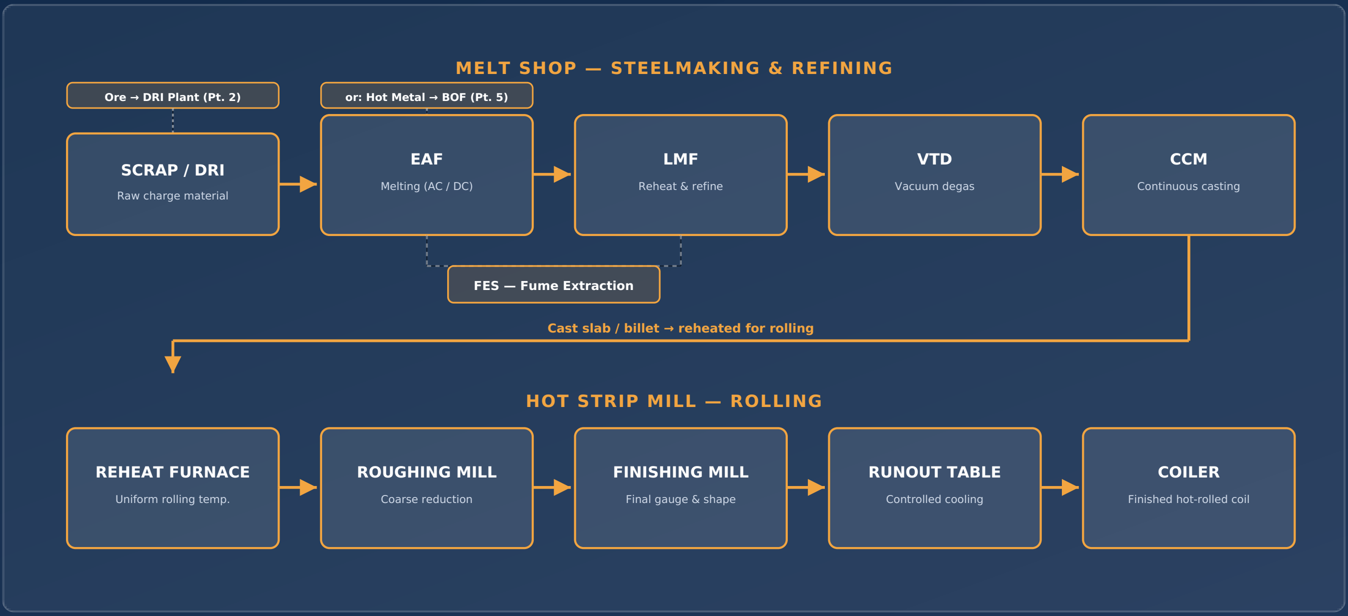
REFERENCE BOOK

The Hot Mill

A complete reference — from scrap and raw iron through melting, refining, casting, and hot rolling to a finished coil

Melt Shop → Refining & Casting → Hot Strip Mill



19 Parts • 154 Pages • Process, Equipment & Electrical Systems Reference

Illustrated Process & Electrical Equipment Guide

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Contents

Nineteen parts — steel production, then the utility plants that make it possible

This reference book combines the full process and electrical-systems documentation for a modern hot mill — from raw ore and charge material through melting, refining, casting, and hot rolling to a finished coil. Parts 1–9 cover the melt shop, including both ironmaking-to-steelmaking pathways (DRI→EAF, in both AC and DC, and BOF); Part 10 is the hot strip mill overview; Parts 11–15 are deep dives into each hot strip mill stage. Parts 16–19 cover the auxiliary and utility plants — compressed air, oxygen, water treatment, and cooling water — that every preceding part depends on. Page numbers match the page numbers printed in this book and shown in your PDF viewer.

Detailed Index

Every section in this book, with its page number

Detailed Index

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i How This Guide Works

Follow the process in order — each stage feeds the next

This guide walks through the electric-furnace steelmaking route in the exact order the steel physically travels through the plant. Each stage has its own page with the same layout, so you can move through the shop step by step:

WHAT'S ON EVERY STAGE PAGE

1. **A simple diagram** showing the equipment and the material flowing through it.
2. **A "How It Works" walk-through** — numbered, plain-language steps in the order they happen.
3. **An Electrical Equipment table** — every major powered / electrically-controlled item at that stage and what it does.
4. **A parameters panel** — typical temperatures, power levels, and cycle times.
5. **A safety note** — the main hazard to be aware of at that stage.

THE ROUTE AT A GLANCE

01	DRI — Direct Reduced Iron production	Raw material
02	EAF — Electric Arc Furnace	Melting
03	LMF — Ladle Metallurgy Furnace	Refining
04	VTD — Vacuum Tank Degasser	Degassing
05	CCM — Continuous Casting Machine	Casting
06	FES — Fume Extraction System	Emission control (runs alongside EAF & LMF)

WHY THIS ROUTE?

DRI (sponge iron) is used instead of scrap or in addition to it, as a clean, consistent iron source charged into the EAF. This route is common where good scrap supply is limited but natural gas or coal-based DRI plants are available.

READING THE DIAGRAMS

■ Vessel / equipment ■ Electrical item ■ Material flow

BEFORE YOU START

This is a training and orientation guide, not an operating or electrical-safety manual. Always follow site lockout/tagout, PPE, and permit-to-work procedures issued by your plant.

1 DRI — Direct Reduced Iron

Stage 1 of 5 | Turning iron ore into solid metallic iron, without melting it

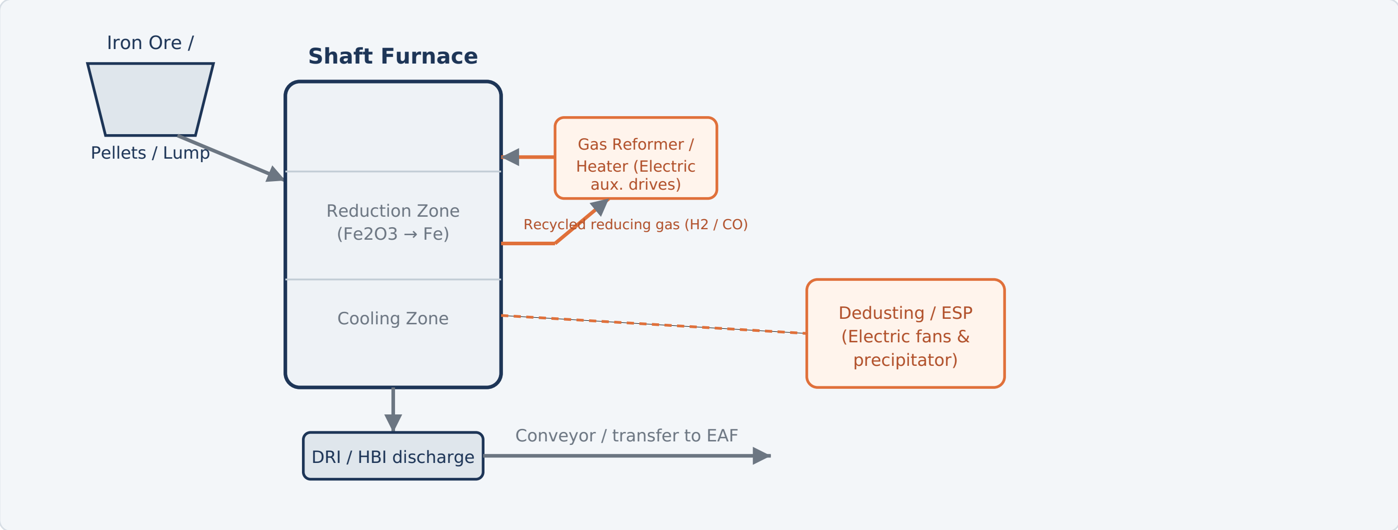


Fig. 1 — Vertical shaft furnace: iron ore is reduced by hot gas (not melted) into solid DRI, then transferred to the EAF.

HOW IT WORKS

1. **Charge:** Iron ore pellets or lump ore are loaded into the top of the vertical shaft furnace.
2. **Reduce:** Hot reducing gas (hydrogen and carbon monoxide) flows upward through the descending ore, stripping oxygen from the iron oxide — this happens below the melting point, so the iron stays solid.
3. **Cool & discharge:** The reduced iron (now metallic "sponge iron") moves down into a cooling zone and is discharged at the bottom as DRI, or briquetted while hot into HBI (Hot Briquetted Iron) for easier handling.
4. **Transfer:** DRI/HBI is conveyed — hot, where the plant is integrated — directly to the EAF as furnace charge, supplementing or replacing scrap steel.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
MV switchgear & motor control center (MCC)	Distributes and protects power to all motor-driven equipment in the DRI plant.
Process gas compressor motors	Circulate and pressurize the reducing gas through the furnace loop.
Gas heater / reformer electrical controls	Regulate temperature of the reducing gas entering the shaft furnace.
Feed & discharge conveyor drives	Electric motors moving ore in and DRI/HBI out of the furnace.
Briquetting press drive motors	Power the rollers that press hot DRI into HBI briquettes.
Dedusting fans & electrostatic precipitator (ESP)	Capture dust generated during material handling and gas cleaning.
PLC / DCS automation system	Monitors and controls gas flow, temperature, and material flow rates.

TYPICAL PARAMETERS

Process temperature	**800–1050 °C**
Metallization	**90–94%**
Reducing gas	**H2 + CO**
Product form	**DRI / HBI**
Furnace state	**Solid-state (no melting)**

KEY HAZARD

Reducing gas is flammable and can displace oxygen. Hot DRI is pyrophoric (can reignite in air) — handling and storage follow strict inerting and cooling procedures.

IRONMAKING, NOT STEELMAKING

A DRI plant is the direct counterpart to a blast furnace — it makes *iron*, not steel. Turning DRI into steel is still the EAF's job (Parts 3–4). See Part 2 for a full deep dive into the shaft furnace, reduction chemistry, and why pyrophoricity drives most of this plant's safety design.

2

EAF — Electric Arc Furnace

Stage 2 of 5 | Melting DRI/scrap into liquid steel using electric arcs

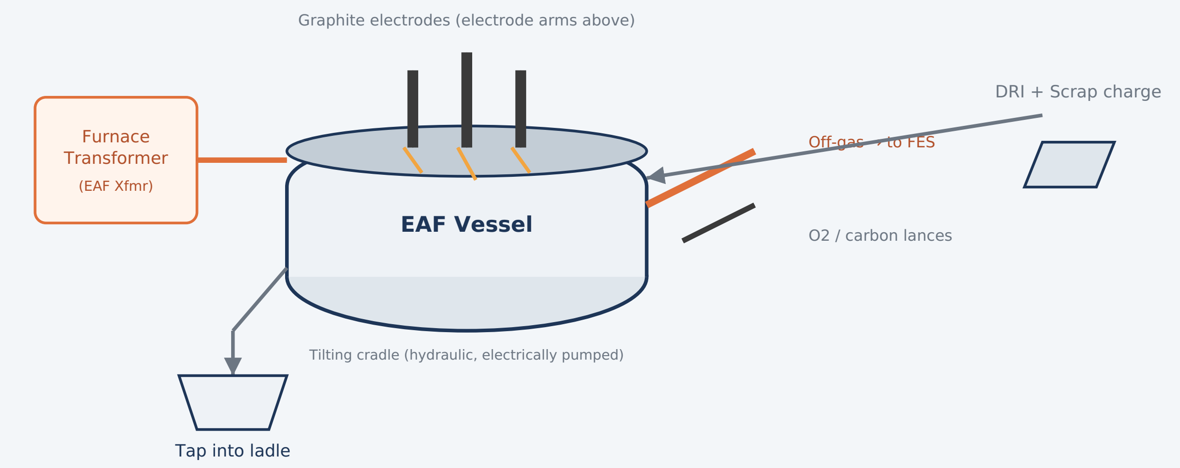


Fig. 2 — EAF: graphite electrodes strike an arc through DRI/scrap to melt the charge into liquid steel, then the furnace taps into a ladle.

HOW IT WORKS

1. **Charge:** DRI/HBI (and/or scrap) is loaded into the furnace shell, either through the open roof by crane bucket or via a continuous DRI feed chute.
2. **Power on:** Three graphite electrodes lower and strike an arc, generating intense heat (well above 3000°C at the arc) that melts the charge.
3. **Refine & oxidize:** Oxygen and carbon are injected to remove impurities, speed melting, and adjust carbon content; a foamy slag layer protects the arc and refractory.
4. **Tap:** Once the bath reaches temperature and composition targets, the furnace tilts and liquid steel is tapped into a waiting ladle, leaving most of the slag behind.
5. **Repeat:** The furnace tilts back, and the next charge begins — a "heat" typically cycles in well under an hour.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
EAF (furnace) transformer	Steps down high voltage to the very high current needed to drive the melting arc — the single largest electrical load in the shop.
HV incoming switchgear / circuit breaker	Connects the furnace transformer to the plant's high-voltage supply and provides protection.
Series reactor	Limits current surges and stabilizes the arc, especially during melt-down.
Electrode arms & electrode regulation system	Automatically raise/lower each electrode to maintain a stable arc length.
Graphite electrodes & holders	Conduct current from the arms into the furnace to create the arc.
Secondary bus tubes / flexible water-cooled cables	Carry very high current from the transformer to the electrode arms.
Static VAR Compensator (SVC) / harmonic filters	Reduce voltage flicker and harmonics caused by the fluctuating arc load.
Oxygen/carbon lance manipulators	Electrically driven positioning of injection lances and burners.
Tilting & hydraulic pump motors	Electric motors drive the hydraulic system that tilts the furnace for slagging and tapping.
Level 1/2 automation (PLC/DCS)	Controls power input, electrode position, and process sequencing.

TYPICAL PARAMETERS

Tap temperature	**~1600–1650 °C**
Transformer rating	**30–150+ MVA**
Arc voltage	**Very high current, low V**
Tap-to-tap time	**~35–60 min**
Output	**Liquid steel + slag**

KEY HAZARD

Extreme electrical current, arc flash, and molten metal splash. Access to the furnace area during melting is restricted, and interlocks prevent roof/door access while powered.

TWO WAYS TO RUN THIS FURNACE, ONE OTHER ROUTE ENTIRELY

This overview covers EAF melting generically — see Part 3 for the classic **AC** three-phase design and Part 4 for the **DC** single-electrode alternative. EAF also isn't the only way to make liquid steel: the **Basic Oxygen Furnace (BOF)** refines molten iron from a blast furnace instead, with no electricity needed at all. See Part 5 for a full deep dive, including why almost all new steelmaking capacity is now built as EAF rather than BOF.

3 LMF — Ladle Metallurgy Furnace

Stage 3 of 5 | Fine-tuning temperature and chemistry away from the EAF

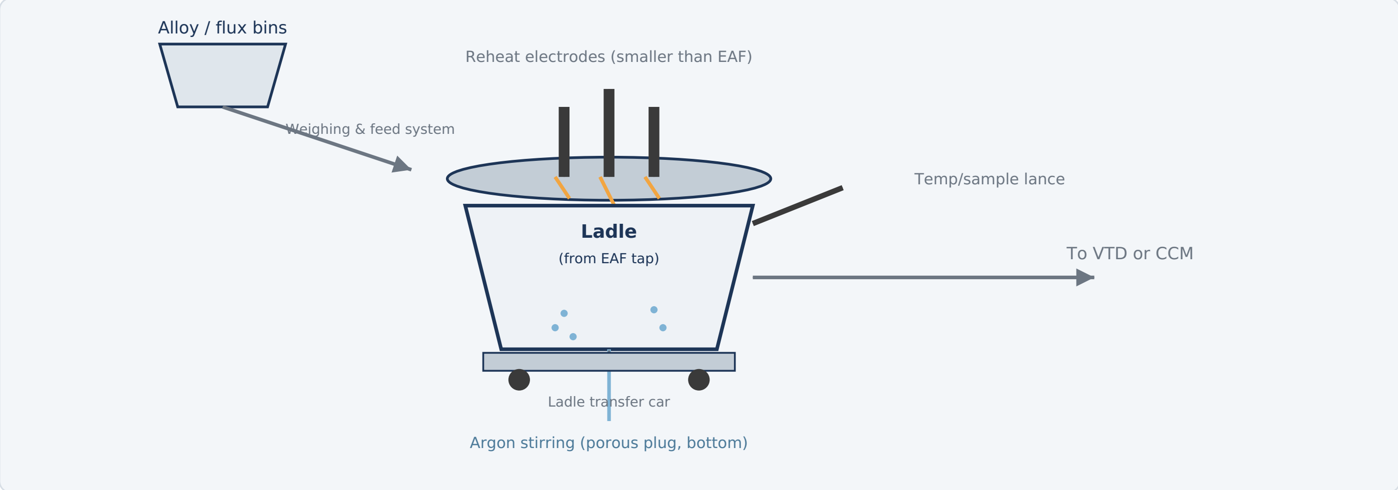


Fig. 3 — LMF: the ladle of liquid steel from the EAF is reheated with electrodes and trimmed with alloys under argon stirring.

HOW IT WORKS

1. **Position ladle:** The tapped ladle is moved on a transfer car to the LMF stand and a lid with electrodes lowers over it.
2. **Reheat:** Electrodes strike an arc through the slag layer, gently reheating the steel to compensate for temperature lost since tapping — without the aggressive oxidation of the EAF.
3. **Stir:** Argon gas injected through a porous plug in the ladle bottom stirs the bath, homogenizing temperature and chemistry and helping non-metallic inclusions float into the slag.
4. **Trim chemistry:** Ferroalloys and flux additions are metered in to hit the exact target composition for the grade of steel being made.
5. **Release:** Once temperature and chemistry are confirmed by sampling, the ladle moves on to vacuum degassing (VTD) or straight to the caster (CCM).

TYPICAL PARAMETERS

Treatment time	**20–45 min**
Transformer rating	**~10–25 MVA**
Stirring gas	**Argon**
Purpose	**Temp + chemistry control**

KEY HAZARD

Overhead electrode lid movement and hot ladle transfer create crush and burn risks; stay clear of the ladle car travel path during moves.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
LMF transformer	Supplies power for the reheating arc — sized much smaller than the EAF transformer.
Electrode arms & regulation system	Positions and controls the three reheating electrodes.
Graphite electrodes	Carry current to strike the reheating arc over the ladle.
Argon flow control valves & flow meters	Electrically controlled stirring gas rate through the porous plug.
Alloy/flux weigh-feed system	Motor-driven bins, conveyors, and load cells that dose precise alloy additions.
Temperature & sampling lance manipulator	Electrically driven arm that dips thermocouples and sample probes into the bath.
Ladle transfer car drive	Electric motor that moves the ladle car between EAF, LMF, VTD, and CCM.
PLC automation system	Sequences reheating, stirring, and alloy additions to hit target chemistry.

4 VTD — Vacuum Tank Degasser

Stage 4 of 5 | Pulling dissolved gases out of the steel under vacuum

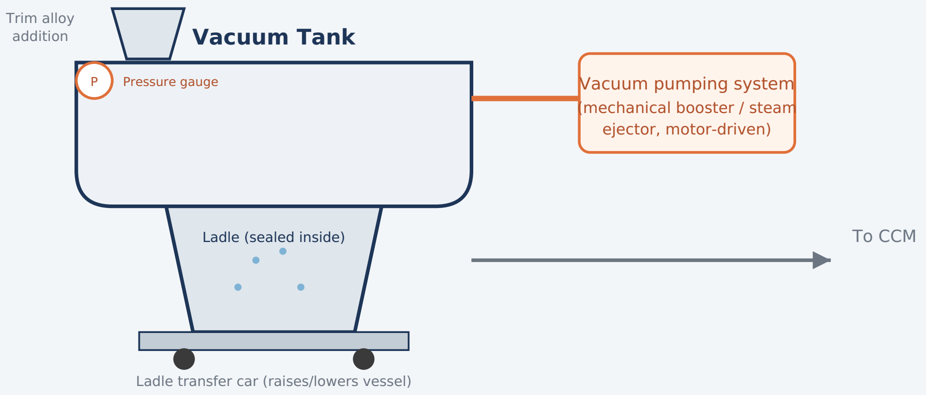


Fig. 4 — VTD: the ladle is enclosed in a vacuum vessel while argon stirring exposes fresh metal surface to the low-pressure environment, pulling out dissolved hydrogen, nitrogen, and oxygen.

HOW IT WORKS

1. **Seal the ladle:** The ladle (still on its transfer car) is raised into the vacuum tank and the vessel lid seals around it.
2. **Draw vacuum:** Vacuum pumps evacuate the tank to a low absolute pressure, dramatically lowering the solubility of gases in the steel.
3. **Stir under vacuum:** Argon continues bubbling through the bath, constantly exposing fresh steel surface to the vacuum so dissolved hydrogen, nitrogen, and some oxygen (as CO) escape.
4. **Trim & finish:** Final alloy trims can be added under vacuum or immediately after breaking vacuum, since the bath is now very clean.
5. **Release:** Vacuum is broken, the lid lifts, and the ladle moves directly to the caster (CCM).

TYPICAL PARAMETERS

Vacuum level	**~0.5–5 mbar**
Treatment time	**15–25 min**
Gas removed	**H_2, N_2, some O_2**
Use case	**Clean / low-H steel grades**

KEY HAZARD

High-pressure vacuum equipment and sudden pressure changes when breaking vacuum; lid and lifting mechanisms present pinch-point hazards.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Vacuum pump motors (mechanical booster pumps)	Electrically driven pumps that evacuate the vacuum vessel to process pressure.
Steam ejector control valves (if steam-assisted)	Electrically actuated valves controlling the vacuum staging.
Argon flow control system	Electronically regulated stirring gas supply during evacuation.
Vacuum vessel lid hoist/seal drive	Electric motor that raises, lowers, and seals the tank lid over the ladle.
Ladle car hydraulic lift drive	Electrically pumped hydraulics that raise the ladle into the sealed vessel.
Alloy trim feeder	Motorized metering system for final composition adjustments.
Pressure transmitters & instrumentation	Electronic sensors tracking vacuum level throughout the cycle.
PLC automation system	Sequences pump-down, stirring, and alloy addition against the vacuum profile.

5 CCM — Continuous Casting Machine

Stage 5 of 5 | Solidifying liquid steel into billets, blooms, or slabs

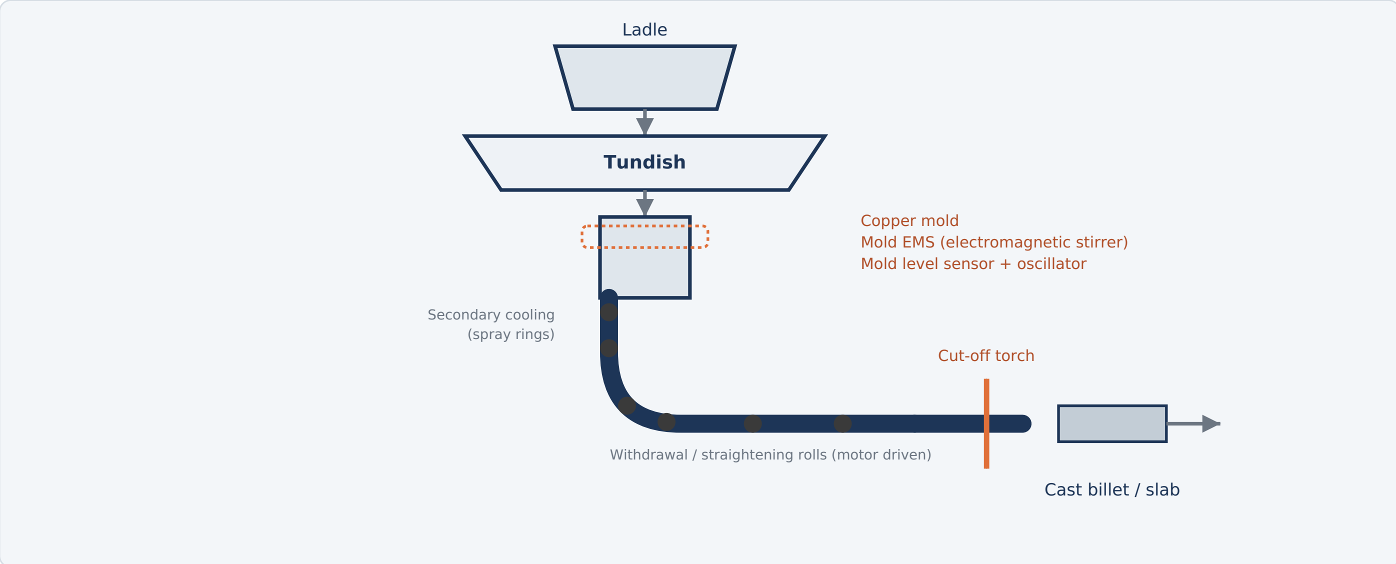


Fig. 5 — CCM: steel flows from ladle to tundish to a water-cooled mold, solidifies as it is withdrawn through support rolls, and is cut to length.

HOW IT WORKS

1. **Ladle to tundish:** The ladle sits above a tundish — a holding vessel that buffers steel flow and distributes it to one or more strands.
2. **Tundish to mold:** Steel flows (rate controlled by a stopper rod or slide gate) into a water-cooled copper mold, where the outer shell begins to solidify almost instantly.
3. **Withdraw:** The partially-solid strand is continuously withdrawn downward (and curved toward horizontal on curved casters) by driven support rolls, while an oscillating mold motion prevents sticking.
4. **Secondary cooling:** Water sprays along the strand finish solidifying the steel from the outside in as it travels through the roll containment segments.
5. **Cut to length:** Once fully or near-fully solid, a torch or shear cuts the continuous strand into billets, blooms, or slabs of the required length.

TYPICAL PARAMETERS

Casting speed	**0.8–5+ m/min**
Mold length	**~0.7–1 m**
Product	**Billet / bloom / slab**
Output	**Solid semi-finished steel**

KEY HAZARD

Breakouts (molten steel escaping a thin shell) are the primary hazard; strict mold-level and cooling monitoring exists specifically to prevent them.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Mold oscillation drive	Electric/servo drive that oscillates the mold vertically to prevent the shell from sticking.
Mold electromagnetic stirrer (M-EMS)	Electromagnetic coils around the mold that stir the liquid pool for a cleaner, more uniform strand.
Strand electromagnetic stirrer (S-EMS)	Additional stirring coils lower down the strand to improve internal quality.
Mold level control system	Electromagnetic or radioactive-source sensor with electric actuator controlling steel flow into the mold.
Withdrawal & straightening roll motors	Electric drives that pull the strand through the machine at a controlled casting speed.
Secondary cooling spray control system	Electrically controlled pumps and valves adjusting water spray along the strand.
Cut-off torch / shear drive	Electrically positioned and triggered cutting unit that segments the strand to length.
Tundish & ladle car drives	Motorized cars positioning the ladle and tundish over the strand(s).
Level 2 casting automation	Coordinates casting speed, cooling, and cutting in real time.

6 FES — Fume Extraction System

Runs alongside melting & refining | Capturing and cleaning furnace off-gas and dust

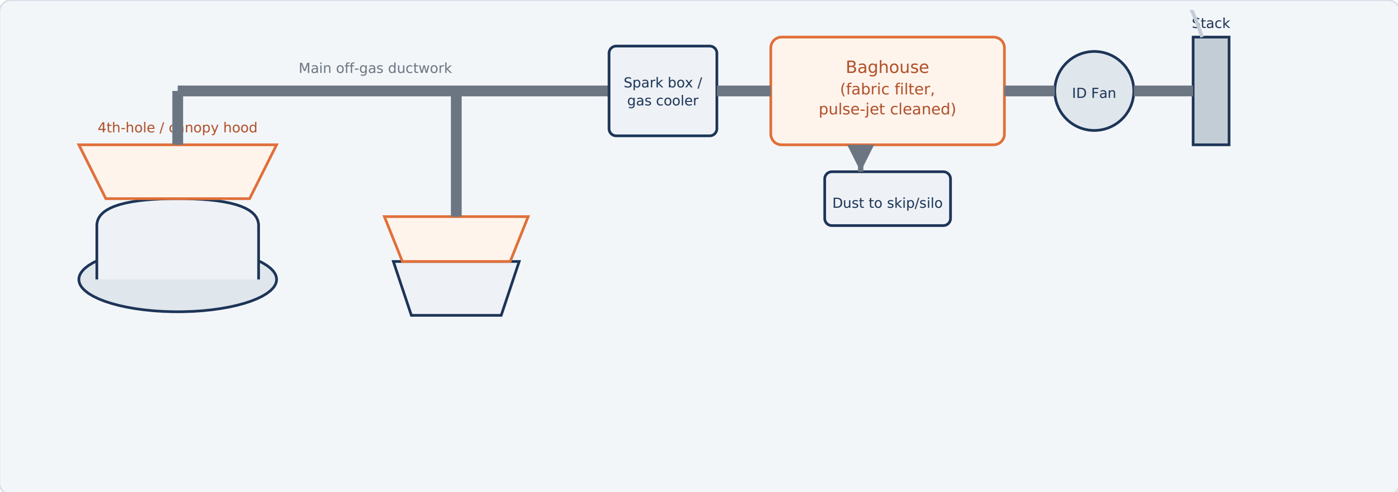


Fig. 6 — FES: hoods above the EAF and LMF capture fume and dust, which is ducted through a gas cooler and baghouse before clean air exits the stack.

HOW IT WORKS

1. **Capture:** Hoods positioned over the EAF (direct shell evacuation / "4th hole" plus a roof canopy hood) and over the LMF ladle stand capture fume, dust, and hot gas at the source.
2. **Duct & cool:** Large induced-draft ductwork carries the hot gas through spark arrestors and coolers, dropping its temperature to protect downstream filter media.
3. **Filter:** The cooled gas passes through a baghouse (fabric filter), where fabric bags trap fine particulate dust; bags are periodically pulse-jet cleaned with compressed air.
4. **Exhaust:** An induced draft (ID) fan pulls the whole system, maintaining negative pressure at the hoods and pushing cleaned air out through the stack.
5. **Dispose:** Collected dust drops into conveyors, skips, or silos for handling, recycling, or disposal as required by environmental regulations.

TYPICAL PARAMETERS

Gas inlet temp	**Up to ~1000+ °C**
Filter type	**Pulse-jet fabric baghouse**
Fan type	**Induced draft (ID)**
Purpose	**Emission & dust control**

KEY HAZARD

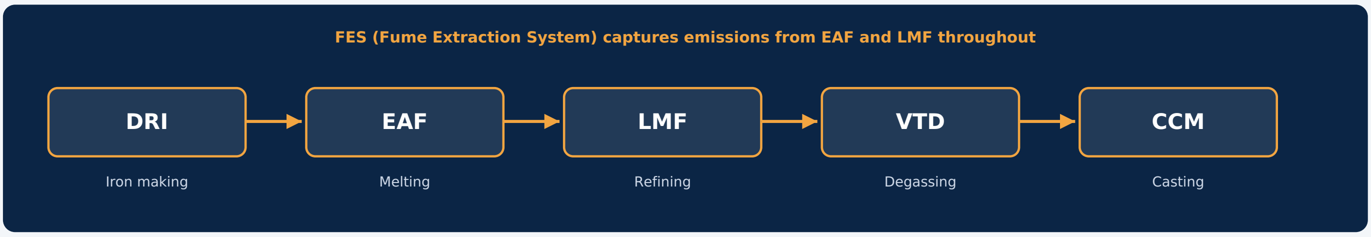
Hot gas ducts, combustible CO in EAF off-gas, and dust explosion potential in the baghouse — spark detection and suppression systems are standard.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Induced draft (ID) fan & motor	The primary driving force of the whole system — pulls fume through the ducts and baghouse and out the stack.
Variable frequency drives (VFDs)	Adjust fan speed to match extraction demand and save energy during idle periods.
Damper actuators	Electrically driven dampers balance airflow between the EAF hood, LMF hood, and building evacuation.
Baghouse pulse-jet cleaning system	Electrically timed solenoid valves fire compressed-air pulses to shake dust off the filter bags.
Rotary airlock / screw conveyor motors	Discharge collected dust from the baghouse hoppers into skips or silos.
MV switchgear / motor control center	Powers and protects the large ID fan motor(s) and auxiliary drives.
Stack emission monitoring system	Continuous electronic instrumentation tracking particulate and gas emissions for compliance.
PLC automation system	Coordinates fan speed, damper position, and bag cleaning cycles with furnace operating state.

Process Summary

The full route, and every stage's electrical equipment at a glance



ELECTRICAL EQUIPMENT — ALL STAGES

STAGE	MAJOR ELECTRICAL EQUIPMENT	PRIMARY ROLE
DRI	Process gas compressors, gas heater controls, conveyor drives, briquetting press, dedusting/ESP	Move & process gas and material for solid-state iron reduction
EAF	Furnace transformer, HV switchgear, series reactor, electrode regulation, SVC/harmonic filters, tilting drives	Deliver massive arc power to melt the charge
LMF	LMF transformer, electrode regulation, argon flow control, alloy weigh-feed system, ladle car drive	Reheat and fine-tune steel chemistry
VTD	Vacuum pump motors, lid hoist/seal drive, ladle car hydraulic lift, argon flow control	Remove dissolved gases under vacuum
CCM	Mold oscillator, M-EMS/S-EMS coils, mold level control, roll drive motors, spray control, cut-off drive	Solidify and cut steel into semi-finished shapes
FES	ID fan & VFDs, damper actuators, baghouse pulse-jet system, rotary airlock motors, emission monitors	Capture, filter, and exhaust furnace fume and dust

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

Iron ore is reduced (not melted) into **DRI** using hot reducing gas, then charged into an **EAF** where electric arcs melt it into liquid steel. The steel is tapped into a ladle and moved to the **LMF**, where it is reheated and its chemistry is precisely trimmed under argon stirring. If the grade requires very low dissolved gas content, the ladle goes to the **VTD**, where vacuum pumping strips out hydrogen and nitrogen. Finally, at the **CCM**, the steel flows continuously through a water-cooled mold and cooling rolls, solidifying into billets, blooms, or slabs that are cut to length. Throughout melting and refining, the **FES** captures furnace fume and dust, cleans it through a baghouse, and exhausts it safely — keeping the shop's emissions under control.

PART 2 — THE DRI PLANT

Why Direct Reduction? The Chemistry Without Melting

An ironmaking route, not a steelmaking one — and the reason it's central to low-carbon steel

A DRI plant is often mentally filed next to the EAF and BOF, but it belongs in a different category entirely: it's an **ironmaking** process, not a steelmaking one — the direct equivalent of a blast furnace, not of an EAF. Its job is turning iron ore into metallic iron; turning that iron into steel is still someone else's job afterward, almost always an EAF's. What makes direct reduction different from a blast furnace is that the ore **never melts**. Instead, a reducing gas — a mixture of hydrogen and carbon monoxide — strips oxygen away from solid iron oxide pellets at a temperature well below iron's melting point, leaving behind a solid, porous pellet of metallic iron known as sponge iron.

$$Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O \quad \bullet \quad Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

Both reactions run solid-state, at ~800-1050°C — well below iron's ~1538°C melting point

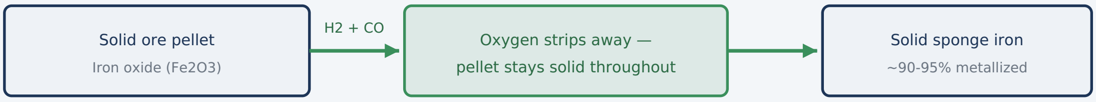


Fig. 1 — Unlike a blast furnace, which melts everything, direct reduction removes oxygen while the pellet stays solid and porous — hence "sponge iron."

IRONMAKING, NOT STEELMAKING — THE DISTINCTION THAT MATTERS

STAGE	BLAST FURNACE ROUTE	DIRECT REDUCTION ROUTE
Ironmaking (ore → iron)	Blast furnace, melts ore with coke	**DRI plant** — this guide
Steelmaking (iron → steel)	BOF, refines liquid hot metal	EAF, melts & refines DRI/scrap

WHY THIS IS THE LEADING LOW-CARBON STEEL PATHWAY

A blast furnace needs coking coal both as fuel and as the chemical reducing agent — a major source of the steel industry's carbon emissions. DRI plants typically use natural gas instead, reformed into a hydrogen/CO mixture, and can run on very high concentrations of hydrogen or even pure hydrogen from electrolysis. Since hydrogen reduction produces water vapor instead of CO2, a hydrogen-based DRI plant paired with an EAF is currently the most credible large-scale route to substantially lower-carbon steel.

TYPICAL PARAMETERS

Reduction temperature	**~800-1050 °C**
Metallization	**~90-95%**
Reducing gas	**H2 + CO (natural gas or H2-based)**

1 The Direct Reduction Process

Ore descends, gas rises — a continuous counter-current exchange, no melting anywhere in it

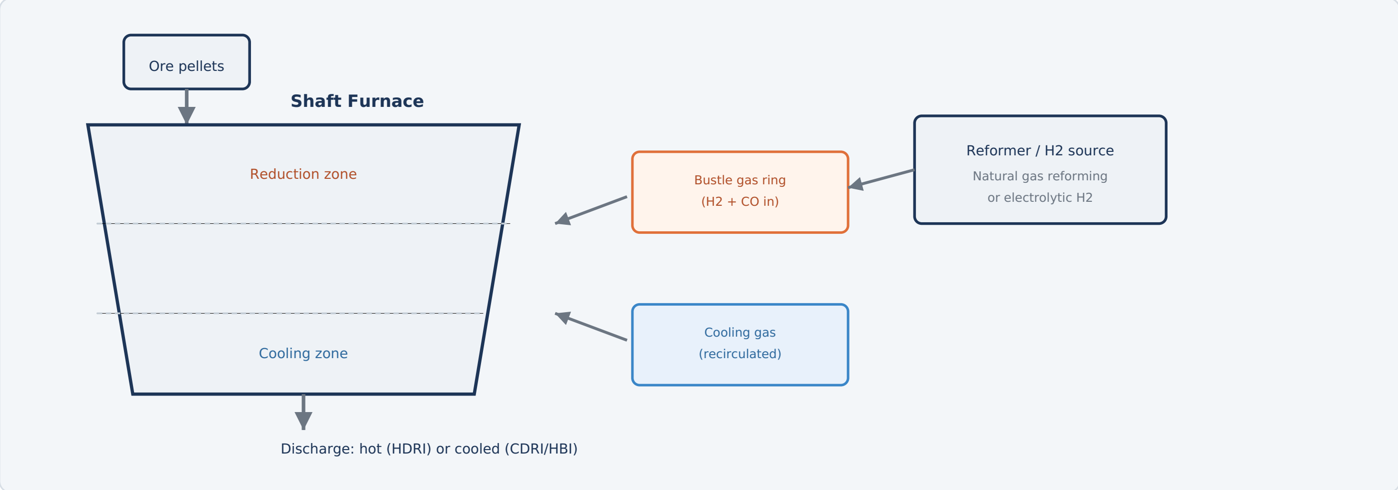


Fig. 2 — Ore pellets descend by gravity through the shaft while reducing gas, injected at the bustle ring, flows upward counter-current through them; a cooling zone at the bottom brings the product to a safe handling temperature before discharge.

HOW IT WORKS

1. **Feed ore pellets:** Iron oxide pellets (or lump ore) load continuously into the top of the shaft furnace.
2. **Reduce, counter-current:** As pellets descend, hot reducing gas injected at the bustle ring flows upward through them, stripping oxygen away without melting the iron.
3. **Control metallization:** Gas composition, temperature, and pellet residence time together set how completely the ore reduces to metallic iron.
4. **Cool (or don't):** A separate cooling gas zone at the bottom of the shaft can cool the product before discharge — or the plant can skip this and discharge hot for direct EAF charging.
5. **Discharge:** DRI exits as cooled DRI (CDRI), hot DRI (HDRI) for direct charging, or gets briquetted into HBI for safe storage and shipment.
6. **Recycle the gas:** Spent reducing gas is cleaned, re-heated, and largely recirculated back into the process rather than wasted.

TYPICAL PARAMETERS

Reduction zone temp.	**~800–1050 °C**
Residence time	**Several hours**
Product forms	**CDRI, HDRI, or HBI**

2 The Shaft Furnace & Counter-Current Reduction

Two ways to get the reducing gas — and why the choice defines a plant's carbon footprint

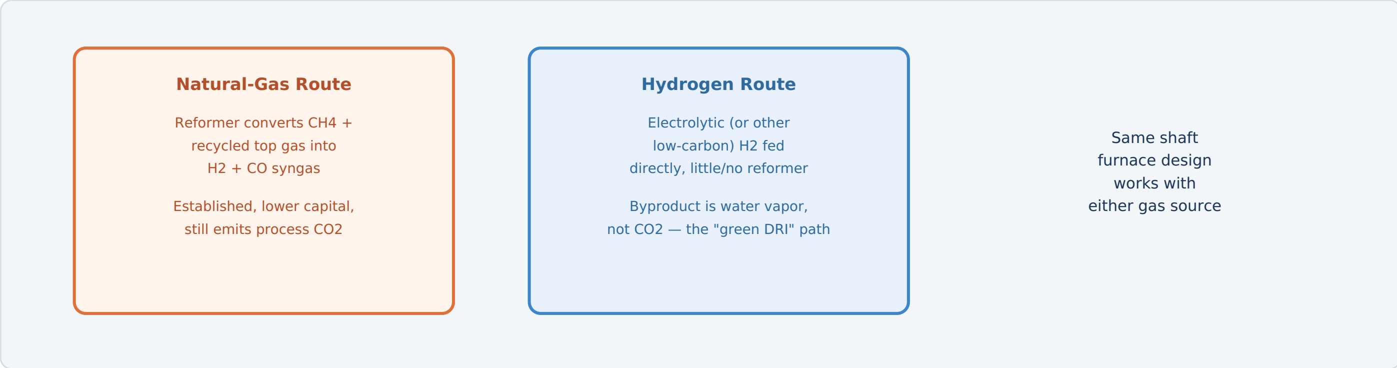


Fig. 3 — The shaft furnace itself is largely agnostic to gas source; what changes is the upstream gas supply system, which is where a plant's carbon footprint is actually decided.

WHY GAS COMPOSITION CONTROLS PRODUCT QUALITY

Metallization degree — how completely the ore has been reduced to metallic iron — and residual carbon content in the DRI both depend on the reducing gas composition, temperature, and how long pellets spend in the reduction zone. Carbon content is actually a deliberate, useful target rather than an impurity to eliminate: carbon carried into the EAF acts as a fuel and reductant there too, reducing the electrical energy the EAF needs to melt and refine the charge.

THE BUSTLE GAS RING

Reducing gas enters through a ring of injection ports around the furnace shell — the "bustle" — positioned at the boundary between the reduction and cooling zones. This geometry is what makes the counter-current exchange work: gas injected here flows upward through descending, hotter pellets in the reduction zone, while a separate cooler gas stream handles the zone below.

ELECTRICAL & CONTROL EQUIPMENT

Reformer burners (NG route)	**High-temp process heating**
Electrolyzer power supply (H2 route)	**Major DC electrical load**
Bustle gas blowers	**High-volume, continuous**
Gas composition analyzers	**Continuous, feeds Level 2 model**

3 Electrical Equipment in Detail

From the reformer or electrolyzer feeding the shaft to the briquetting press finishing the product

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
ORE HANDLING & FEED		
Pellet/ore conveyor & feed system	Continuously feeds ore pellets or lump ore into the top of the shaft furnace.	Continuous duty, weight-controlled
Furnace charging distributor	Distributes incoming ore evenly across the shaft's cross-section for uniform reduction.	Motorized, position-controlled
REDUCING GAS SUPPLY		
Reformer burners & controls (natural gas route)	Convert natural gas and recycled top gas into H2/CO reducing gas.	High-temperature catalytic process
Electrolyzer power supply (hydrogen route)	Large rectified DC power system driving water electrolysis to produce hydrogen.	Major standalone electrical load
Bustle gas blowers/compressors	Deliver reducing gas into the furnace at the bustle ring at the required pressure and flow rate.	High-volume, continuous
Process gas recycle compressors	Recompress and recirculate spent reducing gas after cleaning.	Continuous duty
COOLING & PRODUCT HANDLING		
Cooling gas circulation fans/compressors	Circulate cooled gas through the lower shaft zone to cool product before discharge.	Where cold product is made
Hot DRI conveyor system	Enclosed, inert-atmosphere conveyor carrying hot DRI directly to an adjacent EAF.	Sealed against air ingress
Briquetting press drive (HBI)	High-force hydraulic or mechanical press that compacts hot DRI fines into dense briquettes.	Very high force, continuous cycling
INSTRUMENTATION & AUTOMATION		
Gas composition analyzers	Continuously monitor reducing and top gas composition, feeding process control.	Multiple sample points
Temperature sensors (multi-zone)	Track furnace temperature profile through reduction and cooling zones.	Continuous monitoring
Level 1/2 automation	Coordinates gas flow, temperature, and pellet descent rate to hit target metallization consistently.	Ties the whole furnace profile together

4 Hot Charging & the Pyrophoricity Problem

A product with a genuine tendency to catch fire on its own — and two very different ways to manage it

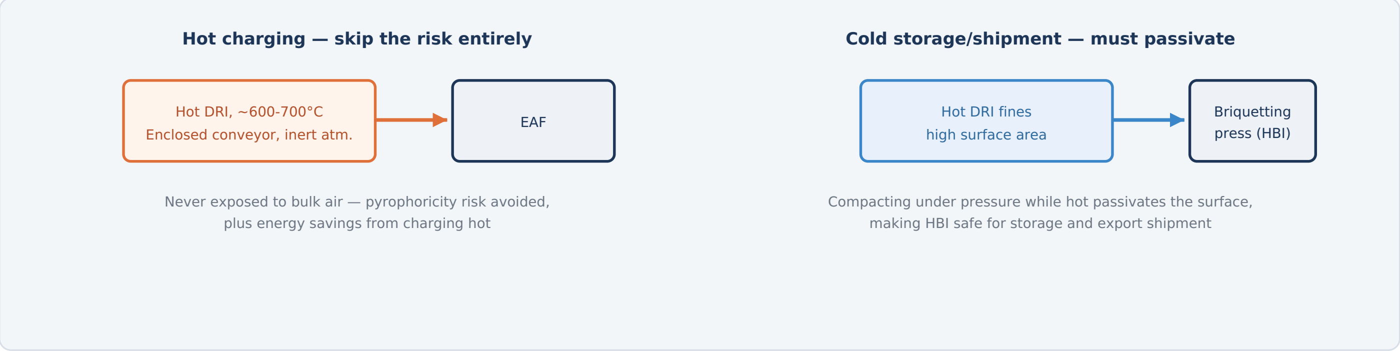


Fig. 4 — DRI's huge internal surface area (from the reduction process itself) makes it reoxidize readily in air, releasing heat; hot charging avoids the exposure entirely, while briquetting into HBI passivates the surface for anything that has to be stored or shipped cold.

WHY DRI IS GENUINELY PYROPHORIC

The same porous, sponge-like structure that makes direct reduction efficient — lots of surface area for the reducing gas to reach — also makes DRI eager to reoxidize the moment it's exposed to air or moisture. That reoxidation is exothermic; in a bulk pile or a ship's cargo hold, the heat can build faster than it escapes, leading to self-heating that has, in real documented industry incidents, progressed to fire. Water contact specifically adds a second, compounding hazard: the reaction between metallic iron and water can generate hydrogen gas, so applying water to a DRI fire or a wetted stockpile without following the product-specific emergency procedure can create a fire/explosion risk rather than simply cooling it. This isn't a theoretical hazard — it's the reason DRI and HBI have dedicated maritime shipping classifications and handling rules.

TWO WAYS PLANTS MANAGE IT

APPROACH	HOW IT WORKS
Hot charging	DRI moves directly from the furnace to an adjacent EAF on an enclosed, inert-atmosphere conveyor, never accumulating in bulk in open air — the safest option, and it saves EAF energy too.
Briquetting (HBI)	Hot DRI fines are compacted under high pressure into dense briquettes, which passivates the surface enough to be handled, stored, and shipped like an ordinary bulk commodity.

ELECTRICAL EQUIPMENT

Hot DRI conveyor	**Enclosed, inert atmosphere**
Briquetting press drive	**High-force, continuous cycle**
Storage pile temperature monitoring	**Detects self-heating**

KEY HAZARD

Cold DRI fines that aren't briquetted require strict moisture control and ventilated storage specifically to prevent self-heating. Water and moisture contact must be managed per the product-specific DRI/HBI emergency procedure — not ordinary active cooling — because the reaction between metallic iron and water can generate hydrogen gas, adding a fire/explosion hazard on top of the reoxidation risk.

5 Metallization vs. Throughput — The Balancing Act

Push the pellets through faster and they come out less reduced — quality and production rate pull against each other

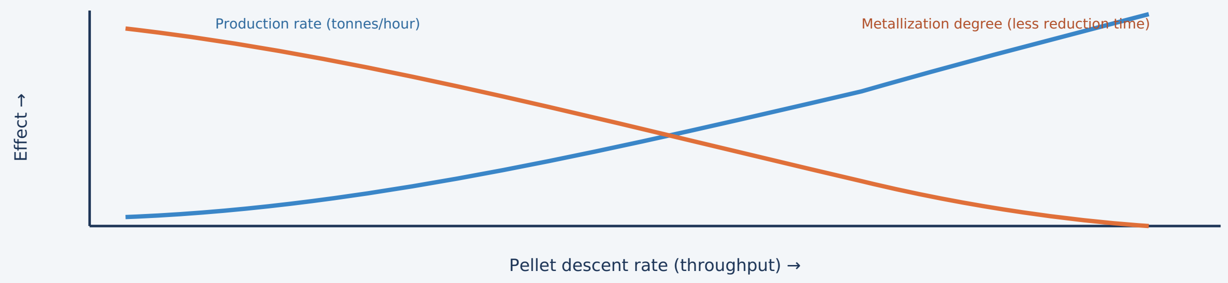


Fig. 5 — Faster pellet descent raises production rate but shortens residence time in the reduction zone, leaving more unreduced oxide behind — the two curves cross at whatever operating point a plant's target grade requires.

WHY UNDER-METALLIZED DRI IS A REAL COST, NOT JUST A TECHNICALITY

DRI that hasn't fully reduced still carries unreacted iron oxide — material that behaves like slag-forming gangue once it reaches the EAF rather than contributing usable iron units. Lower metallization means an EAF needs more energy and flux per tonne of actual steel produced from that charge, pushing the throughput trade-off back downstream rather than eliminating it.

TYPICAL PARAMETERS

Target metallization	**~90–95%**
Control variables	**Gas flow, composition, temp.**
Throughput control	**Discharge/descent rate**

HOW THE BALANCE IS ACTUALLY HELD

Rather than fixing descent rate and hoping, the Level 1/2 automation continuously coordinates gas flow rate, gas composition, and furnace temperature profile against the target metallization for that grade, adjusting pellet descent rate (via the discharge mechanism) to hold quality at the highest throughput that still meets specification.

6

Safety & Protection Systems

A flammable, toxic process gas and a product that wants to reoxidize the moment you look away

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Process gas leak detection	H2/CO concentration detected outside the intended gas path	Alarms operator; can trigger isolation and area evacuation — CO is toxic, H2 is highly flammable
Furnace atmosphere/pressure control	Furnace internal pressure drifts outside safe range	Adjusts gas flow to prevent air ingress or uncontrolled gas release
Hot DRI conveyor atmosphere monitoring	Inert atmosphere integrity compromised on the hot charging conveyor	Alarms operator; can halt conveyor to prevent bulk air exposure
Storage pile temperature monitoring	Self-heating detected in stored cold DRI	Triggers inspection and follows the product-specific emergency procedure — not automatic water application, since water contact can generate hydrogen
Water/moisture contact procedure	Any DRI fire, wetting, or suspected moisture ingress	Follows the product-specific DRI/HBI emergency response; ordinary firefighting water is not assumed safe by default
Reformer/electrolyzer process interlocks	Abnormal temperature, pressure, or gas composition	Trips the affected gas-generation system per its own safety design
Briquetting press guarding	Personnel access near the press while cycling	Physical guards and interlocked access points
Emergency stop (E-stop)	Manual activation anywhere on the platform	Removes power from drives per the defined E-stop category

WHY WATER CONTACT IS TREATED AS ITS OWN DISTINCT HAZARD, NOT JUST "MORE REOXIDATION"

Beyond ordinary reoxidation heating, the reaction between metallic iron and water can generate **hydrogen gas** — so an uncontrolled or well-intentioned application of water to a DRI fire or a wetted stockpile can create a compounding fire/explosion hazard rather than simply cooling it. Every water/moisture response for DRI and HBI must follow the product- and site-specific emergency procedure rather than generic firefighting assumptions.

WHY PYROPHORICITY SHAPES NEARLY EVERY DESIGN CHOICE ON THIS PAGE

Almost every major equipment decision in a DRI plant — hot charging versus briquetting, inert atmosphere conveyors, storage pile ventilation — traces back to the same underlying fact: this product actively wants to react with air, and every system here exists either to prevent that exposure or to manage it safely when it can't be avoided.

✓ Summary

What the DRI plant hands off to the EAF, and a recap of the process

FACTOR	NATURAL-GAS DRI	HYDROGEN ("GREEN") DRI
Reducing agent	H_2 + CO from reformed natural gas	Electrolytic (or other low-carbon) H_2
Reaction byproduct	Mix of H_2O and CO_2	Water vapor only
Maturity	Established, widespread today	Emerging, scaling rapidly
Role in decarbonization	Lower than blast furnace, not zero	Leading credible low-carbon steel pathway

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

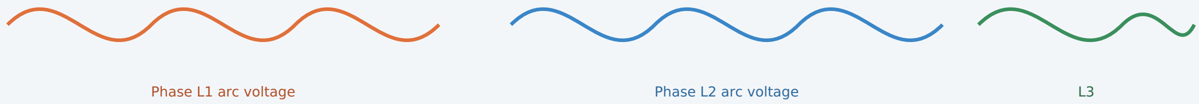
A **DRI plant** is an ironmaking process, not a steelmaking one — the direct counterpart to a blast furnace, feeding the EAF the same way a blast furnace feeds a BOF. Iron ore pellets descend through a **shaft furnace** while a hydrogen/CO **reducing gas**, injected counter-current at the bustle ring, strips away oxygen at ~800–1050°C — well below iron's melting point, so the pellet stays solid and porous throughout. That same porosity that makes reduction efficient also makes the product genuinely **pyrophoric**, which is why plants either **hot-charge** DRI directly into an adjacent EAF on a sealed, inert-atmosphere conveyor, or compact it into **HBI briquettes** that are safe to store and ship. Because the reducing gas can come from reformed natural gas or from low-carbon hydrogen using the same shaft furnace design, DRI-EAF is currently the clearest large-scale path toward substantially lower-carbon steelmaking — which is exactly why it keeps coming up alongside the EAF-versus-BOF conversation covered elsewhere in this book.

Why Three-Phase AC? The Original Melting Arc

No rectifier, no DC busbars — just the plant's native power, stepped down and arced directly

Before any conversion to direct current existed as an option, electric steelmaking already worked — and it worked using exactly the three-phase AC power already available at the plant. An AC EAF's furnace transformer simply steps grid voltage down to the very high current, low voltage an arc needs, and three graphite electrodes — one per phase — carry that current down to strike three independent arcs on the charge. No rectifier, no DC smoothing reactor, no specially engineered bottom electrode: **the AC arc furnace is the original technology this entire guide series' melting chapter is built around**, refined over more than a century, and it still represents the majority of EAFs operating in the world today.

Three independent arcs, one per phase, each re-igniting at every current zero-crossing



100-120 re-strikes per second, per phase — inherently more electrically "noisy" than a continuous DC arc

Fig. 1 — Each phase's arc extinguishes and re-strikes every time its AC current crosses zero; three phases doing this independently and somewhat unevenly is the source of AC's characteristic flicker and arc noise.

WHY AC REMAINED THE STANDARD FOR SO LONG

ADVANTAGE	WHY IT MATTERS
Lower capital cost	No rectifier, smoothing reactor, or DC busbar system to design, install, and maintain.
Simpler maintenance	Fewer specialized power-electronics components; a global supply chain of parts and expertise built up over decades.
Proven across all scales	From small foundry furnaces to full-scale steelmaking shops, AC EAF technology scales well in both directions.

TYPICAL PARAMETERS

Tap temperature	**~1600–1650 °C**
Transformer rating	**30–150+ MVA**
Electrode consumption	**~1.8–2.2 kg/tonne**
Tap-to-tap time	**~35–60 min**

1 The AC Melting Process

Charge, strike three arcs, refine, and tap — the same basic cycle as any EAF, run on native three-phase power

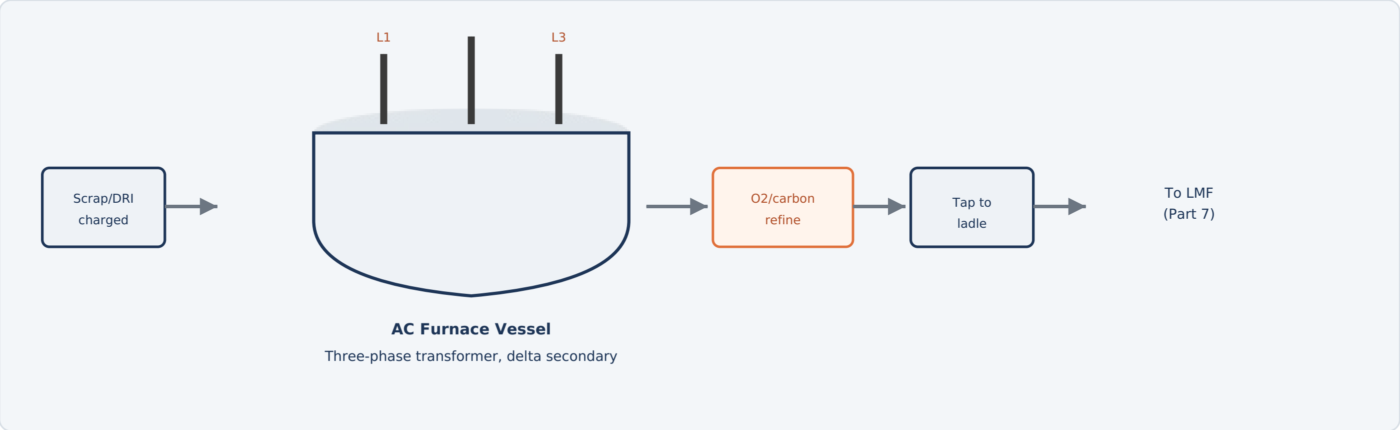


Fig. 2 — A three-phase transformer with a delta-connected secondary feeds three electrodes directly; each strikes its own arc on the charge, and the cycle from charge to tap repeats every heat.

HOW IT WORKS

1. **Charge:** Scrap and/or DRI load into the furnace shell, typically through the open roof by crane bucket.
2. **Power on:** Three electrodes lower and strike three independent arcs, one per phase, generating intense heat that melts the charge.
3. **Refine & oxidize:** Oxygen and carbon injection removes impurities, speeds melting, and adjusts carbon content; a foamy slag layer protects the arc and refractory.
4. **Tap:** Once the bath reaches temperature and composition targets, the furnace tilts and steel taps into a waiting ladle.
5. **Repeat:** The furnace tilts back for the next charge — a full heat typically cycles in well under an hour.

TYPICAL PARAMETERS

Tap temperature	**~1600–1650 °C**
Electrodes	**3, one per phase**
Secondary connection	**Delta, typically**
Output	**Liquid steel + slag**

2 Three-Phase Arc Behavior & Electrode Regulation

Three arms, three arcs, one shared, ever-changing scrap pile underneath all of them

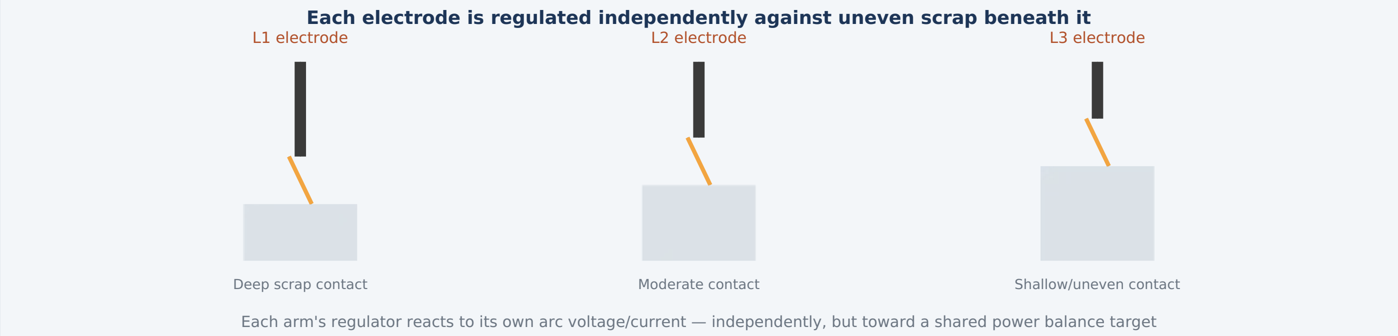


Fig. 3 — Scrap piles are never uniform beneath all three electrodes at once; each electrode's automatic regulator has to react to its own local arc conditions while the system as a whole still targets balanced power across all three phases.

WHY THIS IS A THREE-WAY PROBLEM, NOT THREE SEPARATE ONES

A single DC electrode only has to manage its own arc length. An AC furnace's three electrodes are electrically coupled through the same delta-connected secondary — current in each phase depends on the voltage across it relative to the other two — so an imbalance at one electrode (say, from a locally deep scrap pocket letting that arc run short and hot) doesn't just affect that phase's melting rate, it shifts the electrical balance for the whole furnace. The automatic electrode regulation (AER) system has to continuously reposition all three electrode arms, independently but coordinated, to keep power draw as balanced as scrap conditions allow.

THE DELTA SECONDARY CONNECTION

Rather than each electrode connecting to one phase and a neutral, AC furnace transformers typically connect their secondary in delta, with each electrode fed from the junction between two phases. This means each electrode's arc current is actually a function of two line voltages, not one — part of why three-phase arc behavior is a genuinely more complex control problem than it might first appear.

ELECTRICAL & CONTROL EQUIPMENT

Electrode arms (×3)	**Independent hydraulic positioning**
AER system	**Per-phase voltage/current sensing**
Secondary connection	**Delta, 3-phase**

KEY POINT

Severe, sustained phase imbalance is itself a power-quality problem — it's not just an internal furnace issue, it can show up as measurable imbalance back on the supply grid.

3

Electrical Equipment in Detail

No rectifier or DC busbars — but three of nearly everything else

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
INCOMING POWER & TRANSFORMER		
HV incoming switchgear / circuit breaker	Connects the furnace transformer to the plant's high-voltage supply and provides protection.	Rated to grid voltage, typically 33–138 kV class
Furnace transformer (3-phase)	Steps grid voltage down to the high-current AC the arcs need; secondary typically delta-connected to the three electrodes.	30–150+ MVA typical
On-load tap changer (OLTC)	Motor-driven mechanism that changes transformer taps under load to shift secondary voltage across melting phases.	Electrically driven, multi-position
Series reactor	Limits current surges and stabilizes the arc, especially during melt-down when arcs are least stable.	Per-phase or common, tap-adjustable
ELECTRODE & ARC REGULATION		
Electrode arms & hydraulic positioning (×3)	Independently raise/lower each of the three electrodes to maintain stable arc length.	One system per phase
Automatic electrode regulation (AER) system	Continuously measures each phase's arc voltage/current and repositions that electrode to hold target power balance across all three.	Three-channel, coordinated control
Graphite electrodes & holders (×3)	Conduct current from the arms into the furnace to create each phase's arc.	Consumable, replaced as they wear
Secondary bus tubes / flexible water-cooled cables	Carry very high current from the transformer secondary to each electrode arm.	Delta-connected, water-cooled
POWER QUALITY & AUXILIARY SYSTEMS		
Static VAR Compensator (SVC) / STATCOM	Actively counteracts voltage flicker and harmonics caused by the fluctuating three-phase arc load.	Sized to furnace transformer rating
Harmonic filters	Passive filtering for specific harmonic frequencies generated by arc re-ignition.	Tuned to dominant harmonics
Oxygen/carbon lance manipulators	Electrically driven positioning of injection lances and burners.	Motorized positioning
Tilting & hydraulic pump motors	Electric motors drive the hydraulic system that tilts the furnace for slagging and tapping.	Standard EAF auxiliary drive
Level 1/2 automation (PLC/DCS)	Controls transformer tap, electrode regulation targets, and process sequencing.	Coordinates all three phases together

4 Managing Flicker & Electrode Consumption

Push the arc harder and you melt faster — at the cost of a less stable arc and a hungrier electrode

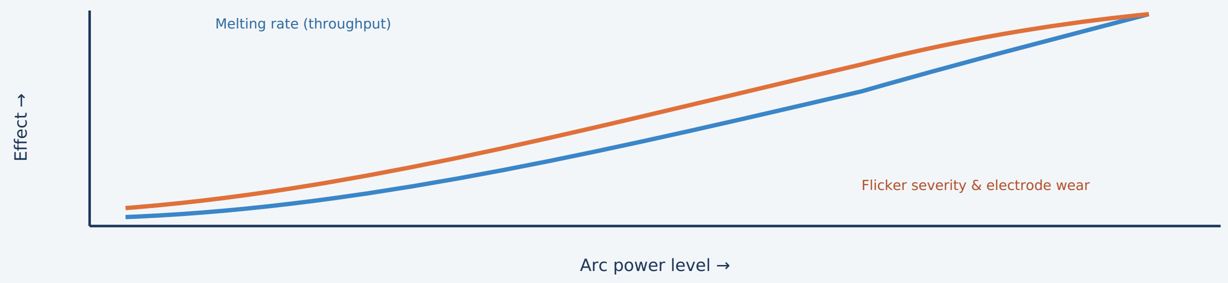


Fig. 4 — Running the arc harder melts faster, but both grid flicker and graphite electrode consumption climb right alongside it — the standard AC toolkit exists to push that trade-off as far right as the grid and budget allow.

THE STANDARD AC TOOLKIT

TOOL	WHAT IT DOES
SVC / STATCOM	Injects or absorbs reactive power in real time to counteract the voltage flicker the arc load creates, protecting other customers on the same grid.
Transformer tap changer	Selects a lower voltage tap during bore-in (when arcs are least stable, buried in scrap) and higher taps once a flat, molten bath lets the arc run more steadily.
Foamy slag practice	A thick foamy slag layer submerges the arc, reducing radiant heat loss to the roof and walls, stabilizing the arc, and reducing both flicker and electrode consumption at the same time.
Series reactor tap selection	Adjusts arc stability and short-circuit current limiting to suit current melting conditions.

TYPICAL PARAMETERS

Electrode consumption	**~1.8–2.2 kg/tonne**
SVC response	**Sub-cycle, real-time**
Tap changer steps	**Multiple, load-dependent**

KEY POINT

This is the same underlying trade-off the DC EAF guide frames as an advantage of rectifying to DC — the toolkit here is how AC furnaces manage that same physics without a rectifier at all.

5 Safety & Protection Systems

Three energized electrode circuits, a tilting vessel, and the same molten-metal hazards as any EAF

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Transformer differential protection	Internal fault detected via primary/secondary current mismatch	Trips the transformer off-line immediately
Phase imbalance protection	Sustained current imbalance beyond safe threshold across the three phases	Alarms operator; can reduce power or trip if severe
Electrode arm collision/position monitoring	Arm position conflicts with furnace structure or another arm	Blocks motion and alarms before contact occurs
Roof/door access interlock	Access attempted while electrodes are energized	Blocks access; requires confirmed power-off state
Series reactor/SVC protection	Abnormal current or thermal condition in power-quality equipment	Trips the affected system per its own protection scheme
Tilting hydraulic pressure monitoring	Loss of pressure during tilt/tap sequence	Halts tilt motion in a controlled, fail-safe position
Emergency stop (E-stop)	Manual activation anywhere on the platform	Removes power from drives per the defined E-stop category

WHY THREE ENERGIZED CIRCUITS CHANGE THE PROTECTION PICTURE

Every protection scheme on this furnace has to account for three electrode circuits that are electrically coupled through a shared delta secondary — a fault or imbalance on one phase is never purely a single-phase event, which is why phase imbalance monitoring sits alongside the more familiar arc-flash and molten-metal hazards shared with every EAF design.

6 AC vs. DC — When Each Makes Sense

Not a question of which is better — a question of which trade-offs fit a given furnace

Replacing three unstable AC arcs with a single continuous DC arc — the subject of the next chapter, Part 4 — buys real advantages: lower electrode consumption, lower grid flicker, and stronger, more uniform bath stirring. None of that makes AC obsolete, though — it means the two technologies suit different situations, and the choice usually comes down to furnace size, grid strength, and how much the added rectifier complexity is worth paying for.

WHERE AC STILL WINS

FACTOR	WHY IT FAVORS AC
Capital cost	No rectifier, smoothing reactor, or DC busbar system — meaningfully lower equipment cost, especially at smaller furnace sizes.
Maintenance simplicity	Fewer specialized power-electronics components and a broader, more mature global supply chain of spares and expertise.
Strong grid connections	Where the local grid can comfortably absorb AC flicker, the extra investment in DC's flicker reduction has less value.
Small-to-mid furnace sizes	Below the size range where DC's per-tonne electrode savings offset its added capital cost, AC often remains the more economical choice.

Fig. 5 — Both technologies do the same job; the right choice depends on furnace size, grid conditions, and how much rectifier complexity a given project is willing to pay for.

✓

Summary

What the AC EAF hands off downstream, and a recap of the process

FACTOR	AC EAF	DC EAF (PART 4)
Electrodes	Three, one per phase	One (plus bottom electrode)
Power conversion	None — native 3-phase AC	Rectified to DC
Capital cost	Lower	Higher
Electrode consumption	~1.8–2.2 kg/t	~1.0–1.3 kg/t
Downstream route	Grade-dependent secondary metallurgy, then casting	Same grade-dependent routing as AC

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

An **AC EAF** melts scrap and/or DRI using nothing but the plant's native **three-phase power** — a furnace transformer steps grid voltage down, and three graphite electrodes, each connected via a **delta secondary**, strike three independent arcs directly on the charge. Because each phase's current crosses zero 100–120 times a second, every arc repeatedly extinguishes and re-strikes, creating the characteristic electrical noise and grid **flicker** this technology is known for — managed day to day through **SVCs**, transformer tap selection, and **foamy slag practice**. Because the three electrodes are electrically coupled through the same secondary, **automatic electrode regulation** has to balance all three arcs simultaneously against whatever uneven scrap pile happens to sit beneath each one, a genuinely three-way control problem. None of this makes AC obsolete next to DC — it remains the lower-capital-cost, more broadly proven choice across most of the world's EAF fleet. AC and DC EAFs can feed the same downstream secondary-metallurgy and casting facilities either way — the actual treatment route is selected for the grade and plant configuration, not fixed by which furnace type made the steel.

Why DC? The Case for a Single-Electrode Furnace

Same job as an AC EAF — melting scrap/DRI into liquid steel — with a fundamentally different electrical design

A conventional AC electric arc furnace strikes three separate arcs — one per electrode, one per phase — down onto the charge. Those arcs are inherently unstable: they extinguish and re-strike 100 or 120 times a second as the AC current crosses zero, and the three arcs interact with each other unevenly as scrap melts unevenly beneath them. A **DC EAF** first converts the incoming AC power to direct current, then strikes a single, continuous, far more stable arc from one large graphite electrode (the cathode) down to the bath, with the circuit completed by one or more electrodes built into the furnace bottom (the anode). That single design change — adding a rectifier and a bottom electrode — is what drives almost every electrical difference covered in this guide.

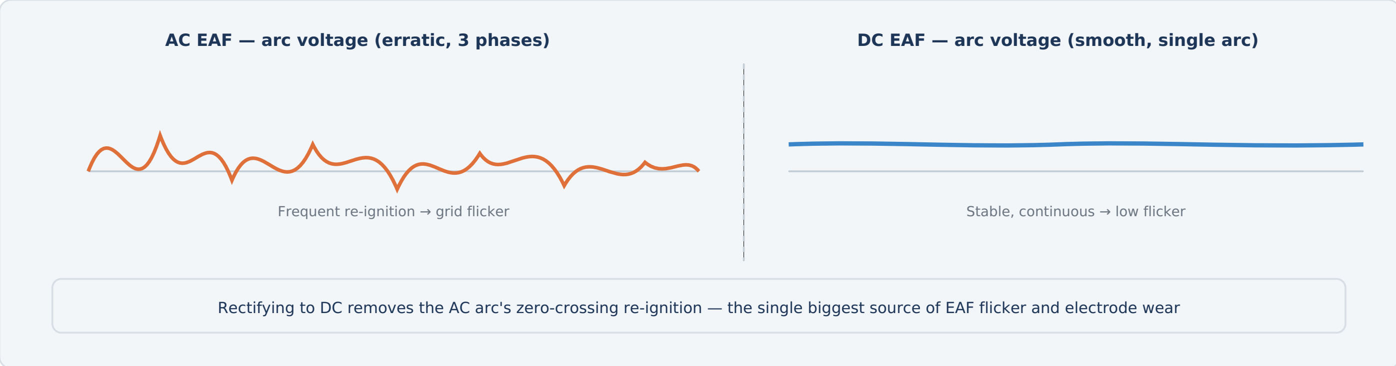


Fig. 1 — Conceptual arc voltage behavior: the AC arc's three phases repeatedly extinguish and re-strike; the DC arc, once struck, burns continuously and far more smoothly.

THE FOUR BENEFITS THAT JUSTIFY THE EXTRA ELECTRICAL COMPLEXITY

BENEFIT	WHY IT HAPPENS
Lower electrode consumption	One electrode instead of three means no electrode-to-electrode arcing losses and less total graphite surface exposed to oxidation; typically ~1.0–1.3 kg of graphite per tonne of steel versus ~1.8–2.2 kg/t for AC.
Lower flicker	A single, continuously burning DC arc doesn't repeatedly extinguish and re-strike the way an AC arc does at every current zero-crossing, so the load on the grid is far steadier — a major benefit on weaker grids or where flicker limits constrain furnace size.
Stronger, more uniform bath stirring	Unidirectional current flowing from the top electrode down through the bath to the bottom electrode generates a Lorentz-force stirring pattern that homogenizes temperature and chemistry faster than AC's less consistent stirring.
Quieter, more consistent operation	Without constant re-ignition, arc noise and mechanical vibration on the furnace structure are both reduced.

WHAT IT COSTS YOU

A DC EAF needs a full AC-to-DC conversion system (rectifier, smoothing reactor, DC busbars) and a specially engineered bottom electrode capable of carrying tens of kiloamps back through the furnace shell — both add significant capital cost and new categories of electrical equipment that a conventional AC furnace simply doesn't have.

WHERE DC EAFS ARE TYPICALLY USED

Mid-to-large melt shops where electrode cost savings and grid flicker limits justify the added rectifier investment — commonly in the ~50–150+ tonne furnace size range, and often favored on weaker regional grids where AC EAF flicker would be a problem.

1 The Melting Process

Charge to tap — the metallurgical steps are the same as an AC furnace, but the current path is not

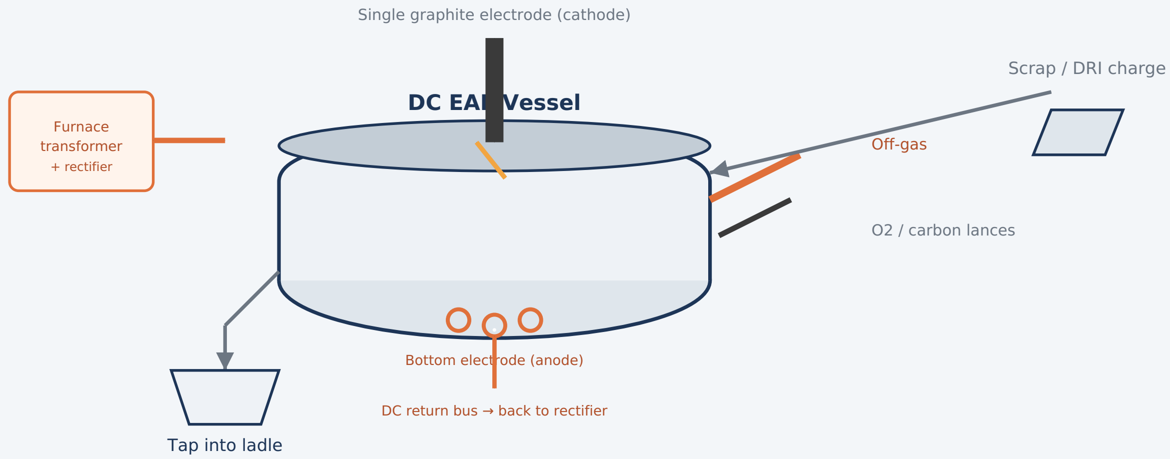


Fig. 2 — Current enters through the single top electrode, arcs to the charge/bath, and returns through the bottom electrode and DC bus back to the rectifier — a closed loop very different from AC's phase-to-phase path.

HOW IT WORKS

1. **Charge:** Scrap and/or DRI is loaded into the furnace shell, same as an AC furnace.
2. **Bore-in:** The single electrode lowers and strikes an arc, initially burning a hole down into the loose scrap pile — this stage is actually gentler on refractory than AC's three simultaneous bore-in points.
3. **Melt-down:** As the arc melts a pool beneath the scrap, the current path changes character: at first it arcs mostly to solid scrap, but once a liquid pool forms and reaches the bottom electrode, current can flow through the molten bath itself, closing the DC circuit and enabling strong electromagnetic stirring.
4. **Flat bath & refine:** Once fully molten, oxygen and carbon injection continue exactly as in an AC furnace — removing impurities, adjusting carbon, and building a foamy protective slag layer.
5. **Tap:** The furnace tilts and taps liquid steel into a ladle, just as an AC EAF does — the downstream secondary-metallurgy and casting equipment can be the same regardless of which furnace made the steel, subject to grade and plant routing.

TYPICAL PARAMETERS

Tap temperature	**~1600–1650 °C**
DC arc current	**~40–80 kA**
DC arc voltage	**~800–1400 V**
Electrode diameter	**~600–800 mm (single, large)**
Furnace size	**~50–150+ t**

2 The DC Power Supply System

From grid power to arc current — everything that has to happen before the arc even strikes

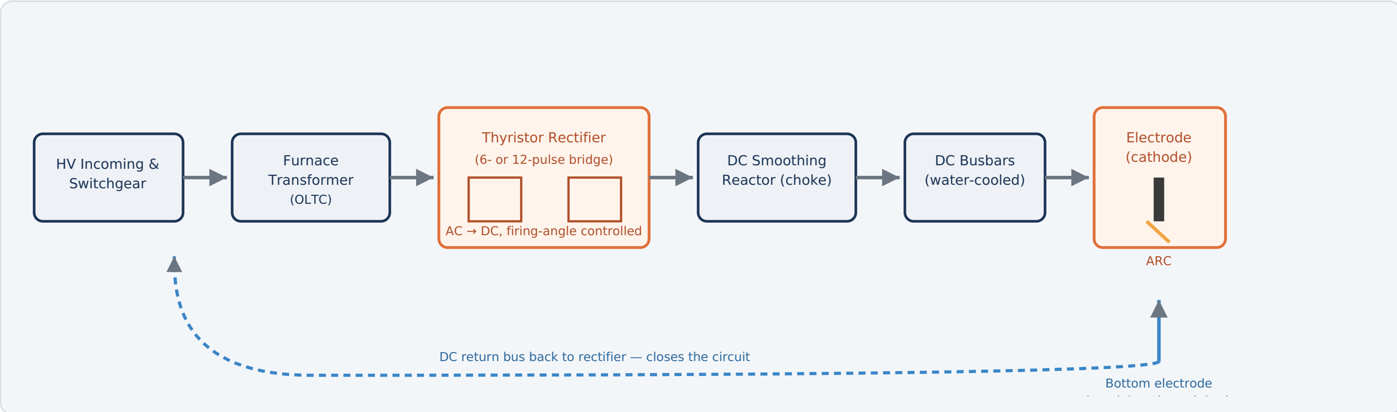


Fig. 3 — Grid power is stepped down, rectified to DC, smoothed, and delivered to the electrode; the circuit only closes once current returns through the bath and bottom electrode to the rectifier's DC return bus.

WHAT EACH STAGE DOES

STAGE	ROLE
HV incoming & switchgear	Connects the furnace to the utility grid and provides the primary circuit breaker and protection for the whole installation.
Furnace transformer (with OLTC)	Steps grid voltage down to the AC level the rectifier needs; an on-load tap changer allows coarse voltage adjustment for different melting phases without interrupting operation.
Thyristor rectifier bridge	Converts 3-phase AC into DC using phase-controlled thyristors. Adjusting the firing angle changes the output DC voltage almost instantly, giving very fast, precise arc power control — much faster than tap-changing alone.
DC smoothing reactor (choke)	A large inductor that smooths the rectifier's output ripple and limits how fast current can change, protecting both the arc's stability and the thyristors from damaging current spikes.
DC busbars	Heavy, typically water-cooled copper conductors carrying tens of kiloamps from the rectifier to the electrode arm.
Electrode (cathode) & arc	Current leaves the DC system here and crosses the arc gap to the charge or bath.
Bottom electrode (anode) & return bus	Current re-enters the electrical system through the furnace bottom and returns via a separate DC bus back to the rectifier, completing the loop (detailed on the next page).

3 Electrical Equipment in Detail

Everything from the incoming breaker to the electrode regulation system

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
INCOMING POWER & TRANSFORMER		
HV circuit breaker & switchgear	Primary protection and disconnect point between the utility grid and the furnace installation.	Rated to grid voltage, typically 33–138 kV class
Furnace transformer	Steps grid voltage down to the rectifier's required AC input; sized for the furnace's full melting power.	~60–120 MVA typical for mid-large DC EAFs
On-load tap changer (OLTC)	Motor-driven mechanism that changes transformer taps under load to shift the secondary voltage range for different melting phases (bore-in vs. flat bath).	Electrically driven, multi-position
RECTIFIER & DC CIRCUIT		
Thyristor rectifier bridge	Banks of series/parallel thyristors (SCRs) that rectify AC to DC; firing-angle control sets output voltage in real time.	6-pulse or 12-pulse (two bridges, 30° phase-shifted) configuration
Thyristor cooling system	Water-cooled heatsinks remove the substantial heat generated by high-current thyristor conduction losses.	Closed-loop water cooling, flow/temperature interlocked
Snubber & protection circuits	RC snubber networks and freewheeling diodes protect thyristors from voltage transients and reverse-recovery stress during switching.	Per-thyristor or per-module protection
DC smoothing reactor (choke)	Large air-core or iron-core inductor in the DC circuit limiting ripple and current rise rate (di/dt).	Often the single heaviest component in the DC circuit
DC busbars (water-cooled)	Carry full arc current — tens of kiloamps — from the rectifier to the electrode arm and from the bottom electrode back to the rectifier.	Laminated copper/aluminum, forced water cooling
DC ground fault detection	Monitors for unintended current paths to ground in the DC circuit — unlike AC, a DC ground fault won't self-clear at a current zero-crossing, so dedicated detection is essential.	Continuous monitoring, trips on threshold
ELECTRODE & ARC REGULATION		
Electrode arm & hydraulic positioning cylinder	Raises/lowers the single graphite electrode to control arc length and power.	Hydraulic actuation, electrically controlled valves
Automatic electrode regulation system	Closed-loop control comparing measured arc voltage/current against setpoints and repositioning the electrode within milliseconds to hold stable arc power.	Works together with rectifier firing-angle control
Arc voltage & current transducers	High-accuracy DC measurement feeding both the electrode regulator and the rectifier firing control loop.	DC current shunts/transducers rated for full arc current
FURNACE-SIDE AUXILIARY DRIVES		
Tilting hydraulic pump motors	Electric motors driving the hydraulics that tilt the furnace for slagging and tapping — same function as an AC furnace.	Standard EAF auxiliary drive
Oxygen/carbon lance manipulators	Electrically positioned injection lances for refining, identical in principle to an AC furnace.	Motorized positioning
Off-gas fan & damper drives	Extract furnace off-gas to the fume extraction system.	Shared with plant FES, see the steelmaking guide
Level 1/2 automation (PLC/DCS + process computer)	Coordinates rectifier firing angle, electrode position, tap changer, and auxiliary drives to hit the target melting power profile for each heat.	Tightly integrated rectifier + electrode control loop

4 Bottom Electrode & Electromagnetic Stirring

The furnace-bottom hardware that closes the DC circuit — and the stirring effect it creates for free

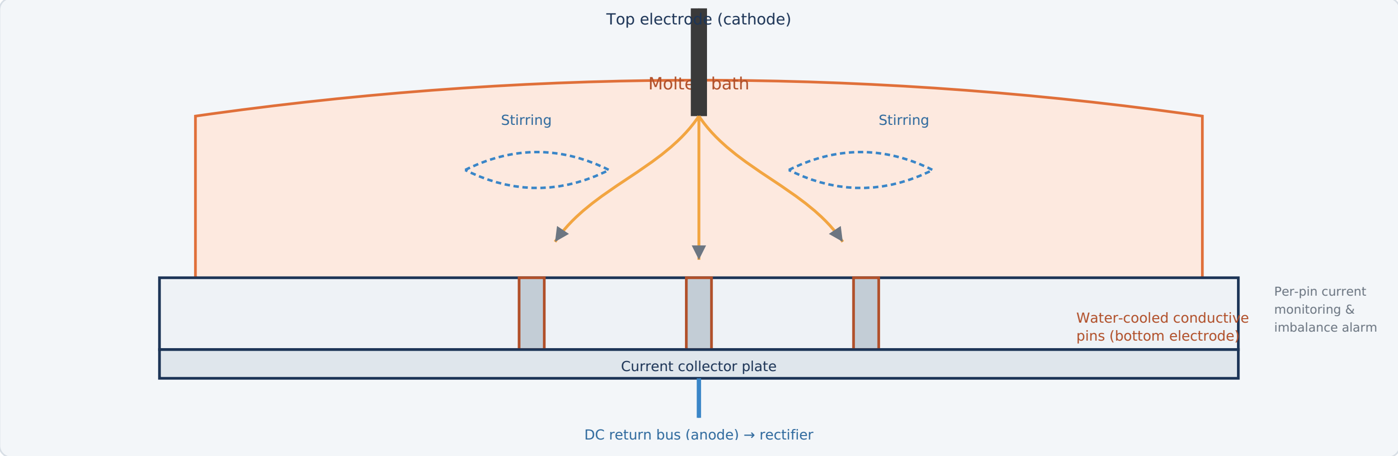


Fig. 4 — Multiple water-cooled conductive pins through the refractory bottom carry return current to a collector plate; the same vertical current flow through the bath produces a strong toroidal stirring pattern.

WHY A BOTTOM ELECTRODE IS NEEDED

A DC circuit must close somewhere — current can't just arc in and disappear. The bottom electrode is the second terminal (the anode) that lets current return from the bath, through the furnace bottom, to the DC bus and back to the rectifier. Two common designs are used:

- **Pin-type (conductive) bottom** — many small water-cooled steel or copper pins are embedded through the refractory lining and wired to a common collector plate. This is the most widely used design because losing one pin doesn't take the furnace out of service — the rest continue carrying current.
- **Single/massive contact design** — fewer, larger conductive elements or a conductive furnace bottom section; simpler but with less built-in redundancy.

THE STIRRING BONUS

Because current flows in one direction from the top electrode straight down through the bath to the bottom electrode, and that current generates its own magnetic field, the interaction between the two (a Lorentz force, J×B) drives a strong, roughly toroidal circulation inside the molten pool — without any extra equipment. This speeds melting, evens out temperature and alloy content, and reduces refractory hot spots compared to AC's less consistent current paths.

BOTTOM ELECTRODE ELECTRICAL EQUIPMENT

Current collector busbars	**Water-cooled, high current**
Per-pin current sensors	**Continuous monitoring**
Pin cooling water system	**Flow/temp interlocked**
Imbalance alarm/trip	**Protects against local burn-through**

KEY HAZARD

If refractory wear thins unevenly around a pin, that pin can carry a disproportionate share of current and overheat — a precursor to a dangerous molten-steel breakout at the furnace bottom. Continuous per-pin current monitoring with an imbalance trip is a standard, safety-critical protection layer, not an optional extra.

5 Power Quality — Harmonics & Flicker

DC solves one grid problem and introduces another — here's how both are managed

Trading three erratic AC arcs for one smooth DC arc dramatically reduces flicker — but the rectifier that makes this possible is itself a large, nonlinear load that injects harmonic currents back into the grid, and it also consumes reactive power depending on how far its thyristors are phased back from full conduction. Both effects need dedicated electrical equipment to manage.

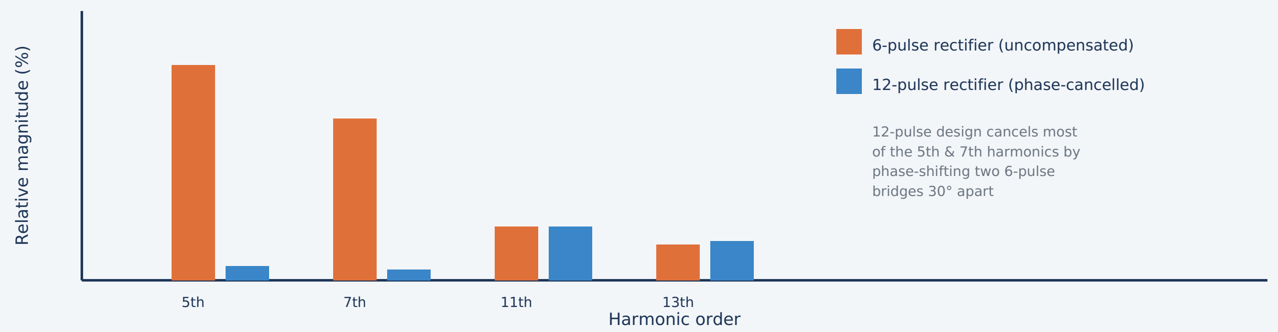


Fig. 5 — A basic 6-pulse rectifier injects strong 5th and 7th harmonics into the grid; feeding two 6-pulse bridges from transformer windings 30° apart (a 12-pulse arrangement) cancels most of that low-order content.

MANAGING HARMONICS

APPROACH	HOW IT WORKS
12-pulse (or higher) rectifier design	Two 6-pulse thyristor bridges are fed from transformer secondary windings wired 30° out of phase (one wye, one delta) and their DC outputs combined — the characteristic 5th and 7th harmonics largely cancel between the two bridges.
Tuned harmonic filters	LC filter banks tuned to the remaining dominant harmonic orders (11th, 13th, etc.) provide a low-impedance path that traps that harmonic current before it reaches the grid.
Active harmonic filters / STATCOM	Power-electronic devices that inject compensating current in real time, covering a broad range of harmonic orders and responding to changing furnace load — used where filter banks alone aren't sufficient.

TYPICAL IMPROVEMENT VS. AC EAF

Flicker (Pst)	**Often 30–50% lower**
Electrode consumption	**~35–45% lower**
New equipment required	**Rectifier, filters, SVC/STATCOM**

KEY POINT

None of these power-quality benefits are automatic — they depend entirely on correctly sized rectifier pulse configuration, filters, and reactive compensation. An undersized or poorly tuned system can still cause grid disturbances despite the DC arc itself being smooth.

MANAGING REACTIVE POWER

Thyristor rectifiers draw reactive power from the grid whenever their firing angle is phased back from zero (which is most of the time, since firing angle is exactly how arc power gets regulated). Static VAR Compensators (SVC) or STATCOMs are commonly installed alongside the rectifier to supply that reactive power locally, correcting the plant's power factor and further smoothing voltage at the point of common coupling with the grid.

6 Safety & Protection Systems

The electrical trip logic unique to (or intensified by) a DC EAF's rectifier and bottom electrode

Many EAF protection systems — tilting interlocks, off-gas monitoring, general electrical safety — are shared with AC furnaces and covered in the main steelmaking guide. The systems below are the ones a DC EAF adds or substantially changes because of its rectifier and bottom electrode.

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
DC ground fault detection	Unintended current path to ground detected anywhere in the DC circuit	Trips the rectifier — DC faults don't self-clear at a zero-crossing the way AC faults can, so this must be an active, continuously monitored system
Bottom electrode current imbalance trip	One or more bottom electrode pins carry disproportionately high current relative to the others	Alarms operator; can automatically reduce power or trip the furnace to prevent local refractory burn-through
Bottom electrode cooling water interlock	Cooling water flow or return temperature to the pins goes outside safe limits	Trips furnace power — loss of pin cooling directly risks a molten steel breakout
Thyristor overcurrent / fuse protection	Current through a thyristor bridge exceeds its rated capability	Fast semiconductor fuses or breakers isolate the affected bridge/module
Thyristor cooling interlock	Cooling water flow or heatsink temperature to the thyristor stacks goes outside limits	Reduces or trips rectifier output before thermal damage occurs
DC reactor overcurrent / thermal protection	Smoothing reactor current or winding temperature exceeds rating	Trips rectifier output
Arc voltage/current limit protection	Arc parameters exceed the safe operating envelope (e.g., a dead short to the bath)	Electrode regulator rapidly raises the electrode and/or rectifier reduces firing angle to cut current
Electrode breakage detection	Sudden loss of expected electrical continuity through the electrode column	Alarms operator and halts electrode feed/lowering sequence
OLTC interlock	Tap change requested while conditions (current, position feedback) are outside safe limits	Blocks the tap change until conditions clear
Emergency stop (E-stop)	Manual activation anywhere on the furnace platform	Removes power from the rectifier and halts auxiliary drives per the defined E-stop category

WHY DC NEEDS ITS OWN PROTECTION PHILOSOPHY

AC protection relies partly on the natural current zero-crossing every half-cycle to help interrupt faults; DC current has no such crossing, so DC-side protection must be faster and more actively engineered — this is why ground fault detection, semiconductor fusing, and bottom electrode monitoring are treated as first-class, continuously active safety systems rather than backups.

✓ Summary

What the DC EAF hands off downstream, and a recap of the process

FACTOR	DC EAF	AC EAF (PART 3)
Electrodes	One (plus bottom electrode)	Three, one per phase
Power conversion	Rectified to DC	None — native 3-phase AC
Capital cost	Higher	Lower
Electrode consumption	~1.0–1.3 kg/t	~1.8–2.2 kg/t
Downstream route	Grade-dependent secondary metallurgy, then casting	Same grade-dependent routing as DC

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **DC EAF** melts scrap and/or DRI exactly like an AC furnace — charge, bore-in, melt-down, flat-bath refining, tap — but gets there through a completely different electrical path. Incoming grid power passes through a **furnace transformer** and a **thyristor rectifier** that converts it to DC, smoothed by a large **DC reactor** before reaching a single graphite **electrode (cathode)**. The arc it strikes is far more stable than any of AC's three phase arcs, which is what drives the whole point of the design: lower electrode consumption and much lower grid flicker. The circuit only closes once current returns through the bath to a **bottom electrode (anode)** — typically an array of water-cooled conductive pins — and back to the rectifier; that same current path happens to drive strong electromagnetic bath stirring as a side benefit. The trade-off is a materially more complex electrical plant: rectifier protection, harmonic filters, reactive power compensation, and bottom-electrode monitoring are all systems an AC furnace simply doesn't need — none of which makes AC obsolete, since the two technologies suit different furnace sizes, grid conditions, and capital budgets, covered in full in Part 3.

Why Oxygen Refining? The Chemistry of the Blow

A furnace that needs no external heat source at all — the chemistry supplies it

A blast furnace produces molten pig iron — "hot metal" — that's already liquid, but far from being steel: it carries roughly 4–4.5% carbon along with silicon, manganese, and phosphorus, all of which have to come out before it's usable. The Basic Oxygen Furnace does this with a single, brutally direct method: blow pure oxygen onto the molten bath and let it burn those impurities out. **Carbon oxidizes to CO gas, silicon and manganese oxidize into the slag, and every one of those reactions releases heat** — so much heat that the metallurgical bath needs no burners, no electrodes, and no external heat input during the blow. In fact, the process generates more heat than the bath needs, which is exactly why scrap steel is charged into the vessel first: it acts as a coolant, melting as the blow proceeds and letting the plant recycle scrap for free at the same time. (The BOF *facility* as a whole still consumes substantial external electrical energy elsewhere — for oxygen production, drives, fans, pumps, cooling water, and automation, all covered later in this guide; it's only the metallurgical reaction itself that needs no added heat.)

$$C + \tfrac{1}{2}O_2 \rightarrow CO \quad \bullet \quad Si + O_2 \rightarrow SiO_2 \quad \bullet \quad Mn + \tfrac{1}{2}O_2 \rightarrow MnO$$

All exothermic — every one of these reactions releases heat into the bath

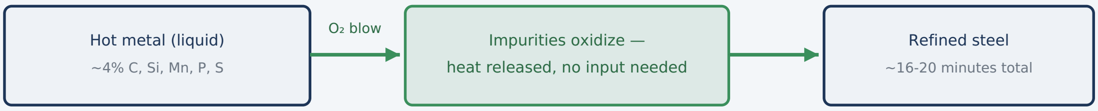


Fig. 1 — Unlike an EAF, which spends electrical energy melting solid scrap from cold, a BOF starts with liquid metal and refines it purely through exothermic oxidation — the reactions themselves are the heat source.

WHY "BASIC"?

The vessel is lined with basic (alkaline) refractory — dolomite or magnesia — rather than acid refractory, specifically because removing phosphorus requires a basic slag chemistry. This detail is where the process gets its name, distinguishing it from earlier oxygen-blowing predecessors that used acid linings and couldn't remove phosphorus effectively.

TYPICAL PARAMETERS

Hot metal charge	**~1300–1350 °C, ~4–4.5% C**
Blow time	**~16–20 min**
Heat size	**~150–350 t typical**
External heat input	**None — fully exothermic**

1 The Oxygen Steelmaking Process

Charge, blow, sample, and tap — one of the fastest refining cycles in all of steelmaking

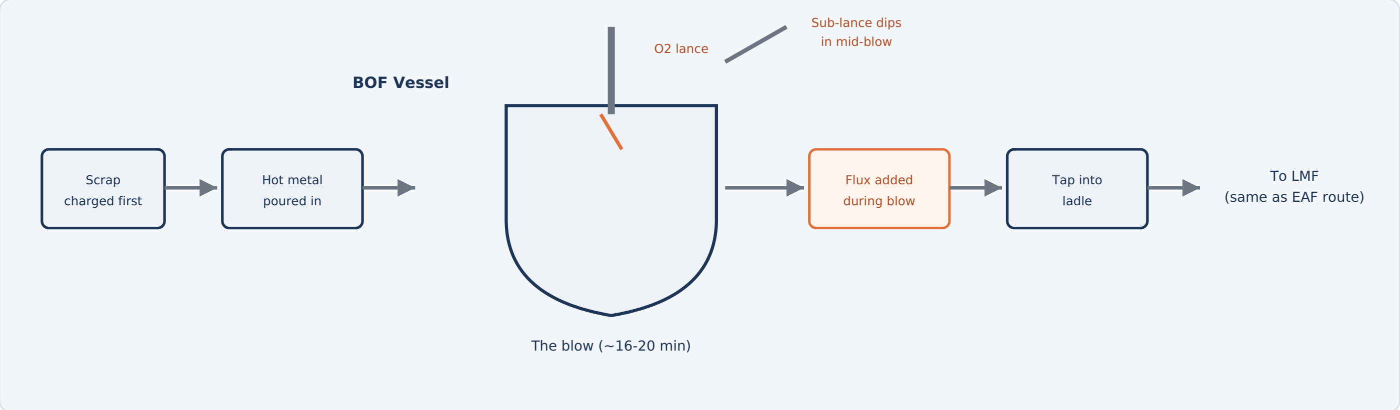


Fig. 2 — Scrap and hot metal are charged, the lance blows oxygen for roughly 16–20 minutes while flux builds a basic slag, a sub-lance checks progress mid-blow, and the finished steel taps into a ladle bound for the same downstream refining route as EAF-produced steel.

HOW IT WORKS

1. **Charge scrap, then hot metal:** The vessel tilts to receive scrap first (acting as a coolant and recycling feedstock), then molten pig iron from the blast furnace.
2. **Lower the lance, start the blow:** A water-cooled oxygen lance descends into the vessel and blows pure oxygen onto the bath at supersonic velocity, without touching the metal.
3. **Build a basic slag:** Lime and other flux additions build a basic slag during the blow that absorbs phosphorus and other oxidized impurities — sulfur removal is normally handled mainly before the BOF (hot-metal pretreatment) and/or during secondary metallurgy, since the strongly oxidizing blow conditions favor dephosphorization over desulfurization.
4. **Check progress mid-blow:** A retractable sub-lance dips in partway through to measure temperature and estimate carbon content without interrupting the blow.
5. **End the blow on target:** The lance withdraws once temperature and carbon both hit their targets simultaneously.
6. **Tap the steel:** The vessel tilts to pour refined steel into a waiting ladle through a tap hole, leaving the slag behind to be poured off separately.
7. **Hand off downstream:** The tapped steel proceeds to whatever secondary-metallurgy treatment the grade requires — often the LMF, VTD, or CCM route covered elsewhere in this guide — before casting.

TYPICAL PARAMETERS

Blow time	**~16–20 min**
Tap-to-tap time	**~30–40 min**
Oxygen purity	**>99.5%**
Scrap ratio	**~15–30% of charge**

2 The Oxygen Lance & Supersonic Blowing

Accelerating oxygen to Mach 2 so it penetrates the bath instead of just skating across the surface

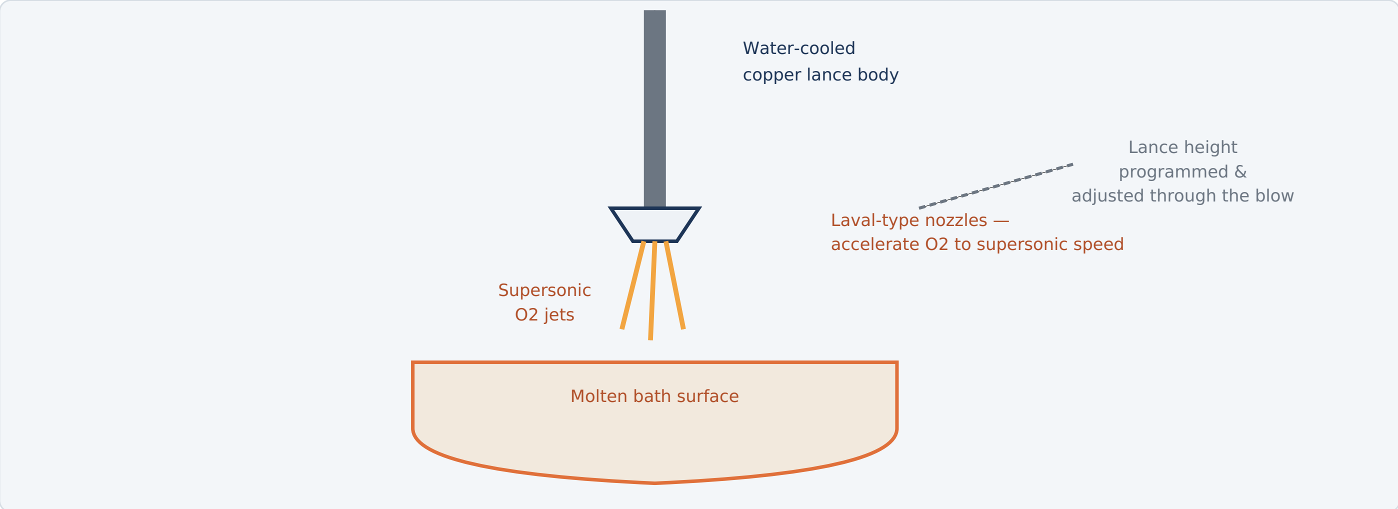


Fig. 3 — Multi-hole Laval nozzles accelerate oxygen to roughly Mach 2 as it exits the lance tip; lance height above the bath is continuously programmed through the blow to balance penetration against splashing.

WHY SUPERSONIC VELOCITY MATTERS

A slow-moving oxygen stream would simply skate across the bath surface, oxidizing only a thin layer and reacting far too slowly to refine hundreds of tonnes of metal in minutes. Laval-type nozzles — the same convergent-divergent nozzle geometry used in rocket engines — accelerate the oxygen jet to roughly Mach 2 as it exits the lance tip, giving it enough momentum to penetrate deep into the bath and drive intense mixing, which is what makes a 16–20 minute blow time possible at all.

WHY LANCE HEIGHT IS A PROGRAMMED, NOT FIXED, VARIABLE

LANCE POSITION	EFFECT
High ("soft blow")	Gentler surface impact, promotes early slag formation, less splashing — used early in the blow.
Low ("hard blow")	Deeper penetration, faster decarburization — used mid-blow, but risks splashing and refractory wear if held too long.

Most blow programs raise the lance again near the end to gently finish the heat without violent splashing as the bath reaches its final, lowest-carbon and hottest condition.

ELECTRICAL & CONTROL EQUIPMENT

Lance hoist drive	**Motorized, position-programmed**
Oxygen flow control valves	**High-capacity, precision metered**
Lance cooling water system	**Closed-loop, continuous**
Air separation unit (ASU)	**On-site oxygen production**

KEY POINT

The oxygen itself comes from an on-site air separation plant — a major electrical facility in its own right, running large cryogenic air compressors continuously to keep the BOF supplied.

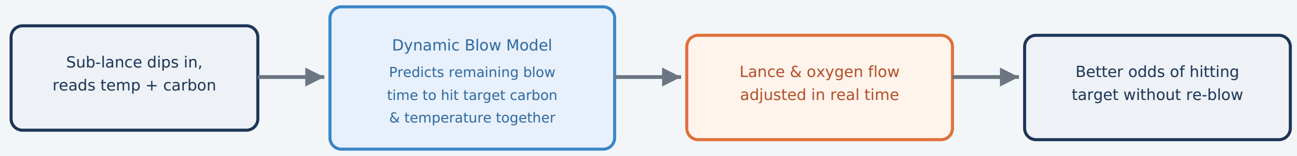
3 Electrical Equipment in Detail

From the trunnion drive that tilts a 300-tonne vessel to the sub-lance that reads it mid-blow

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
VESSEL & CHARGING		
Vessel tilting (trunnion) drive	Large geared motor drive that tilts the entire vessel for charging, sampling, and tapping.	Very high torque, precision angle control
Scrap charging crane	Overhead crane that loads scrap into the vessel before hot metal.	Heavy-duty, dedicated to charging bay
Hot metal charging crane/car	Positions and tilts the torpedo car or ladle to pour molten pig iron into the vessel.	Coordinated with vessel tilt sequence
OXYGEN BLOWING SYSTEM		
Oxygen lance hoist drive	Raises, lowers, and precisely positions the lance per the programmed blow profile.	Motorized, closed-loop position control
Oxygen flow control valves	Meter oxygen flow rate from the supply system to the lance throughout the blow.	High-capacity, fast-response
Air separation unit (ASU)	On-site cryogenic plant producing the high-purity oxygen the process consumes continuously.	Major standalone electrical facility
Lance cooling water system	Circulates cooling water through the lance body to prevent it melting in the furnace.	Closed-loop, continuously monitored
SUB-LANCE & AUTOMATION		
Sub-lance hoist drive	Rapidly inserts and withdraws the measurement probe mid-blow without stopping oxygen flow.	Fast cycle, precision positioning
Temperature/carbon sensor probes	Consumable-tip sensors providing a rapid temperature and carbon estimate each time the sub-lance dips.	Single-use probe tips
Level 1/2 automation (dynamic blow model)	Combines sub-lance readings with a real-time process model to predict the exact moment to end the blow.	Coordinates lance position, oxygen flow, and flux additions
FLUX, OFF-GAS & TAPPING		
Flux weighing & charging system	Electrically controlled bins and conveyors that add lime and other flux during the blow on a programmed schedule.	Automated, weight-verified
Off-gas (OG) hood & ID fan system	Captures the large volume of hot, CO-rich furnace gas; often includes gas cooling and can recover the gas for its fuel value.	Very high flow capacity
Tap hole & slag door mechanisms	Electrically/hydraulically actuated systems controlling steel tapping and separate slag removal.	Sequenced with vessel tilt

4 The Sub-Lance & Dynamic End-Point Control

Hitting target carbon and temperature at exactly the same instant, without stopping to check



Missing the end-point means a costly re-blow or a coolant correction — the model exists to avoid both

Fig. 4 — The sub-lance takes a fast reading mid-blow without halting oxygen flow; a dynamic model uses that single data point plus the process history to predict exactly when carbon and temperature will both hit target simultaneously.

WHY HITTING BOTH TARGETS AT ONCE IS HARD

Carbon content falls throughout the blow while temperature rises — but they don't move in lockstep, and the relationship depends on the specific hot metal chemistry and scrap ratio of that heat. Stopping too early leaves excess carbon, requiring a costly re-blow that adds time and wears the vessel lining faster. Stopping too late overshoots temperature and undershoots carbon, requiring coolant additions to correct — also costly. The sub-lance and dynamic model exist specifically to let operators end the blow once, correctly, without guessing.

A RECURRING IDEA ACROSS THIS GUIDE SERIES

This is the same fundamental pattern seen in the DC EAF's power supply control, the LMF's temperature and chemistry loop, and the CCM's dynamic solidification model: a fast, targeted measurement feeds a real-time model that adjusts the process before the outcome is locked in, rather than measuring only after the fact and hoping the result lands on target.

ELECTRICAL EQUIPMENT

Sub-lance hoist	**Fast-cycle, precision**
Sensor probe tips	**Consumable, single-use**
Dynamic blow model	**Level 2 automation**

KEY POINT

Before sub-lance technology, operators relied on flame color and sound to judge blow progress — genuinely an experience-based skill. Dynamic end-point control replaced that judgment with repeatable, real-time data.

5 Safety & Protection Systems

Pure oxygen, molten metal eruptions, and a vessel that tilts hundreds of tonnes of liquid steel

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Slopping/eruption detection	Bath eruption risk detected via sound, vibration, or off-gas signature during peak decarburization	Automatically raises the lance and can reduce oxygen flow to calm the bath
Oxygen system cleanliness enforcement	Standard procedure for all oxygen-contact equipment	Strict no oil/grease policy — pure oxygen is a severe fire accelerant with hydrocarbons
Oxygen leak detection	Abnormal oxygen concentration detected in the work area	Alarms operator; isolates affected supply section
Hot metal charging interlock	Charging sequence or vessel position not confirmed safe	Blocks pour until conditions are verified — moisture contact with hot metal is explosive
Vessel tilt position interlock	Tilt angle not correct for the commanded operation (charge, blow, tap)	Blocks the next step until position is confirmed
Off-gas CO monitoring	Carbon monoxide detected outside the intended gas path	Alarms operator; can trigger area evacuation procedures
Refractory wear monitoring	Lining thickness approaches minimum safe limit after repeated heats	Schedules a reline before failure risk becomes significant
Emergency stop (E-stop)	Manual activation anywhere on the platform	Removes power from drives per the defined E-stop category

WHY OXYGEN CLEANLINESS IS TREATED SO STRICTLY

Pure oxygen doesn't just support combustion, it dramatically accelerates it — even trace hydrocarbon contamination on a valve or fitting can ignite violently in a pure-oxygen environment. Every component in the oxygen supply path is cleaned and maintained to oxygen-service standards specifically because of this.

6 BOF vs. EAF — Why New Mills Choose Electric

BOF still produces roughly a third of the world's steel — but very little new BOF capacity has been announced in recent years

BOF and EAF aren't competing add-ons to the same plant — they're two entirely different starting points for making liquid steel, requiring two entirely different upstream supply chains. A BOF is only ever one part of a full **integrated steelworks**, typically drawing on an upstream blast-furnace ironmaking supply chain — coking coal, iron-bearing burden (sinter, pellets, and/or lump ore in plant-specific combinations), and the blast furnace itself — to supply hot metal. An EAF needs none of that — it melts scrap and/or DRI directly with electricity. That single structural difference is what's driving most new steelmaking capacity decisions toward EAF as of this writing.

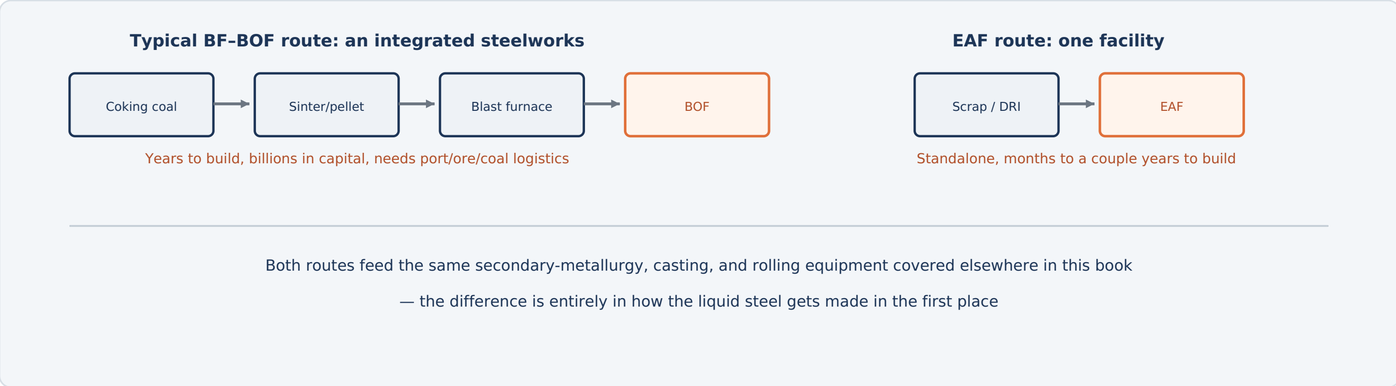


Fig. 5 — A BOF is inseparable from an entire integrated steelworks upstream of it; an EAF is a complete melting solution on its own, which is the root of nearly every advantage driving new investment toward it.

THE MAIN REASONS NEW CONSTRUCTION FAVORS EAF

FACTOR	WHY IT FAVORS EAF
Capital cost & build time	An integrated BF-BOF works costs billions and takes years to build; a standalone EAF mini-mill can be built for a fraction of the cost in a fraction of the time.
Feedstock flexibility	EAF runs on scrap and/or DRI, both available on the open market; BOF is locked to a captive blast furnace supply chain that itself depends on ore and coking coal.
Carbon emissions	The BF-BOF route is one of the most carbon-intensive processes in heavy industry; EAF using scrap emits dramatically less CO2 per tonne, and improves further with lower-carbon electricity.
Scale & location flexibility	Blast furnaces need very large, continuous scale to be efficient; EAF mini-mills can be sized and sited close to scrap supply and regional demand.
Operational flexibility	A blast furnace runs continuously for years between relines; an EAF can start, stop, and scale output with market demand far more easily.

WHY BOF HASN'T DISAPPEARED

Existing integrated steelworks represent enormous sunk capital and often produce grades or volumes where BOF still holds real advantages, so BOF continues to supply a large share of global steel today. The shift isn't that BOF has stopped working — it's that relatively few new BOF-route projects are being announced compared with EAF, because every one of the structural advantages above applies specifically to *new* investment decisions. (This is a market trend, not a fixed law — it's worth checking current data if the specific pace of the shift matters for your purposes.)

✓ Summary

What the BOF hands off downstream, and a recap of the process

FACTOR	BOF (INTEGRATED ROUTE)	EAF (ELECTRIC ROUTE)
Heat source	Fully exothermic — oxidation of impurities	Electrical arc energy
Required upstream supply	Hot metal from an integrated blast-furnace route (coke/reducing-agent + site-specific burden)	None — scrap/DRI direct
Typical new-build choice	Rare for new capacity	Dominant for new capacity
Downstream route	Grade-dependent secondary metallurgy, then casting	Same grade-dependent routing as BOF

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **Basic Oxygen Furnace** refines molten pig iron from a blast furnace into steel primarily using **oxygen** — a water-cooled lance blows a supersonic jet onto the bath, and the resulting exothermic oxidation of carbon, silicon, and manganese supplies all the heat the metallurgical reaction needs, with scrap charged in first specifically to absorb the excess; the surrounding facility still draws substantial external electricity for oxygen production, drives, and auxiliaries. A **sub-lance** dipping in mid-blow feeds a **dynamic end-point model** that predicts when to stop, improving the odds of hitting target carbon and temperature together without a corrective re-blow, in a single ~16–20 minute blow. The tapped steel then proceeds through whatever secondary-metallurgy treatment its grade requires — such as LMF or VTD, the same equipment covered elsewhere in this book — before continuous casting in the CCM, regardless of whether the steel started as BOF or EAF. What's changed isn't the chemistry, which still works exactly as it always has — it's that BOF's dependence on an entire integrated steelworks upstream of it makes EAF the overwhelmingly preferred choice for nearly every new steelmaking investment being made today.

PART 6 — FUME EXTRACTION SYSTEM

Why a Fume Extraction System? Capturing What the Furnace Doesn't Keep

Melting scrap and DRI releases far more than steel — dust, heavy metals, and hot gas all have to go somewhere

Every tonne of steel melted in an EAF also produces dust — fine particulate loaded with iron oxide, zinc, lead, and other metals volatilized out of the scrap, along with hot off-gas containing carbon monoxide and other combustion products. Left uncontrolled, that fume would fill the melt shop, endanger workers, and violate air quality regulations almost everywhere in the world. A **Fume Extraction System (FES)** — also called a dedusting system — exists to capture that fume at its source, cool it to a temperature filtration equipment can survive, strip out the particulate, and release clean air through a stack, all while keeping careful account of exactly how much it's emitting.

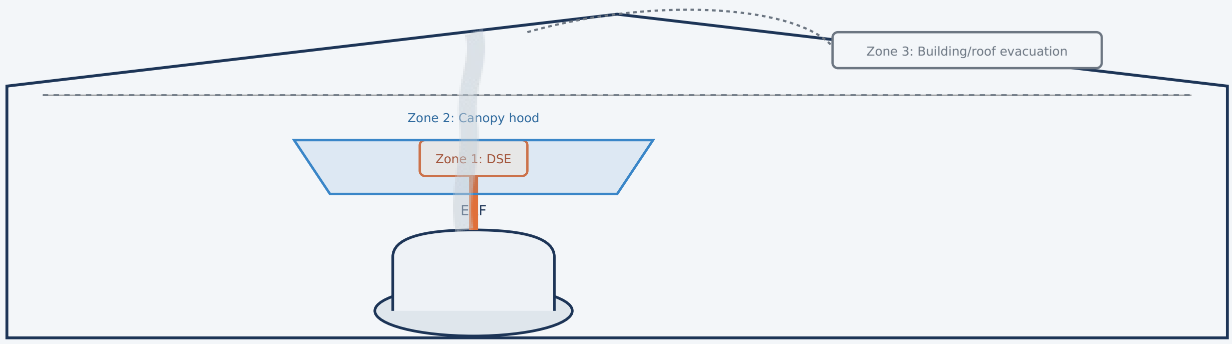


Fig. 1 — Fume escapes at more than one point, so effective capture needs three coordinated zones, not just one duct at the furnace.

THE THREE CAPTURE ZONES

ZONE	WHAT IT CAPTURES	WHEN IT MATTERS MOST
Zone 1 — Direct Shell Evacuation (DSE / "4th hole")	The hottest, most concentrated gas, drawn directly from inside the furnace shell through a dedicated fourth port alongside the three electrode ports.	Continuously during melting and refining, whenever the roof is on
Zone 2 — Canopy (roof) hood	Fugitive fume that escapes around the roof ring, electrode gaps, and slag door — inevitable even with good DSE draft.	Charging, tapping, and any time the roof is off or cracked open
Zone 3 — Building/roof evacuation	Whatever fume still escapes the first two zones and rises into the building — the last line of defense before it would otherwise vent through roof monitors into the outside air uncontrolled.	Continuously, at lower intensity, as a backstop

WHAT'S ACTUALLY IN THE FUME

EAF dust is a fine particulate rich in iron oxide, along with zinc and lead volatilized from galvanized and painted scrap, plus smaller amounts of other metals. In many jurisdictions it's classified as a hazardous waste requiring specialized handling and recycling rather than ordinary landfill disposal.

TYPICAL PARAMETERS

Total system airflow	**~600,000–1,200,000 Nm³/h**
Stack particulate	**Often <10 mg/Nm³**
ID fan motor power	**~3–8 MW**

1 The Fume Extraction Process

From furnace to stack — capture, cool, filter, exhaust, and dispose

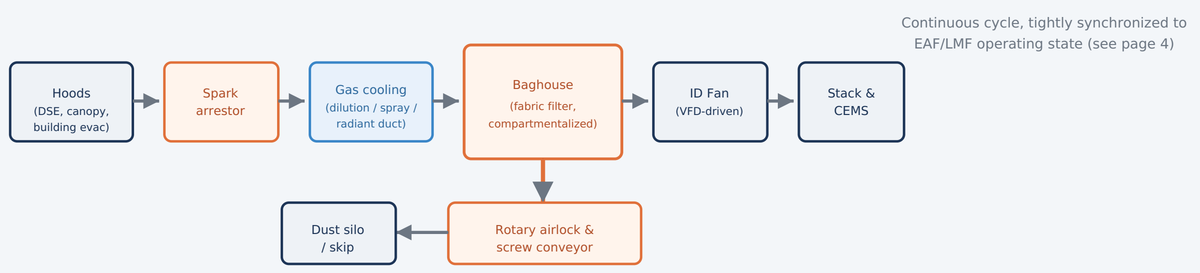


Fig. 2 — Gas is captured, cleared of sparks, cooled, filtered, and exhausted through a stack fitted with continuous emissions monitoring, while trapped dust drops out through a separate handling path.

HOW IT WORKS

1. **Capture:** Hoods at all three zones pull fume into the ductwork at the moment and location it's produced — most of it through direct shell evacuation while the furnace is powered on.
2. **Remove sparks:** A settling chamber or spark arrestor lets heavy embers and large particles drop out of the gas stream before they can damage ductwork or ignite dust further downstream.
3. **Cool the gas:** Radiant cooling duct sections, evaporative water spray, and/or dilution air brought in through control dampers bring gas temperature down from several hundred degrees to a level the filter fabric can survive continuously.
4. **Filter:** The cooled gas passes through the baghouse, where fabric bags trap fine particulate — including the heavy metals — down to sub-micron size.
5. **Exhaust:** The induced draft (ID) fan — the system's primary motive force — pulls the cleaned gas out through the stack, where continuous emissions monitoring confirms the plant is within its permitted limits.
6. **Handle the dust:** Collected dust drops from baghouse hoppers through rotary airlock valves onto screw conveyors, into a silo or skip for transport to a specialized recycler.

TYPICAL PARAMETERS

Gas temp at capture	**Up to ~1000+ °C locally**
Gas temp at baghouse inlet	**~120–200 °C (cooled)**
Filter area	**Tens of thousands of m²**
Number of bags	**Several thousand to 10,000+**

2 Gas Cooling & Capture Zone Control

Protecting the filter fabric, and shifting draft to wherever the furnace is actually leaking fume

COOLING THE GAS BEFORE IT REACHES THE BAGS

METHOD	HOW IT WORKS
Radiant cooling duct	Long runs of uninsulated (or water-jacketed) ductwork that let hot gas lose heat to the surrounding air simply by traveling further before reaching the baghouse.
Evaporative (spray) cooling	Fine water mist injected directly into the gas stream absorbs heat as it flashes to steam — a compact, fast-acting way to drop temperature sharply.
Dilution air	Ambient air drawn in through control dampers mixes with and cools the hot gas — simple and reliable, though it increases total gas volume the fan and baghouse must handle.

DAMPER SEQUENCING BY FURNACE OPERATING STATE

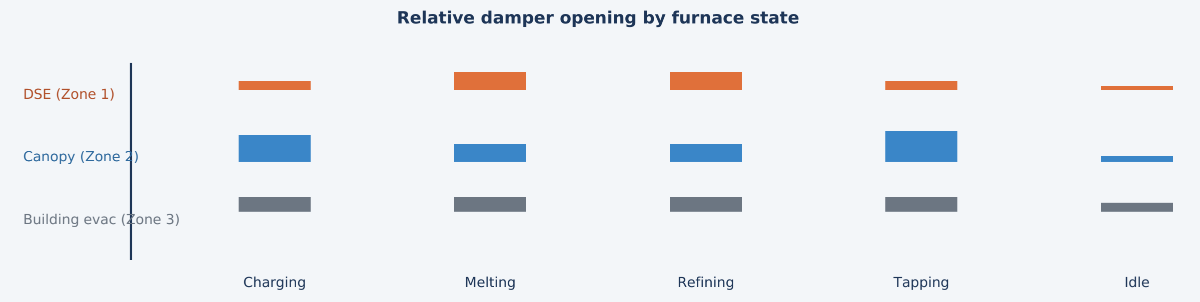


Fig. 3 — Damper positions shift automatically with furnace state: DSE dominates during melting/refining when the roof is sealed; canopy draft opens wide during charging and tapping when the roof is open or off.

Because a single ID fan drives the whole system, the total available draft has to be allocated intelligently between zones rather than run wide open everywhere at once — that wastes energy and still might not capture fume where it's actually escaping at that moment. Electrically actuated dampers, positioned in response to a signal from the furnace control system indicating its current operating state (charging, melting, refining, tapping, or idle), continuously reallocate draft to the zone that needs it most.

COOLING & ZONE CONTROL EQUIPMENT

Water spray pump motors	**Evaporative cooling**
Dilution air dampers	**Electrically/pneumatically actuated**
Zone damper actuators	**Position feedback, 4–20 mA control**
Furnace-state interface	**Digital signal from EAF/LMF PLC**

3 The Baghouse in Detail

Fabric filtration — simple physics, but a system that has to clean itself without ever fully stopping

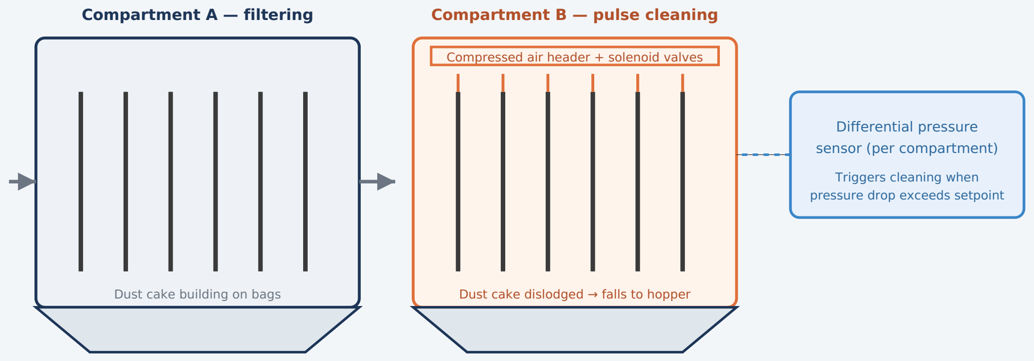


Fig. 4 — While one compartment keeps filtering gas online, another is briefly isolated and pulse-cleaned with compressed air — differential pressure across each compartment decides when cleaning is actually needed.

WHY A DUST CAKE IS A FEATURE, NOT A PROBLEM

As dust builds up on the fabric surface, that layer — the "dust cake" — actually becomes the primary filtering medium, trapping particles even finer than the fabric weave alone could. The goal of cleaning isn't to strip the bags completely bare; it's to remove enough cake to control pressure drop (and therefore energy use) while leaving a thin base layer that keeps filtration efficiency high.

COMPARTMENTALIZED, ONLINE CLEANING

A large baghouse is divided into many separate compartments, each independently isolated by dampers. This lets the system clean a subset of compartments at a time — briefly taking them offline for a pulse-jet or reverse-air cycle — while the rest keep filtering, so the whole system never has to stop.

FILTER MEDIA TEMPERATURE RATINGS

Polyester	**~130–150 °C continuous**
Aramid (e.g., Nomex)	**~200–220 °C continuous**
PTFE / fiberglass	**~250 °C+ continuous**

TYPICAL CLEANING PARAMETERS

Pulse air pressure	**~6–8 bar**
Cleaning trigger	**ΔP ~1,000–1,500 Pa**

4

Electrical Equipment in Detail

From the ID fan's medium-voltage motor down to the last hopper level sensor

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
ID FAN & MOTOR DRIVE SYSTEM		
Induced draft (ID) fan & motor	The single largest electrical load in the FES — a large centrifugal fan whose motor pulls gas through the entire system from hoods to stack.	~3–8 MW, medium-voltage motor (typically 4.16–13.8 kV class)
Variable frequency drive (VFD)	Controls fan speed continuously to match draft demand to furnace operating state, saving substantial energy versus running at fixed speed.	Medium-voltage VFD sized to full motor rating
Motor protection relays	Protect the large ID fan motor from overcurrent, ground fault, and thermal overload conditions.	Dedicated MV protection relay per motor
Fan bearing vibration/temperature sensors	Continuously monitor bearing health on a machine that runs nearly continuously at high load.	Vibration + RTD temperature monitoring
DAMPER & CAPTURE ZONE CONTROL		
Zone damper actuators (DSE, canopy, building evac)	Electrically or pneumatically position each zone's damper in response to furnace operating state signals.	Position feedback (potentiometer or digital encoder)
Dilution air damper actuators	Modulate ambient air intake for gas cooling and total volume control.	4–20 mA control signal
Draft/pressure transmitters	Measure negative pressure at key points (furnace roof, main duct) to confirm adequate capture draft.	Continuous electronic monitoring
GAS COOLING		
Evaporative spray pump motors	Deliver fine water mist into the gas stream for rapid, compact cooling.	Motor-driven high-pressure pumps
Gas temperature sensors (multi-point)	Monitor temperature through the cooling path and at the baghouse inlet, feeding automatic cooling control.	Thermocouples/RTDs at each cooling stage
BAGHOUSE CLEANING SYSTEM		
Compressed air compressor plant	Supplies the high-pressure air used for pulse-jet bag cleaning across the whole baghouse.	Motor-driven compressors, receiver tanks, air dryers
Pulse-jet solenoid valves	Fire in sequence, row by row, to blast compressed air back through the bags and dislodge the dust cake.	One valve per bag row, PLC-sequenced
Differential pressure sensors (per compartment)	Trigger cleaning cycles based on actual pressure drop rather than a fixed timer, optimizing bag life and cleaning air use.	Continuous monitoring, compartment-level
Compartment isolation dampers	Take one compartment offline for cleaning or maintenance while the rest of the baghouse keeps filtering.	Motor or pneumatic actuated
DUST HANDLING & EMISSIONS MONITORING		
Rotary airlock valve motors	Discharge collected dust from baghouse hoppers into conveyors without breaking the system's negative pressure.	Variable-speed motor drive
Screw conveyor motors	Move dust from the airlocks to a silo or skip for transport.	Standard industrial motor drive
Hopper level sensors	Detect dust buildup in hoppers to prevent overfilling or blockage.	Ultrasonic or capacitance level sensors
Continuous emissions monitoring system (CEMS)	Electronic opacity/particulate sensors in the stack, continuously verifying compliance with permitted emission limits.	Optical or triboelectric particulate monitors
Level 1/2 automation (PLC/DCS)	Coordinates fan speed, damper sequencing, cooling, and bag cleaning against real-time furnace operating state.	Tightly integrated with EAF/LMF control system

5 Draft & Flow Control — The Balancing Act

Too little draft and fume escapes capture; too much and you're burning energy for nothing

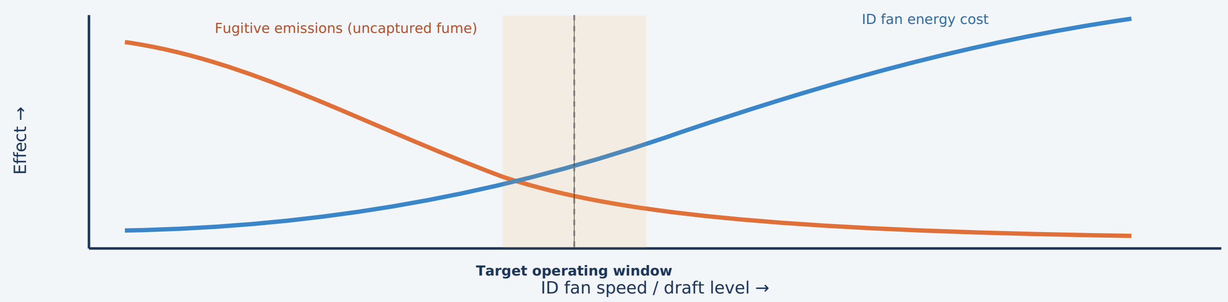


Fig. 5 — Draft that's too low lets fume escape capture (a visible haze in the shop); draft that's too high wastes fan energy and pulls excess dilution air through the baghouse. VFD control keeps the system in the target window as conditions change.

WHY FIXED-SPEED OPERATION DOESN'T WORK WELL

Fume generation isn't constant — it spikes sharply during charging (scrap bucket dumped in) and tapping, and drops during quieter refining periods. A fan running at a single fixed speed has to be sized for the worst case, wasting enormous energy the rest of the time. Modern systems instead use VFD-driven fan speed control, continuously adjusted against real-time draft/pressure feedback and the furnace's current operating state, to track demand rather than over-provision for it constantly.

WHAT OVER-DRAFTING ACTUALLY COSTS

CONSEQUENCE	WHY IT HAPPENS
Wasted fan energy	Fan power scales roughly with the cube of airflow — small increases in draft demand large increases in energy.
Excess dilution air	Pulling more air than necessary through leaks and open dampers dilutes the gas stream, increasing total volume the baghouse must filter.
Reduced baghouse life	Higher volumes mean higher air-to-cloth ratios across the filter media, accelerating wear and increasing cleaning frequency.
Furnace chilling (extreme cases)	Excessive draft at the DSE port can pull enough cold air into the furnace to measurably affect arc stability and energy efficiency.

CONTROL INPUTS TO THE FAN SPEED LOOP

Furnace operating state	**From EAF/LMF PLC**
Duct/roof pressure	**Continuous transmitters**
Baghouse pressure drop	**Compartment sensors**
Opacity/particulate trend	**From CEMS**

KEY POINT

A visible brown haze inside the melt shop is usually the first sign that draft has fallen out of the target window — not a cosmetic issue, but a signal that capture efficiency has dropped and worker exposure is rising.

6 Safety & Protection Systems

Fire, dust explosion, and toxic gas hazards unique to moving huge volumes of hot, dirty gas

An FES moves enormous volumes of hot gas laden with combustible dust and carbon monoxide through a network of ducts and equipment that workers must periodically access for maintenance. Its protection systems reflect those specific hazards.

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Spark/ember detection	Optical (IR/UV) sensors detect a hot spark or ember in the duct, typically just ahead of the baghouse	Triggers fast-acting water spray or CO_2 injection to extinguish it before it reaches the filter media
Duct/baghouse fire suppression	Confirmed fire or sustained high temperature inside ductwork or a baghouse compartment	Automatic suppression system activates; affected section can be isolated by dampers
Dust explosion venting/suppression	Rapid pressure rise inside a baghouse compartment consistent with a dust deflagration	Explosion vents relieve pressure outward; suppression systems can inject agent to arrest the event; isolation valves block flame propagation to adjacent compartments
Carbon monoxide monitoring	CO concentration in ductwork or confined spaces exceeds safe threshold	Alarms operator; blocks confined-space entry permits until levels clear
ID fan motor protection	Overcurrent, ground fault, or thermal overload on the large MV motor	Trips the motor off-line; alarms operator
Fan bearing vibration trip	Vibration exceeds safe threshold, indicating developing mechanical failure	Alarms and can automatically reduce speed or trip the fan before catastrophic failure
Rotary airlock/conveyor guarding & interlocks	Access panel opened while equipment is running	Stops the equipment before it can be accessed
Confined space entry interlock	Entry into ductwork, baghouse compartments, or hoppers requested	Requires isolation, purging, and atmosphere testing (CO, oxygen) before permit is issued
Emergency stop (E-stop)	Manual activation anywhere along the system	Removes power from the affected section's drives per the defined E-stop category

WHY THIS SYSTEM CARRIES REAL FIRE AND EXPLOSION RISK

Combustible metal dust, hot embers from the furnace, and carbon monoxide from incomplete combustion are all present simultaneously in an FES — a combination that, without dedicated detection and suppression systems, would make baghouse fires and dust explosions a routine occurrence rather than a rare one.

With Effective FES vs. Without — Summary

What a properly controlled dedusting system changes, and a recap of the process

FACTOR	POORLY CONTROLLED / UNDERSIZED FES	WELL-CONTROLLED FES
Worker exposure	Visible haze in the shop, elevated exposure to metal-bearing dust and CO	Fume captured at source before it enters the occupied space
Regulatory compliance	Risk of exceeding permitted stack emission limits	Continuous monitoring confirms compliance, often well under limits
Energy use	Either under-drafted (missing fume) or fixed at maximum speed (wasting energy)	VFD-driven, state-matched draft minimizes energy for a given capture target
Baghouse life	Shortened by excess volume, poor temperature control, or infrequent cleaning	Extended by right-sized flow, effective cooling, and pressure-triggered cleaning
Fire/explosion risk	Elevated without spark detection and suppression	Actively managed with dedicated detection and suppression systems

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **Fume Extraction System** captures melt shop emissions at three coordinated zones — **direct shell evacuation** at the furnace itself, a **canopy hood** for fugitive fume during charging and tapping, and **building evacuation** as a backstop — with electrically actuated dampers shifting draft between them as the furnace moves through its operating states. That hot, dust-laden gas passes through a spark arrestor and one or more **cooling stages** before reaching a **baghouse**, where fabric bags (helped by their own accumulated dust cake) filter out fine particulate, periodically cleaned compartment-by-compartment with compressed-air pulses without ever taking the whole system offline. A large, **VFD-driven ID fan** — the system's single biggest electrical load — pulls the cleaned gas out through a stack fitted with continuous emissions monitoring, while collected dust is discharged through rotary airlocks to a silo for specialized recycling. Getting the draft level right, continuously, across every furnace operating state is the central control challenge — and the fire, dust explosion, and CO hazards inherent to the gas being handled are why the system's safety interlocks are treated as seriously as the ones on the furnace itself.

PART 7 — LADLE METALLURGY FURNACE

Why an LMF? Splitting Melting from Refining

Why mills don't just tap the EAF straight into the caster

An EAF is excellent at one thing: melting scrap and DRI as fast as possible. It is a poor place to do careful metallurgy — the bath is turbulent, the atmosphere is oxidizing (needed to remove carbon and other impurities), and every extra minute spent refining in the EAF is a minute its enormous transformer and electrode system sit idle from a melting standpoint. The **Ladle Metallurgy Furnace** (also called a Ladle Furnace, LF) exists to take over everything after tapping: reheating, homogenizing, desulfurizing, precisely alloying, and cleaning up inclusions — in a smaller, cheaper-to-run station, so the EAF can go back to melting the next heat sooner.

THE FIVE JOBS AN LMF DOES

FUNCTION	HOW IT'S ACHIEVED
Reheat	Three graphite electrodes strike an arc that reheats the bath to compensate for the temperature lost during tapping and transport — without the aggressive oxygen blowing used in the EAF.
Homogenize	Argon gas injected through the ladle bottom stirs the bath, evening out temperature and composition from top to bottom.
Desulfurize	Under a reducing (low-oxygen), basic slag — the opposite atmosphere from the EAF — sulfur partitions out of the steel and into the slag far more effectively.
Alloy trim	With a calmer bath and precise temperature control, ferroalloys can be added and allowed to dissolve accurately to hit tight composition specifications.
Inclusion modification	Calcium treatment (via wire injection) converts hard, angular alumina inclusions into soft, liquid calcium-aluminate inclusions that won't clog the caster nozzle or weaken the final product.

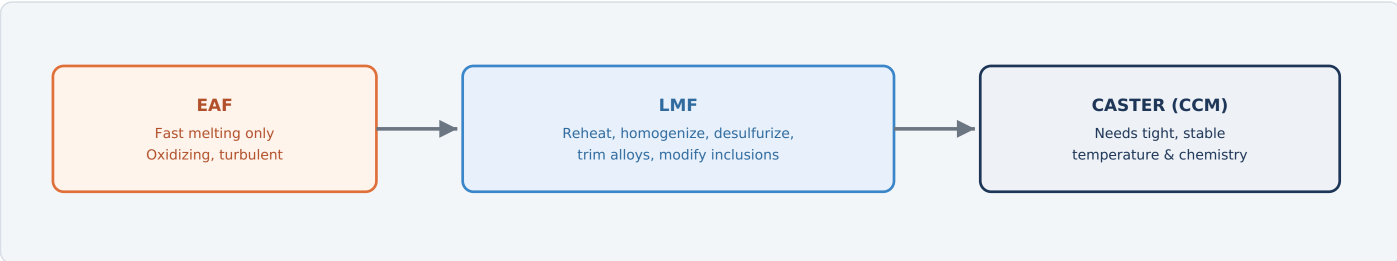


Fig. 1 — Decoupling melting (EAF) from refining (LMF) lets each station do what it's best at — and lets the EAF's tap-to-tap cycle shrink since it no longer has to wait through careful refining work.

THE ECONOMIC ARGUMENT

An EAF transformer and electrode system represent enormous fixed capital and power demand charges. Every minute it spends idling through slow refining chemistry is a minute it isn't melting. Moving refining to a smaller, lower-power LMF station lets the EAF's tap-to-tap time shrink substantially — often the single biggest productivity lever in a melt shop.

TYPICAL PARAMETERS

Treatment time	**20–45 min**
Transformer rating	**~10–25 MVA**
Temperature control target	**±5–10 °C at release**
Typical sulfur reduction	**~0.02–0.03% → <0.005%**

1 The LMF Process

From ladle arrival to release — reheating, stirring, and refining under a controlled atmosphere

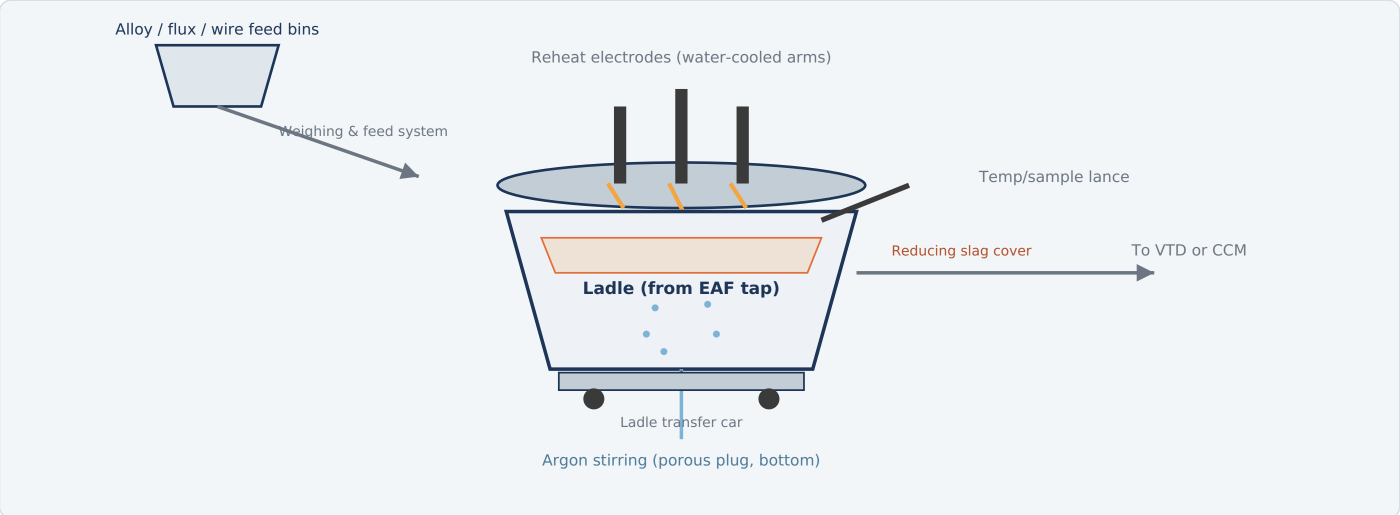


Fig. 2 — The tapped ladle sits under a lid with three electrodes; reheating happens through the slag cover while argon stirring and alloy additions bring the bath to target temperature and chemistry.

HOW IT WORKS

1. **Arrive & position:** The ladle, freshly tapped from the EAF, is moved on its transfer car into the LMF stand, and a water-cooled lid with three electrodes lowers over it.
2. **Strike the arc & reheat:** The electrodes lower until their tips sit just at or below the slag surface — a submerged-arc condition that traps most of the arc's radiant heat under the slag instead of losing it up to the roof, improving thermal efficiency and reducing electrode wear compared to an open arc.
3. **Stir continuously:** Argon injected through a bottom porous plug rises through the bath as a plume of fine bubbles, driving a steady circulation that mixes temperature and composition top to bottom and helps inclusions float up into the slag.
4. **Build a reducing slag:** Lime, fluorspar, and deoxidizers are added to create a low-oxygen, basic slag — the conditions sulfur needs to move out of the steel and into the slag.
5. **Trim alloys:** Once composition is roughly known from an initial sample, precise ferroalloy additions bring the bath to the exact target analysis.
6. **Sample & adjust:** An automated lance dips in for temperature and a steel sample; results feed back into further small corrections until both temperature and chemistry are within specification.
7. **Release:** The lid lifts and the ladle moves on — either directly to the caster, or to a vacuum degasser first if the grade needs extra-low dissolved gas content.

TYPICAL PARAMETERS

Treatment time	**20–45 min**
Reheat rate	**~3–5 °C/min**
Electrode diameter	**~350–500 mm (3, AC)**
Stirring gas	**Argon**
Slag condition	**Basic, reducing (low O2)**

2 The Electrical Power Supply System

A smaller, gentler cousin of the EAF's power system — three-phase, submerged-arc, no separate return bus

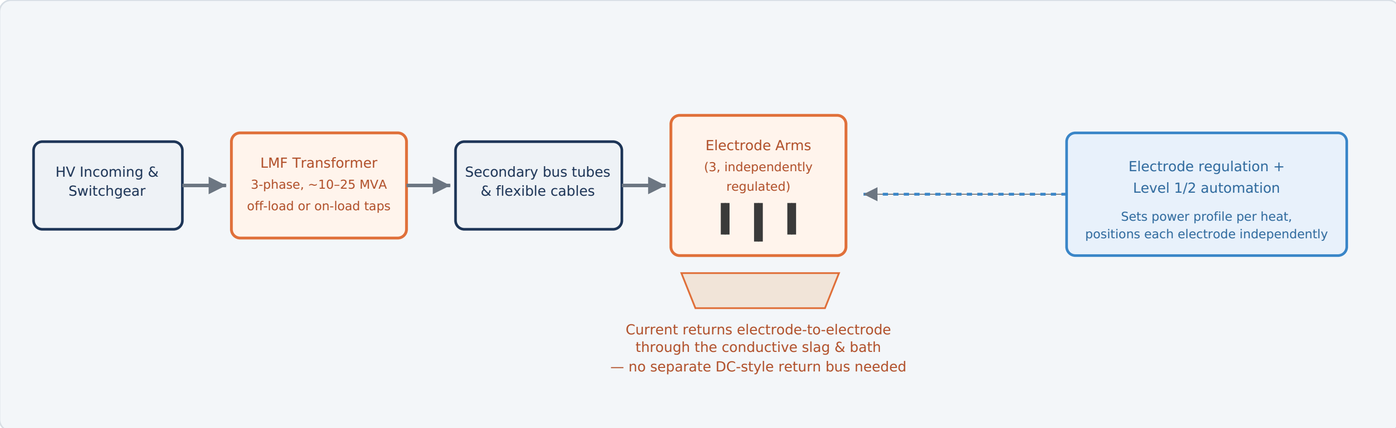


Fig. 3 — Like the EAF, an LMF runs on 3-phase AC power, but at far lower current and power density; current returns through the conductive slag and steel bath between electrodes rather than through a separate return path.

WHAT EACH STAGE DOES

STAGE	ROLE
HV incoming & switchgear	Connects the LMF to the plant's high-voltage supply, typically sharing the same substation infrastructure as the EAF.
LMF transformer	Steps voltage down to the level needed for arc reheating; rated far lower than the EAF transformer since the LMF only needs to reheat and hold temperature, not melt solid charge.
Secondary bus tubes & flexible cables	Carry current from the transformer to each of the three electrode arms, exactly as on an EAF but at a smaller scale.
Electrode arms (3, AC)	Independently positioned graphite electrodes that strike and maintain the submerged arc into the slag-covered bath.
The bath/slag as the return path	Unlike a DC EAF, there's no separate return busbar — current simply flows from each energized electrode, through the conductive molten slag and steel, to the other two electrodes, exactly as in a 3-phase AC EAF.
Electrode regulation & automation	Continuously adjusts each electrode's position to hold a stable arc and deliver the power profile the process model has calculated for that heat.

3 Electrical Equipment in Detail

Transformer through to alloy dosing — every powered system on the LMF stand

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
TRANSFORMER & INCOMING POWER		
HV switchgear & circuit breaker	Primary protection and disconnect between plant supply and the LMF transformer.	Shares substation with EAF in most shops
LMF transformer	Steps down to arc reheating voltage; often has more voltage taps than an EAF transformer to fine-tune power at low, stable heat-holding loads.	~10–25 MVA, 3-phase
Tap changer	Coarse voltage adjustment — on-load or off-load depending on installation — to match transformer output to the required power level.	Motor-driven if on-load
ELECTRODE & ARC REGULATION		
Electrode arms & hydraulic positioning cylinders	Raise/lower each of the three electrodes independently.	Electrically controlled hydraulic valves
Automatic electrode regulation system	Closed-loop control per phase, comparing arc voltage/current to setpoint and repositioning the electrode to hold a stable submerged arc.	Independent control loop per electrode
Graphite electrodes & holders	Conduct current from the arms into the slag-covered bath.	~350–500 mm diameter, smaller than EAF electrodes
Secondary bus tubes / water-cooled cables	Carry current from the transformer to each electrode arm.	Water-cooled, lower current rating than EAF equivalents
Roof/lid hoist drive	Electrically driven mechanism that raises and lowers the water-cooled roof carrying the electrodes.	Motor or hydraulic hoist
ARGON STIRRING SYSTEM		
Argon supply station	Regulates plant argon supply pressure down to process level before it reaches the flow control panel.	Pressure-reduced from header supply
Mass flow controllers	Electronically set and monitor argon flow rate to each porous plug, switching between soft-stir and vigorous-stir modes.	Two-plug systems common for redundancy
Back-pressure sensors	Detect a rise in back-pressure that signals a porous plug is becoming blocked, before stirring is lost entirely.	Continuous monitoring, alarms operator
ALLOY, FLUX & WIRE FEED SYSTEMS		
Alloy/flux weigh-feed system	Motor-driven bins, conveyors, and load cells that dose precise ferroalloy and flux additions per the chemistry model's recipe.	Multiple bins, gravimetric dosing
Calcium wire feeder	Motor-driven feed rollers inject calcium-silicon (CaSi) wire into the bath at a controlled speed for inclusion modification.	Speed-controlled feed motor, wire straightener
Temperature & sampling lance manipulator	Electrically driven arm that dips disposable thermocouple tips and sampling probes into the bath on demand.	Automated positioning, single-use consumable tips
HANDLING & AUTOMATION		
Ladle transfer car drive	Electric motor moving the ladle car between EAF, LMF, VTD, and caster positions.	Precision stopping for electrode/lance alignment
Level 1/2 automation (process model)	Calculates power input, argon flow, and alloy additions in real time to hit target temperature and chemistry by a scheduled release time.	Coordinates with EAF and caster scheduling

4 Argon Stirring in Detail

A small gas flow with an outsized effect on quality — and a system that has to be tuned carefully

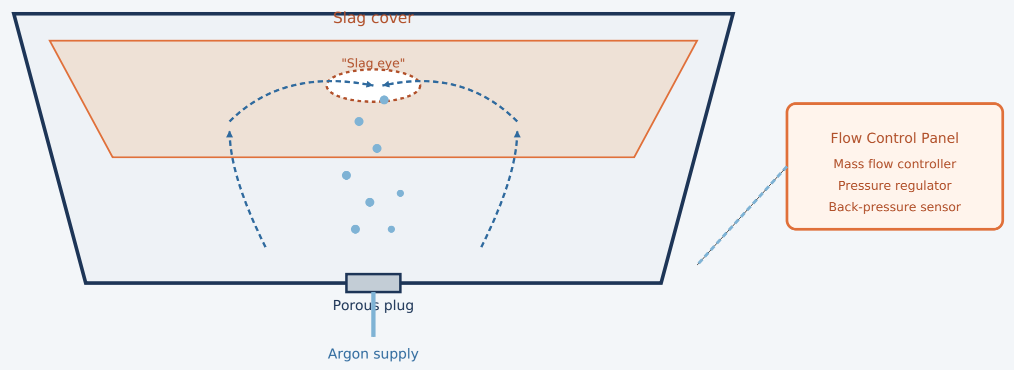


Fig. 4 — Argon rising from a bottom porous plug drives a toroidal circulation pattern through the bath; too much flow opens too large a "slag eye," exposing bath to reoxidation from air.

WHY STIRRING INTENSITY IS A BALANCING ACT

More argon means faster mixing, faster alloy dissolution, and better inclusion removal — but push it too far and the rising gas plume punches a hole clean through the slag layer (the "slag eye"), exposing bare steel to the air above. That reoxidizes the bath, undoing the careful reducing conditions built up for desulfurization and potentially adding fresh inclusions right when the process is trying to remove them. Getting stirring intensity right — and controlling it electrically in real time — is one of the LMF's most important control tasks.

TWO STIRRING MODES

MODE	USED FOR
Soft stir	Low argon flow late in treatment (and often just before release) to promote inclusion flotation without disturbing the slag cover — critical for the cleanest possible steel going to the caster.
Vigorous stir	Higher argon flow used earlier in treatment to rapidly homogenize temperature after reheating and speed alloy/flux dissolution.

ELECTRICAL CONTROL EQUIPMENT

Mass flow controllers	**Set & verify flow electronically**
Pressure regulators	**Step down from header supply**
Back-pressure sensors	**Detect plug blockage early**
Redundant plugs	**Often 2 per ladle**

KEY HAZARD

A blocked or failed porous plug not only stops stirring — it can force operators to use an emergency lance for gas injection through the top, a more hazardous manual operation. Continuous back-pressure monitoring exists specifically to catch blockage early enough to avoid that.

5 Temperature & Chemistry Control

A closed-loop system that has to hit a moving target — the caster's schedule

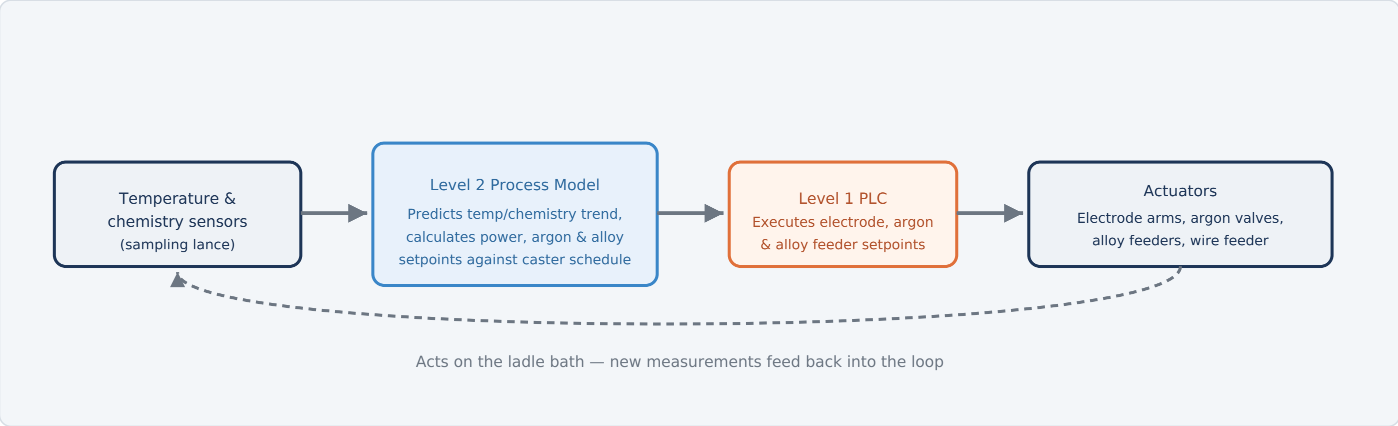


Fig. 5 — The LMF runs as a continuous closed loop: measured temperature and chemistry feed a predictive model, which sets electrode power, argon flow, and alloy dosing to hit both a target analysis and a target release time.

WHY TIMING MATTERS AS MUCH AS CHEMISTRY

The LMF doesn't just have to get temperature and composition right — it has to get them right *at the moment the caster needs the ladle*. Finish too early and the steel cools while it waits, forcing another reheat; finish too late and the caster can run out of steel mid-sequence, a costly interruption. The Level 2 process model continuously recalculates the remaining time needed to hit target temperature given current power input, and adjusts the pace of heating to land the ladle's readiness right on the caster's schedule.

WHAT THE MODEL BALANCES

Electrode power	**Reheat rate vs. refractory wear**
Argon flow	**Mixing speed vs. slag eye/reoxidation**
Alloy timing	**Dissolution time vs. release schedule**
Caster pacing	**Ladle readiness vs. cast sequence**

THE SAMPLING AND WIRE FEED SYSTEMS THAT MAKE IT POSSIBLE

EQUIPMENT	ROLE
Automated sampling lance	Dips a disposable thermocouple/sample tip into the bath on a programmed or on-demand cycle, feeding results directly into the process model without manual intervention.
Calcium wire feeder	Injects CaSi wire at a precisely controlled speed timed near the end of treatment, converting solid alumina inclusions to liquid calcium aluminates right before release.
Alloy trim feeders	Make small, fast corrections based on the latest sample results without needing a full manual addition cycle.

6 Safety & Protection Systems

The interlocks that keep a smaller arc furnace working safely alongside constant ladle traffic

Many protection systems are shared with the EAF (electrical protection, arc flash precautions, tilting/hydraulic interlocks where applicable) and covered in the main steelmaking guide. The systems below are the ones most specific to, or most safety-critical for, LMF operation.

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Ladle car position interlock	Ladle car not correctly aligned/stopped under the electrode roof	Blocks roof lowering and arc initiation until alignment is confirmed
Roof/lid position interlock	Roof not fully seated or seal not confirmed	Blocks power-on to the electrodes until the roof is correctly positioned
Electrode short-circuit / collision protection	Sudden abnormal current spike suggesting electrode contact with the ladle rim or a metallic object	Rapidly raises the affected electrode and trips power
Argon back-pressure / blockage alarm	Porous plug back-pressure rises above normal, indicating partial blockage	Alarms operator; can prompt a switch to a redundant plug
Low argon supply pressure trip	Header supply pressure drops below the minimum needed for reliable stirring	Alarms operator; process model flags reduced stirring effectiveness
Electrode breakage detection	Sudden loss of expected electrical continuity through an electrode column	Alarms operator and halts electrode lowering/feeding sequence
Fume extraction interlock	Local hood extraction airflow falls below setpoint while arcing	Alarms operator; can reduce or trip power depending on plant policy
Over-temperature / refractory monitoring	Shell temperature sensors detect abnormal hot spots suggesting refractory thinning	Alarms operator; heat may be diverted or treatment shortened
Emergency stop (E-stop)	Manual activation anywhere on the LMF platform	Removes power from electrodes and halts auxiliary drives per the defined E-stop category

WHY LADLE TRAFFIC IS A BIGGER FACTOR HERE THAN ON THE EAF

Unlike an EAF, which stays in one place, the LMF sees a continuous stream of ladles arriving and departing on transfer cars between the EAF, LMF, VTD, and caster. Position and alignment interlocks are therefore just as safety-critical here as the electrical protections themselves — most LMF incidents trace back to timing or positioning errors during ladle transfer, not to the arc system itself.

With LMF vs. Without — Summary

What a ladle furnace changes about the whole melt shop, and a recap of the process

FACTOR	TAP STRAIGHT TO CASTER (NO LMF)	WITH LMF
EAF tap-to-tap time	Longer — EAF must fully refine and superheat before tapping	Shorter — EAF taps as soon as melting/rough composition is done
Temperature control at caster	Limited — whatever temperature survives tapping and transport	Precisely controlled, often ±5–10 °C at release
Sulfur content	Limited by EAF's oxidizing conditions, typically higher	Can be reduced well below 0.005% under a reducing slag
Alloy accuracy	Harder to hit tight specs in a turbulent, oxidizing bath	Precise trims possible in a calm, controlled bath
Inclusion cleanliness	Limited control	Calcium treatment + soft stirring produces castable, clean steel
Overall melt shop throughput	Bottlenecked by the EAF doing everything	Higher — EAF, LMF, and caster can each run near their own optimum pace

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **LMF** exists to take over everything an EAF is bad at doing quickly: careful reheating, homogenization, desulfurization, precise alloying, and inclusion cleanup. Electrically, it's a smaller, gentler version of the EAF — a 3-phase transformer feeding three independently regulated graphite electrodes that strike a **submerged arc** into a reducing slag cover, with current returning through the conductive bath itself rather than through any separate busbar. **Argon injected through a bottom porous plug** drives continuous stirring, carefully tuned between vigorous mixing and gentle "soft stir," while **automated sampling and wire-feed systems** feed a closed-loop process model that adjusts power, gas flow, and alloy dosing in real time — not just to hit a target chemistry and temperature, but to have the ladle ready exactly when the caster needs it. The payoff for all this added equipment is a melt shop where the EAF, LMF, and caster can each run near their own best pace instead of the whole line waiting on the slowest step.

Why Vacuum? The Physics Behind Degassing

Some gases simply won't leave the steel until the pressure above it drops

Hydrogen and nitrogen both dissolve into liquid steel during melting and stay there through refining — argon stirring alone helps them escape only slowly, because at atmospheric pressure the steel can hold a fair amount of dissolved gas in equilibrium with the atmosphere above it. A **Vacuum Tank Degasser (VTD)** changes that equilibrium directly: by pulling the pressure above the bath down to a few millibars, it forces dissolved gas out of solution, because the steel simply can't hold as much gas anymore at that pressure — no matter how long you wait at atmospheric pressure, it would never happen on its own. The two gases don't respond equally, though: hydrogen removal is comparatively fast and reliable, while nitrogen removal is far more sensitive to kinetics — dissolved sulfur and oxygen at the steel surface can block the interfacial reaction, so achieving a nitrogen target depends heavily on starting chemistry, stirring intensity, and exposure time, not on vacuum level alone.

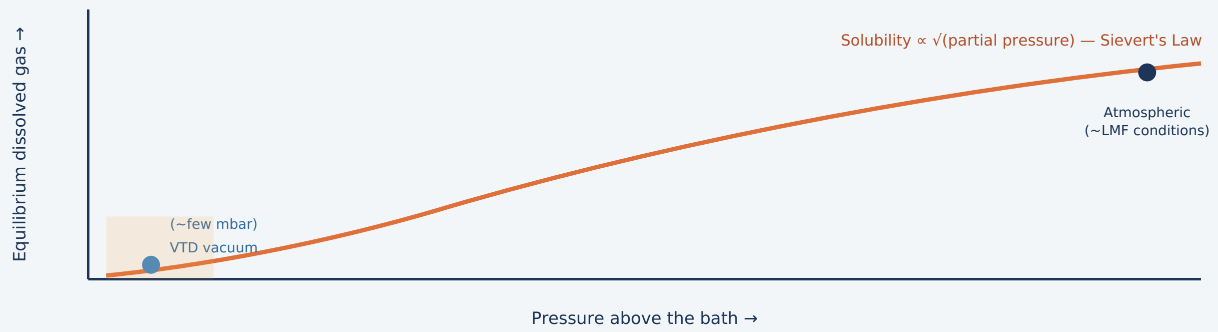


Fig. 1 — Sievert's Law: the equilibrium amount of a diatomic gas (H_2, N_2) that liquid steel can hold falls with the square root of the pressure above it — so dropping pressure to a few millibars pulls dissolved gas out far more effectively than any amount of stirring at atmospheric pressure.

WHO ACTUALLY NEEDS THIS, AND WHY

PRODUCT / GRADE	WHY LOW DISSOLVED GAS MATTERS
Line pipe steel	Hydrogen trapped in the solid steel can cause hydrogen-induced cracking in service, especially in sour (H_2S-bearing) environments.
Heavy forgings & plate	Thick sections cool slowly, giving trapped hydrogen time to collect at internal defects and form dangerous internal cracks known as "flakes."
Bearing steel	Needs both very low hydrogen and exceptional inclusion cleanliness for long fatigue life under repeated rolling contact stress.
Ultra-high-strength automotive steel	High-strength microstructures are especially susceptible to hydrogen embrittlement at even modest dissolved levels.
Stainless & ultra-low-carbon grades	A related vacuum-refining process (VOD, covered later in this guide), run in a specifically equipped installation, pulls carbon down to very low levels without burning up expensive alloying elements.

TYPICAL PARAMETERS

Vacuum level	**~0.5–5 mbar**
Treatment time	**15–25 min**
Typical hydrogen reduction	**~4–6 ppm → <1.5–2 ppm**
Nitrogen reduction	**Possible, but kinetics-limited**
Gas removed	**H_2, N_2, some O_2 (as CO)**

1 The Degassing Process

From sealing the ladle to releasing it back to the caster

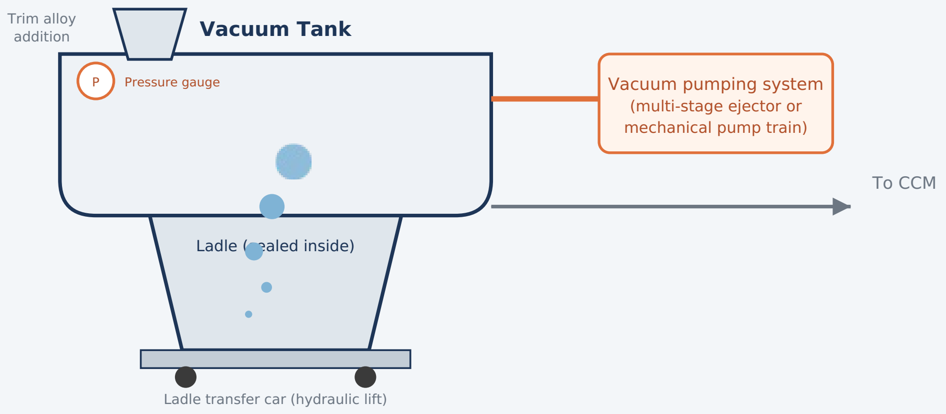


Fig. 2 — The ladle is raised into the sealed vacuum vessel; as pumps draw the pressure down, argon bubbles expand dramatically on their way up, stirring the bath and constantly refreshing the surface exposed to vacuum.

HOW IT WORKS

1. **Seal the ladle:** The ladle, still on its transfer car, is raised into the vacuum tank and the vessel lid seals around it.
2. **Draw vacuum:** Pumps evacuate the tank in a carefully staged sequence — too fast a pump-down can cause dissolved gas to nucleate violently and splash ("spit") molten steel onto the vessel walls.
3. **Stir under vacuum:** Argon continues bubbling through the bath; because bubbles expand enormously as pressure drops, even modest gas flow drives intense circulation, constantly bringing fresh steel to the exposed surface.
4. **Remove dissolved gas:** With so much surface area cycling through the low-pressure environment, hydrogen and nitrogen diffuse out of solution and are carried away by the vacuum system.
5. **Trim & finish:** Final alloy additions can be made under vacuum or immediately after breaking it, since the bath is now especially clean and reactive.
6. **Break vacuum:** Pressure is equalized gradually rather than all at once, avoiding thermal and mechanical shock to both the steel and the vessel.
7. **Release:** The lid lifts and the ladle moves directly to the caster.

TYPICAL PARAMETERS

Vacuum level	**~0.5–5 mbar**
Treatment time	**15–25 min**
Gas removed	**H_2, N_2, some O_2**
Use case	**Clean / low-H steel grades**

2 The Vacuum Pumping System

Reaching a few millibars takes more than one pump — it takes a carefully staged train of them

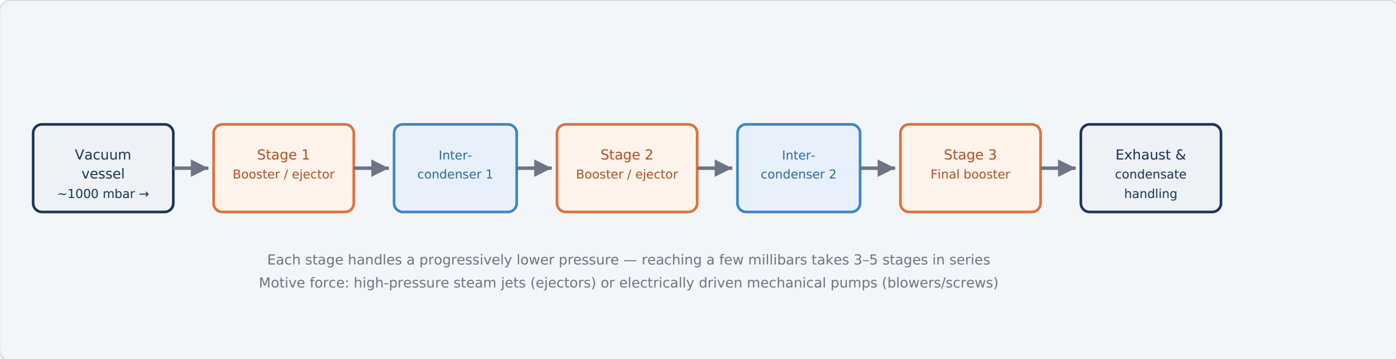


Fig. 3 — A multi-stage train progressively lowers pressure; intercondensers between stages remove condensable vapor so later, smaller stages don't have to handle it, which is what makes reaching a few millibars practical.

TWO WAYS TO BUILD THE TRAIN

TECHNOLOGY	HOW IT WORKS	TRADE-OFFS
Steam ejector train	High-pressure steam nozzles create suction by momentum transfer (a Venturi effect); no moving parts in the ejectors themselves.	Simple, reliable, but consumes large amounts of process steam per cycle and produces contaminated condensate needing treatment.
Mechanical pump train	Electrically driven roots blowers and screw/liquid-ring pumps do the same staged pressure reduction without steam.	Lower energy and water treatment burden, faster response, but higher electrical demand and more rotating equipment to maintain.
Hybrid train	Mechanical booster stages handle the higher-pressure end, with steam ejectors finishing the final low-pressure stages (or vice versa).	Common compromise — balances energy use, capital cost, and response speed.

ELECTRICAL EQUIPMENT (EITHER TRAIN TYPE)

Mechanical pump motors	**Roots blowers, screw/liquid-ring pumps**
Boiler feed & condensate pump motors	**Steam-ejector systems only**
Intercondenser spray water pump motors	**Both train types**
Vacuum control valves	**Electrically/pneumatically staged**

3 Bubble-Driven Stirring Under Vacuum

A third way to stir a ladle in this guide series — and the most dramatic-looking one

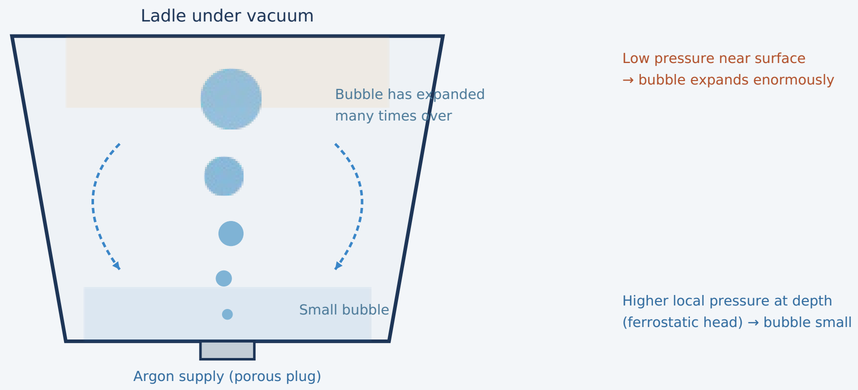


Fig. 4 — Under normal atmospheric pressure an argon bubble barely grows as it rises; under vacuum, the same bubble can expand by two orders of magnitude or more by the time it reaches the surface, driving far more vigorous stirring per unit of gas flow.

THE SAME IDEA, A THIRD MECHANISM

This guide series has now shown bath stirring driven three different ways: a DC EAF's vertical current path creates a Lorentz-force stir; a CCM mold's rotating magnetic field induces a swirl with no current flowing through the steel at all; and here, an LMF-style argon plug drives stirring purely through simple gas expansion — an argon bubble injected at the bottom of a deep ladle sits under significant ferrostatic pressure (the weight of steel above it) plus atmospheric pressure. As it rises toward the vacuum above the bath, the surrounding pressure drops enormously, and by the ideal gas law the bubble's volume expands to match. That expansion does mechanical work on the surrounding steel — violently, compared to the same bubble rising through steel at atmospheric pressure — which is exactly why VTD stirring looks and behaves so differently from LMF stirring even though both use the same porous plug hardware.

THE TRADE-OFF: STIRRING VS. "SPITTING"

The same violent bubble expansion that makes stirring so effective can also throw droplets of steel up onto the vessel walls and lid if argon flow isn't reduced as vacuum deepens — a phenomenon operators call "spitting." Electrically controlled mass flow controllers throttle argon down through the pump-down sequence specifically to manage this, then can increase flow again once the vessel is stable at full vacuum and the risk of a sudden splash has passed.

ELECTRICAL CONTROL EQUIPMENT

Mass flow controllers	**Coordinated with vacuum level**
Vessel pressure transducers	**Continuous, feed the flow schedule**
Back-pressure sensors	**Detect porous plug blockage**

KEY HAZARD

Uncontrolled spitting can damage refractory and coat the vessel's internal instrumentation and lid seal — argon flow scheduling against the pump-down curve is a safety and maintenance issue, not just a process efficiency one.

4 Electrical Equipment in Detail

From the pump train's motors to the alloy lock feeding additions into a sealed vessel

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
VACUUM PUMPING SYSTEM		
Mechanical pump motors (roots blowers, screw/liquid-ring pumps)	Electrically driven pumps that evacuate the vessel to process pressure, staged in series.	Combined load often several hundred kW for a full train
Boiler feed & condensate pump motors	Support the steam supply and condensate return for ejector-based trains.	Steam-ejector systems only
Intercondenser spray pump motors	Circulate cooling water that condenses motive steam and process vapor between pump stages.	Present in both steam and hybrid trains
Vacuum control valves	Electrically or pneumatically staged valves that sequence which pump stages are active during pump-down.	Interlocked to the pressure control sequence
VESSEL & LADLE HANDLING		
Vacuum vessel lid hoist/seal drive	Electric motor that raises, lowers, and seals the lid over the ladle.	Position and seal-pressure feedback
Ladle car hydraulic lift drive	Electrically pumped hydraulics that raise the ladle into the sealed vessel.	Precision positioning for seal alignment
Vessel seal integrity monitoring	Pressure/leak-rate sensors confirming the seal is holding before pump-down proceeds.	Interlocked to the pump-down sequence
ARGON & PRESSURE CONTROL		
Mass flow controllers	Electronically throttle argon flow through the pump-down curve to manage stirring intensity and spitting risk.	Coordinated with vessel pressure transducers
Vessel pressure transducers	Continuously measure absolute pressure inside the vessel, the primary feedback for the whole process.	Wide-range gauges covering atmosphere down to sub-mbar
Back-pressure sensors (porous plug)	Detect plug blockage, as on an LMF.	Continuous monitoring
ALLOY ADDITION & INSTRUMENTATION		
Vacuum-compatible alloy feed system	Hoppers and airlock-style chutes that add alloys without breaking the vessel's vacuum seal.	Motorized dosing with lock-valve sequencing
Temperature & sampling lance manipulator	Electrically driven arm for bath sampling, shared in concept with the LMF's sampling system.	Automated positioning
Off-gas analyzers	Monitor extracted gas composition, useful for tracking decarburization progress on VOD-equipped vessels.	Continuous or periodic sampling
AUTOMATION		
Level 1/2 automation (degassing model)	Sequences pump-down rate, argon flow, and alloy timing, and predicts remaining hydrogen content in real time against a target.	Coordinates every subsystem above

5 Decarburization Under Vacuum (VOD)

A related, specifically-equipped process: pulling carbon down to roughly 100–200 ppm without burning up your alloy

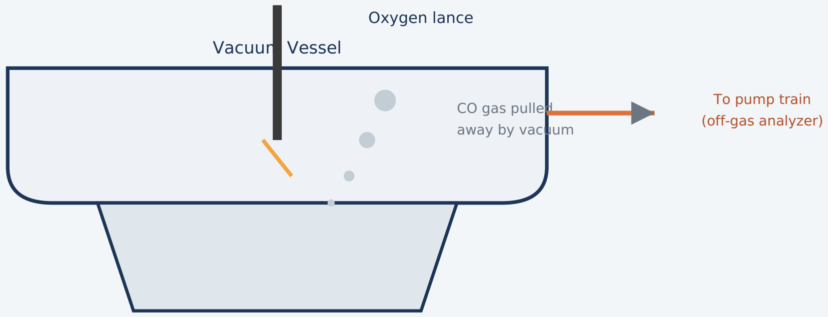


Fig. 5 — Oxygen blown into the vacuum vessel reacts with dissolved carbon to form CO gas; because vacuum continuously removes that CO, the reaction keeps running to much lower carbon levels than would be possible at atmospheric pressure.

WHY THIS NEEDS VACUUM SPECIFICALLY

Removing the last traces of carbon from steel means reacting it with oxygen to form CO gas. At atmospheric pressure, pushing that reaction to very low carbon levels requires blowing in a lot of excess oxygen — but once carbon is scarce, that extra oxygen starts oxidizing other elements in the bath instead, including expensive alloying elements like chromium in stainless steel. Lowering the pressure above the bath shifts the carbon-oxygen equilibrium directly, in the same Sievert's Law spirit as hydrogen removal: a given carbon and oxygen content can support a much lower CO partial pressure requirement under vacuum, so the reaction keeps progressing to very low carbon without needing nearly as much excess oxygen — and without burning up the chromium.

WHY STAINLESS STEEL PRODUCERS CARE MOST

This is precisely the process — often called VOD (Vacuum Oxygen Decarburization) — that lets stainless steel mills hit ultra-low-carbon specifications while retaining the chromium content the grade depends on. Without it, achieving the same carbon level by blowing oxygen at atmospheric pressure would oxidize away enough chromium to be both metallurgically and economically unacceptable.

NOT JUST A LANCE BOLTED ONTO A STANDARD DEGASSER

Running a VOD cycle safely takes more than adding an oxygen lance to a generic vacuum tank: the oxygen-lance manipulator, off-gas handling, refractory, and process interlocks all have to be specifically engineered for controlled oxygen blowing under vacuum. A VTD vessel built without that equipment shouldn't be assumed capable of running VOD.

TYPICAL PARAMETERS

Achievable carbon	**Down to ~0.01–0.02%**
Oxygen lance control	**Electrically positioned manipulator**
Off-gas monitoring	**CO analyzers track progress**

KEY HAZARD

CO generated during VOD treatment is both toxic and flammable; the vacuum system's off-gas path and any venting or afterburning equipment must be designed and monitored specifically for this — the same category of hazard covered in more depth in the Fume Extraction System guide.

6 Safety & Protection Systems

Sealed vessels, staged pressure changes, and the hazards specific to working under vacuum

Many protection systems are shared with the LMF (ladle handling, argon supply, electrical protection) and covered in that guide. The systems below are the ones specific to, or intensified by, operating a sealed vessel under vacuum.

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Pump-down rate control	Pressure falling faster than the safe curve, risking violent gas evolution ("spitting")	Throttles pump stages and argon flow to follow a controlled pressure-vs-time profile
Vessel seal integrity interlock	Lid/seal not confirmed sealed before pump-down begins	Blocks pump-down start until seal pressure confirms a good seal
Pressure equalization control (breaking vacuum)	Vacuum release requested	Admits air/inert gas gradually rather than all at once, avoiding thermal and mechanical shock
CO monitoring (VOD-equipped vessels)	Carbon monoxide detected in the off-gas path or surrounding work area	Alarms operator; blocks confined-space entry permits until levels clear
Steam system safety devices (ejector trains)	Steam pressure/temperature outside safe limits	Standard high-pressure steam plant protections — relief valves, pressure interlocks
Pump motor protection	Overcurrent, thermal overload on mechanical pump motors	Trips affected motor off-line
Ladle car position interlock	Car not correctly aligned under the vessel before lifting	Blocks lid lowering/sealing sequence
Argon back-pressure alarm	Porous plug back-pressure rises, indicating blockage	Alarms operator, as on the LMF
Emergency stop (E-stop)	Manual activation anywhere on the platform	Removes power from drives and can trigger a controlled vacuum release, per defined procedure

WHY STAGED PRESSURE CHANGE MATTERS SO MUCH HERE

Both drawing vacuum and breaking it are moments of real risk if rushed: too-fast pump-down risks spitting molten steel inside the vessel, while too-fast pressure equalization when opening up can shock both the steel and the refractory-lined vessel structure. Nearly every VTD-specific safety system in this table exists to enforce a controlled rate of pressure change in one direction or the other.

✓ With VTD vs. Without — Summary

What vacuum treatment adds to the melt shop, and a recap of the process

FACTOR	LMF ONLY (NO VACUUM)	WITH VTD
Achievable hydrogen	Limited by atmospheric-pressure equilibrium, typically several ppm	Can reach well under 2 ppm for demanding grades
Achievable carbon (VOD)	Limited without excess chromium loss on stainless grades	Ultra-low carbon achievable while retaining chromium
Applicable products	General commodity grades	Line pipe, heavy forgings, bearing steel, ultra-high-strength automotive, stainless
Cycle time	Shorter — no added station	Adds 15–25 minutes plus vessel transfer time
Capital & operating cost	Lower — no vacuum plant needed	Higher — vacuum pumping system, steam or electrical infrastructure, vessel

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **VTD** takes a ladle already reheated and refined at the LMF and seals it inside a vessel that a **multi-stage pumping train** — steam ejectors, mechanical pumps, or a hybrid of both — evacuates down to a few millibars. At that pressure, **Sievert's Law** means the steel simply can't hold as much dissolved hydrogen and nitrogen as it could at atmospheric pressure, so those gases diffuse out — helped enormously by **argon bubbles that expand dramatically** as they rise through the falling pressure, driving far more vigorous stirring than the same gas flow could achieve on an LMF. A related process, **VOD**, runs in an installation specifically engineered for controlled oxygen blowing under vacuum — a dedicated lance system, off-gas handling, and interlocks beyond a standard tank degasser — pulling carbon down to roughly 100–200 ppm without burning away valuable alloying elements, the technique that makes ultra-low-carbon stainless steel possible. Because both drawing and breaking vacuum are moments of real risk — violent splashing on one end, thermal shock on the other — the system's electrical controls are built entirely around enforcing a safe, staged rate of pressure change in both directions.

Why Continuous Casting? The Metallurgical Length Problem

One idea organizes almost everything a CCM's electrical systems do: keep the liquid core ahead of the machine

Before continuous casting became standard, steel was cast into individual ingot molds, allowed to solidify, stripped, and reheated before rolling — batch work with real yield losses and a lot of wasted energy reheating steel that had already been hot once. A **Continuous Casting Machine (CCM)** instead pours a steady stream of liquid steel into a short, water-cooled mold and continuously withdraws it as a solidifying strand, cutting it to length only once it's fully solid. That single change — casting continuously instead of in batches — is what drives modern steelmaking's yield above 96%, but it also creates the central engineering problem every system in this guide exists to solve.

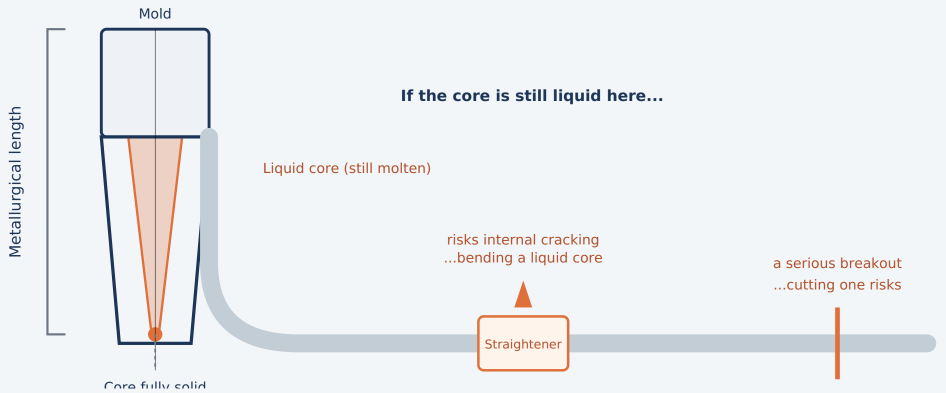


Fig. 1 — The "metallurgical length" is the distance from the mold meniscus to the point the strand's core finally solidifies. Casting speed and cooling are set so that point stays within the caster's designed containment and solidification envelope — for many casters that means fully solid before the straightener, though some machines are specifically designed to bend or straighten a strand with a liquid or mushy core.

INGOT CASTING VS. CONTINUOUS CASTING

FACTOR	INGOT CASTING (BATCH)	CONTINUOUS CASTING
Yield	~85–90% (crop losses, piping defects)	~96–98%
Energy use	Higher — solidified ingots often fully reheated before rolling	Lower — strand can sometimes be hot-charged directly to the rolling mill
Product consistency	Batch-to-batch variation	Continuous, tightly controlled cross-section and quality
Central engineering challenge	Managing solidification shrinkage in a static mold	Managing a moving liquid-to-solid transition in real time — the metallurgical length problem

WHY THIS MATTERS FOR EVERYTHING THAT FOLLOWS

Many of the major electrical systems on a CCM — mold level control, oscillation, electromagnetic stirring, spray cooling, and soft reduction — exist either to control how fast and how evenly the strand solidifies, or to protect the still-thin shell before it's strong enough to support itself. Keep that frame in mind through the rest of this guide.

1 The Casting Process

From ladle to cut strand — a continuous, tightly synchronized sequence

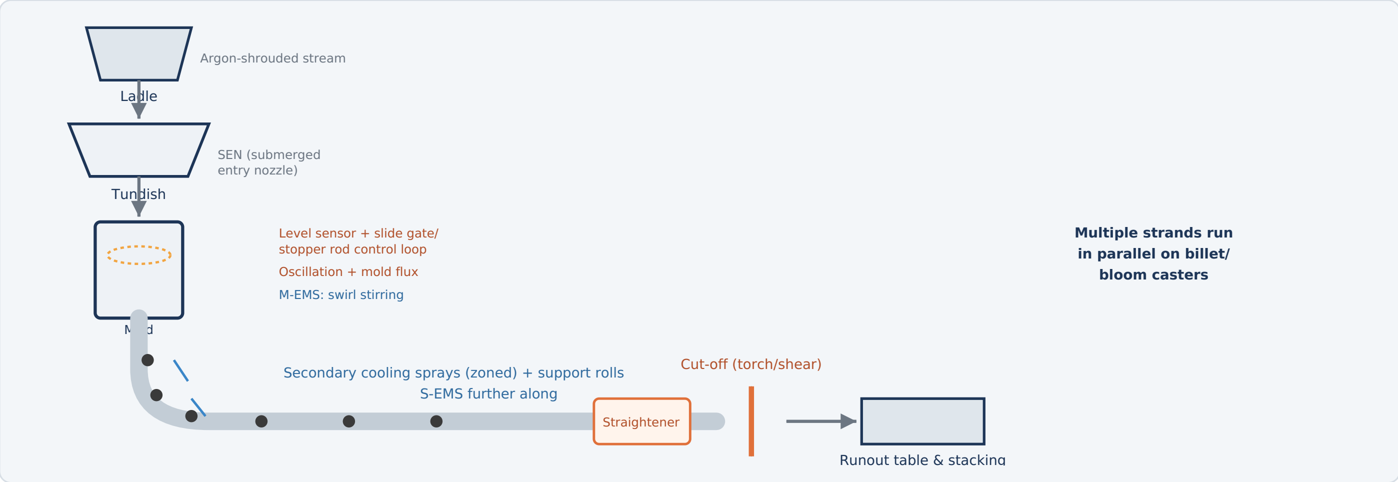


Fig. 2 — Steel flows continuously from ladle to tundish to mold, then the solidifying strand is guided, cooled, stirred, straightened, and finally cut to length once fully solid.

HOW IT WORKS

1. **Ladle to tundish:** Steel flows from the ladle through an argon-shrouded stream into the tundish, a refractory-lined buffer vessel that keeps casting continuous even as ladles are swapped mid-sequence.
2. **Tundish to mold:** A submerged entry nozzle (SEN) delivers steel from the tundish into the water-cooled copper mold below the surface, preventing reoxidation and splashing.
3. **Solidify the shell:** Contact with the mold walls freezes a thin outer shell almost instantly; mold oscillation and a layer of mold flux keep that fragile shell from sticking as it's continuously withdrawn.
4. **Stir near the surface:** Electromagnetic coils around the mold (M-EMS) drive a gentle swirling motion in the still-liquid pool, improving surface quality and helping trapped bubbles and inclusions escape upward into the flux layer.
5. **Cool & support the strand:** Below the mold, rows of water (or air-mist) spray nozzles cool the strand from the outside in, while closely spaced support rolls keep the still-thin shell from bulging under the liquid core's weight.
6. **Stir again, straighten, and finish solidifying:** A second stirrer (S-EMS) further down improves internal quality; withdrawal and straightening rolls pull the strand at a controlled speed and — on curved-mold casters — gradually bend it from curved to horizontal. Depending on the caster's design, that bending may happen only after the core is fully solid, or, on machines specifically engineered for it, while a liquid or mushy core is still supported within calculated strain limits.
7. **Cut to length:** Once fully solid, an oxy-fuel torch (or mechanical shear, on billet casters) cuts the continuous strand into billets, blooms, or slabs, which move off on a runout table for stacking or direct transfer to the rolling mill.

Note: this sequence describes normal, steady-state running. Getting the strand started in the first place — before any solid steel exists in the machine at all — needs the dummy bar system described next.

TYPICAL PARAMETERS

Casting speed	**0.8–5+ m/min**
Mold length	**~0.7–1 m**
Billet section	**~100–200 mm square**
Slab section	**~200–300 mm × 900–2200 mm**
Strands per machine	**1 (slab) to 8 (billet)**

2 Starting the Cast — The Dummy Bar System

A mold with an open bottom needs something to seal against and pull on before any real steel exists

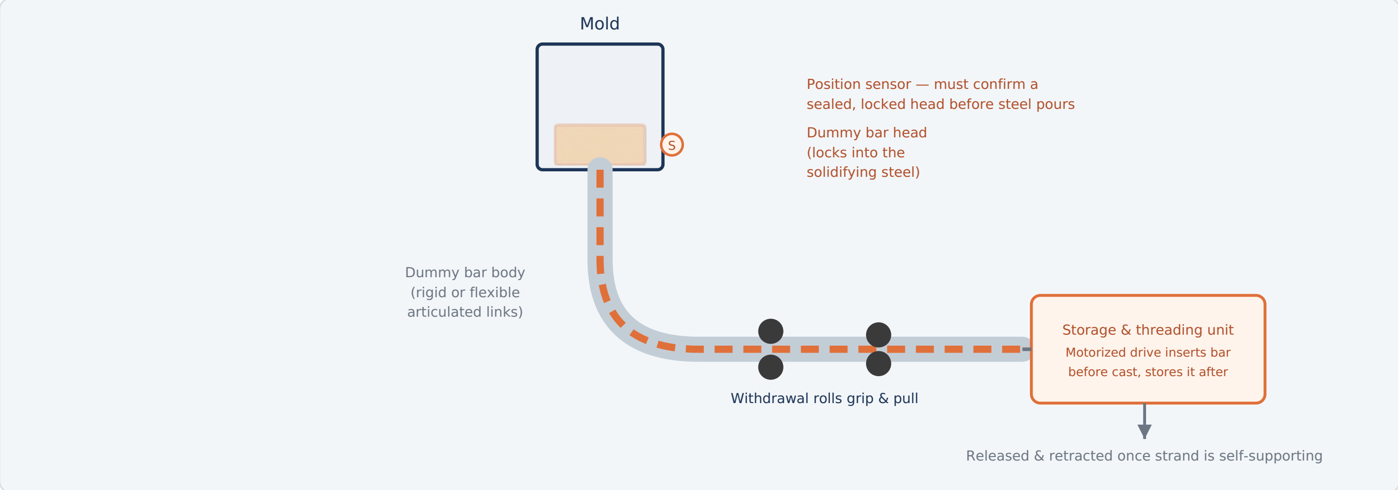


Fig. 3 — Before casting starts, the dummy bar is threaded up through the machine and its head seals the open mold bottom; once real strand builds up enough to support itself, the bar is released and stored for the next cast.

HOW IT WORKS

1. **Thread in:** Before any steel is poured, the dummy bar is driven up from its storage position, through the curved guide and roll segments, until its head sits at the bottom of the empty mold.
2. **Confirm the seal:** Position sensors verify the head is fully seated and locked — this is a hard interlock; the tundish gate cannot open until the seal is confirmed, because pouring onto an unsealed mold bottom would cause an immediate, uncontained breakout.
3. **Pour & lock:** Steel fills the mold around and onto the dummy bar head; the bottom layer solidifies onto it, mechanically locking the head to the new steel shell (often helped by a dovetail or key feature machined into the head).
4. **Begin withdrawal:** Withdrawal rolls, gripping the dummy bar body, pull it — and the newly solidifying shell locked to it — out of the mold at a slow, carefully controlled starting speed.
5. **Ramp to normal speed:** As real strand length builds behind the dummy bar head and the shell thickens, casting speed ramps up to the normal target rate for that section size and grade.
6. **Release & store:** Once enough solid strand exists to be self-supporting, the dummy bar is disconnected — by an unlocking mechanism at the head, or by cutting it free — and withdrawn to storage, cleaned and ready for the next cast start.

ELECTRICAL & CONTROL EQUIPMENT

Threading drive motor	**Inserts bar before cast**
Head position/seal sensors	**Hard-wired pour interlock**
Head lock/release mechanism	**Mechanical or hydraulic**
Storage rack drive	**Coils flexible bar after release**

KEY HAZARD

An unsealed or unconfirmed dummy bar is one of the most dangerous conditions possible at the start of a cast — steel poured onto an open mold bottom has nothing at all to contain it. This interlock is treated with the same severity as breakout protection during normal running.

TWO DUMMY BAR DESIGNS

TYPE	TRADE-OFF
Rigid	A single solid bar matching the machine's exact path shape — simple and robust, but needs dedicated storage space matching that geometry and can't easily serve a different machine radius.
Flexible (articulated)	Hinged chain-like segments that flex around the curved path and coil into a compact storage rack — more mechanically complex, but far more space-efficient, and the modern standard on curved-mold casters.

3 Mold Level Control — The Most Critical Loop

A few millimeters of error here shows up as a surface defect — or worse

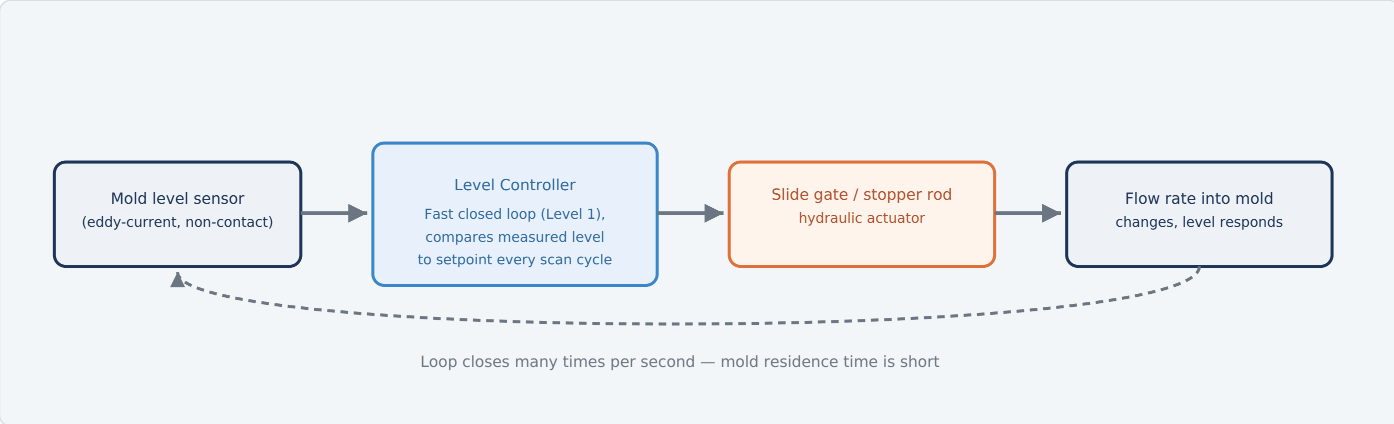


Fig. 3 — A non-contact sensor continuously measures the meniscus level; the controller adjusts a slide gate or stopper rod to hold it within a few millimeters of setpoint, closing the loop fast enough to matter at full casting speed.

WHY THE TOLERANCE IS SO TIGHT

The meniscus — where liquid steel meets mold flux at the top of the mold — is where the strand's surface quality is set. A level that drifts high or low disturbs the delicate balance between flux infiltration, initial shell formation, and mold oscillation timing, producing surface defects like oscillation marks, flux entrapment, or in the worst case an overflow or a shell so thin locally that it can't survive being withdrawn — a direct precursor to a breakout further down the machine.

TWO SENSING TECHNOLOGIES

TECHNOLOGY	HOW IT WORKS
Electromagnetic (eddy-current) sensor	An induction coil mounted around the mold generates a field that interacts with the conductive liquid steel surface; the modern standard, fully non-contact and free of any radioactive material.
Radioactive-source sensor (legacy)	A sealed gamma source and detector pair measure level by attenuation through the steel; still found on some older installations but being phased out in favor of eddy-current systems.

WHAT ELSE LIVES IN THE MOLD

Mold flux (casting powder) is continuously fed onto the meniscus, where it melts and forms a thin liquid layer that lubricates the shell against the mold wall, absorbs floating inclusions, and insulates the steel surface from reoxidation and excessive heat loss — its behavior is directly tied to how stable the level control loop can keep the meniscus.

TYPICAL PARAMETERS

Level control tolerance	**±3–5 mm**
Oscillation frequency	**~100–300 cycles/min**
Oscillation stroke	**~4–10 mm**
Actuator type	**Slide gate or stopper rod**

KEY POINT

Mold oscillation itself is driven by an AC servo motor system with encoder feedback, precisely shaping an asymmetric waveform (a faster downstroke relative to the upstroke — "negative strip time") that helps the flux lubricate the shell without tearing it.

4 Electromagnetic Stirring (M-EMS / S-EMS)

The same Lorentz-force trick seen in the DC EAF bath and LMF ladle — applied here with coils instead of a DC electrode

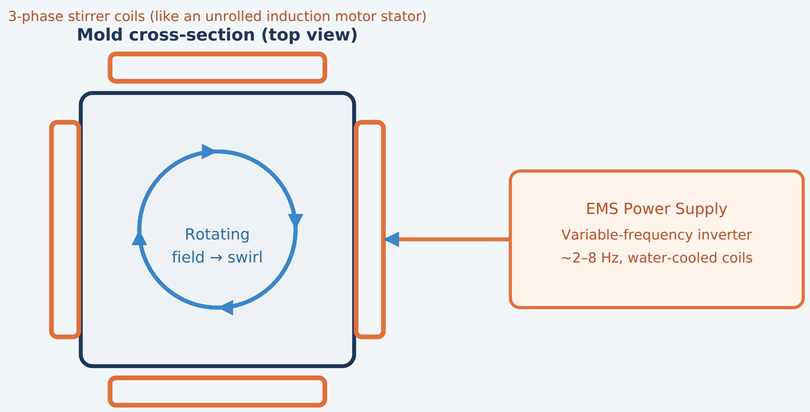


Fig. 4 — Coils arranged around the mold (or lower on the strand for S-EMS) create a rotating magnetic field, exactly like an unrolled induction motor stator; the induced currents in the liquid steel interact with that field to drive a swirling stir — no moving parts touch the steel at all.

TWO STIRRERS, TWO DIFFERENT JOBS

STIRRER	LOCATION	WHAT IT IMPROVES
M-EMS	Around the mold, near the meniscus	Surface quality, bubble/inclusion flotation into the flux layer, finer equiaxed grain structure near the surface
S-EMS / F-EMS	Lower on the strand, in the secondary cooling zone	Internal quality — reduces centerline segregation and porosity by stirring the remaining liquid pool before it fully solidifies

A RECURRING IDEA ACROSS THIS GUIDE SERIES

This is the same underlying physics — a Lorentz force from current interacting with a magnetic field — that drives bath stirring in a DC EAF and (via argon bubbling) in an LMF ladle. Here, though, there's no DC current flowing through the steel at all; the stirring current is *induced* by the coils' alternating field, the same way a motor's rotor currents are induced by its stator — which is why EMS needs its own dedicated variable-frequency power supply rather than tapping into the casting process's DC or arc circuits.

ELECTRICAL EQUIPMENT

EMS power supply	**Variable-frequency inverter**
Stirrer coils	**Water-cooled copper windings**
Typical power	**~100–500 kW per stirrer**
Typical frequency	**~2–8 Hz**

KEY POINT

Stirring intensity is normally scheduled against casting speed and steel grade by the Level 2 process model — too little stirring wastes the quality benefit, too much can disturb the still-thin shell near the meniscus.

5 Electrical Equipment in Detail

From the ladle turret's slewing drive to the torch carriage motor

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
LADLE TURRET & TUNDISH		
Turret rotation drive motor	Rotates the ladle turret's large slewing bearing to swing a full ladle into casting position while the previous one finishes — enables continuous multi-heat sequence casting.	Large geared motor drive on slewing ring
Ladle arm lifting & weighing system	Load cells on each turret arm track remaining steel weight in real time, feeding the sequence-casting schedule.	Per-arm load cell array
Tundish car drive motor	Positions the tundish under the ladle and above the mold(s).	Precision-stopping motor drive
Argon shrouding flow control	Electronically regulated inert gas flow around the ladle-to-tundish and tundish-to-mold streams, preventing reoxidation.	Mass flow controllers
MOLD SYSTEMS		
Mold cooling water pump motors	Circulate high-velocity water through the copper mold's cooling channels — the single highest heat-flux duty on the caster; see Part 19 for the supplying infrastructure.	High-flow, closed-loop, precision-filtered
Mold level sensor & controller	Non-contact eddy-current measurement feeding a fast closed loop to the flow-control actuator (see previous page).	Scan rate fast enough for full casting speed response
Slide gate / stopper rod hydraulic actuator	Physically meters steel flow from tundish to mold based on the level controller's output.	Electro-hydraulic servo valve
Mold oscillation servo drive	AC servo motor with encoder feedback precisely shapes the oscillation waveform (frequency, stroke, and asymmetry).	Closed-loop position control
Mold thermocouple array	Hundreds of embedded thermocouples across the copper mold plates, feeding the breakout prediction system (see safety page).	Typically 100s of points per mold
Mold flux feeder	Automated dispensing of casting powder onto the meniscus at a controlled rate.	Motorized/vibratory feeder
ELECTROMAGNETIC STIRRING		
M-EMS / S-EMS power supplies	Variable-frequency inverters driving the stirrer coils; see previous page for detail.	~100–500 kW per stirrer
Stirrer coil cooling water system	Removes heat from the water-cooled copper windings during continuous operation.	Closed-loop cooling circuit
SECONDARY COOLING & ROLL DRIVES		
Spray header control valves (zoned)	Independently controlled water or air-mist flow to each cooling zone along the strand length.	Electrically actuated, model-driven setpoints
Cooling water pump motors	Supply pressurized water to the spray header network.	Multiple pumps, zone-matched pressure
Segment roll drive motors	Drive the withdrawal and straightening roll pairs that move the strand at controlled casting speed.	Synchronized AC drives across all driven rolls
Soft reduction hydraulic cylinders	Precisely close the roll gap in the final solidification zone; position and force feedback let the system apply a calculated, gradual reduction.	Electro-hydraulic servo control
Support roll/segment (machine) cooling	A separate closed-loop circuit cools roll bearings, segment frames, and hydraulics from the strand's radiant heat — protecting equipment, not shaping the product.	Closed-loop, distinct from spray cooling
CUT-OFF, DEBURRING & AUTOMATION		
Torch cutting machine (TCM) carriage drive	Moves the cutting torch assembly in sync with the moving strand so the cut is straight despite the strand not stopping; see the dedicated cut-to-length page for detail.	Motorized carriage, speed-tracked to strand
Torch ignition & gas flow control	Electrically controlled oxy-fuel ignition and cutting gas (oxygen + fuel) flow.	Automated sequencing
Strand length tracking encoders	Measure strand travel to trigger each cut at the ordered length.	Roller table mounted
Deburring machine drive motor	Powers the rotating grinding/milling head that removes the cut burr from each piece.	See dedicated cut-to-length page
Deburring head positioning actuators	Engage and retract the deburring tool relative to each piece.	Automated per-piece cycle
Level 1/2 automation (dynamic solidification model)	Predicts the solidification profile and metallurgical length in real time, setting spray zone flows, soft reduction, cut timing, and safe casting speed limits.	Coordinates every subsystem above

6 Secondary Cooling & Soft Reduction

Cool enough to keep the shell safe, gently enough to avoid cracking it — and squeeze the last liquid just right

THREE DIFFERENT KINDS OF "COOLING" ON A CASTER

It's easy to lump every water system on a CCM together, but they do three genuinely different jobs:

COOLING TYPE	WHAT IT COOLS	WHY
Mold (primary) cooling	The copper mold itself — indirect, water never touches the steel	The single highest heat-flux point in the whole process; extracts enough heat to freeze a solid shell before the strand ever leaves the mold
Spray (secondary) cooling	The strand surface directly, via nozzle banks in the **spray chamber** below the mold	Continues extracting heat, zone by zone, until the core is fully solid — the balancing act covered on this page
Machine cooling	The caster's own hardware — support roll bearings, segment frames, hydraulics	Protects equipment from the strand's own heat; has nothing to do with shaping the product

The **spray chamber** is simply the enclosed housing containing the secondary-cooling nozzle banks and support rolls — enclosed both to control where the spray goes and to drain spent water to the plant's scale pits. All three cooling duties are supplied by the same mill-wide infrastructure covered in full in Part 19 (Cooling Water System); this page focuses on how spray cooling specifically is controlled.

THE COOLING BALANCING ACT

TOO LITTLE COOLING	TOO MUCH COOLING
Shell stays thin longer, increasing breakout risk and pushing the metallurgical length past the machine's straightener — may force a slower casting speed than the schedule wants.	Sharp temperature swings as the strand alternately cools under sprays and reheats from its own core between zones, creating thermal stress that can crack the surface — especially damaging right at the straightener, where the strand is also under bending stress.

Soft reduction near the end of solidification

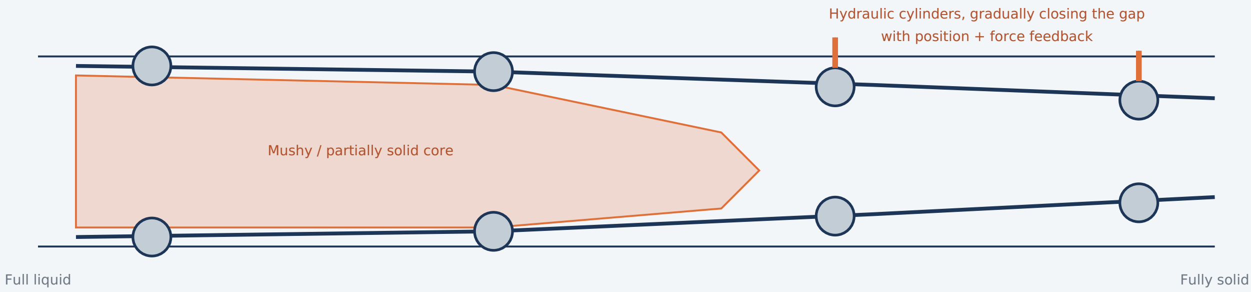


Fig. 5 — In the last few meters before full solidification, hydraulic cylinders gradually narrow the roll gap to compensate for shrinkage and squeeze the mushy zone, reducing centerline porosity and segregation.

WHY SOFT REDUCTION IS A HARD CONTROL PROBLEM

As steel solidifies from the outside in, the last liquid to freeze — right at the strand center — shrinks, pulling solute-rich liquid inward and leaving porosity and segregation behind if nothing counteracts it. Applying a very slight, precisely positioned roll gap reduction exactly where that mushy zone sits pushes the dendritic structure together and compensates for the shrinkage. The hard part: the mushy zone's position isn't fixed — it moves as casting speed, steel grade, and superheat change — so the Level 2 solidification model has to continuously recalculate where it is and command each roll pair's hydraulic cylinder accordingly.

TYPICAL PARAMETERS

Secondary cooling zones	**~5–15 along strand**
Soft reduction total	**~2–6 mm cumulative**
Reduction control	**Position + force feedback**

KEY POINT

Both cooling zone flows and soft reduction targets are recalculated continuously by the same dynamic solidification model that tracks metallurgical length — they aren't independent settings, but two outputs of one real-time model of the whole strand's thermal state.

7 Cut-to-Length — Torch Cutting & Deburring

Segmenting a strand that never stops moving, then cleaning up the edge that cutting leaves behind

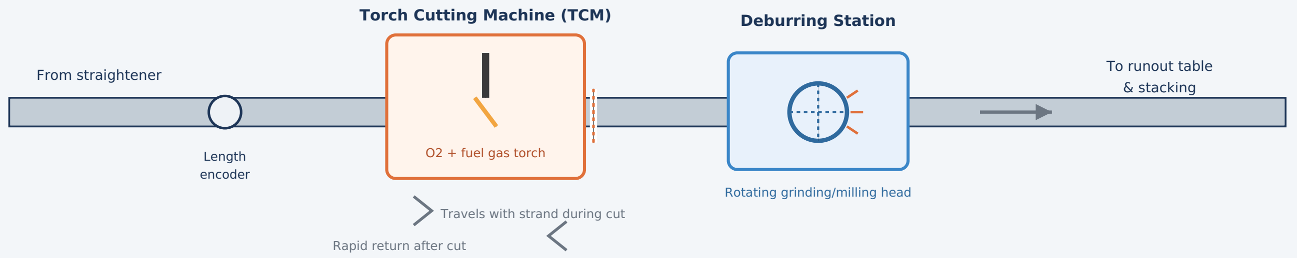


Fig. 6 — The TCM's torch carriage travels with the strand during each cut, then returns rapidly for the next one; a deburring station immediately downstream grinds away the resulting burr before the piece moves on.

THE TORCH CUTTING MACHINE (TCM)

Cutting has to happen without ever stopping the strand, so the TCM is a "flying" cutter: its torch carriage rides on rails alongside the strand and travels at exactly matching speed for the few seconds a cut takes, then releases and snaps back to its starting position, ready for the next one. The cut itself uses oxy-fuel cutting — the torch preheats a small spot to ignition temperature, then a jet of pure oxygen is injected, rapidly oxidizing (burning) the iron and blowing the molten oxide out of the kerf. Larger cross-sections (blooms and slabs) often use multiple torches firing together to complete the cut within the carriage's available travel distance. Some billet casters use a mechanical shear instead of a torch — faster and gas-free, but limited to smaller sections. Strand-length encoders on the roller table track exactly how far the strand has traveled since the last cut, triggering the next one at the ordered length.

THE DEBURRING MACHINE

Oxy-fuel cutting leaves a burr — a ridge of resolidified metal and adhering slag along the underside of the cut — plus dross on the cut face itself. Left in place, that burr can damage handling equipment, cause pieces to nest or stack incorrectly in storage, and even carry through to create defects once the piece is reheated and rolled. A motor-driven grinding or milling head, positioned immediately after the TCM, engages the cut edge of each piece to remove the burr and clean up the face — typically triggered automatically per piece, with sensors detecting burr size to set engagement force and dwell time.

ELECTRICAL EQUIPMENT

TCM carriage drive	**Speed-synced AC servo**
Torch ignition & gas valves	**Electric ignition, solenoid control**
Length tracking encoders	**Triggers each cut**
Deburring head motor	**Rotating grind/mill tool**
Burr detection sensors	**Sets engagement per piece**
Spark/dust extraction fan	**Fire safety for grinding**

KEY HAZARD

Oxy-fuel gas supply (leak/flame safety), hot cut ends, and grinding sparks are all present in the same small area — gas detection, flame monitoring, and dust/spark extraction are treated as standard safety equipment here, not options.

8 Breakout Prevention & Safety Systems

The single most dangerous failure mode on a caster — and the sensor array built to see it coming

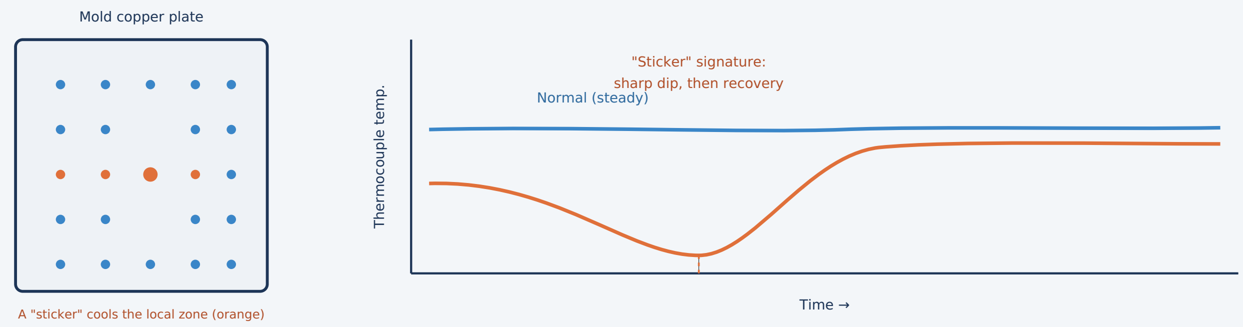


Fig. 6 — A grid of thermocouples across the mold copper plate continuously tracks temperature; a sticker event (shell locally sticking, then tearing free thinner) shows up as a characteristic sharp dip-and-recovery pattern the system is trained to catch before it becomes a breakout.

WHAT A BREAKOUT IS, AND WHY IT'S SO DANGEROUS

If the solidifying shell sticks to the mold wall even briefly, it can tear as it's withdrawn, leaving a locally thin spot that continues down the machine looking fine on the outside — until the liquid core behind it finally breaks through, below the mold where there's no containment. The result is molten steel escaping directly onto machine structure, rolls, and cooling water lines, with serious fire, explosion (from steel contacting water), and personnel injury risk.

HOW PREDICTION WORKS

Because a sticking event locally reduces heat transfer to the mold wall before the shell tears and thins (briefly restoring contact and heat transfer), it leaves a recognizable temperature signature across the thermocouple grid: a sharp, spatially-localized dip followed by a fast recovery. Pattern- recognition software continuously scans the live thermocouple grid for this signature and can automatically slow or stop the cast before the affected section ever reaches the point of no support below the mold.

SAFETY & PROTECTION TABLE

PROTECTION	ACTION
Breakout prediction alarm	Auto-slows or stops the cast
Mold level deviation trip	Alarms; can halt pour
Spray water flow loss	Alarms; may reduce speed
Segment hydraulic pressure loss	Halts affected drives
Roll segment overload	Trips affected drive
Torch flame failure	Halts cutting sequence
Turret position interlock	Blocks rotation if unsafe
Emergency stop (E-stop)	Removes power per zone

Basic vs. Fully Automated Casting — Summary

What modern electrical control adds on top of the basic casting process, and a recap

FACTOR	BASIC CONTROL (FIXED SETPOINTS)	FULLY AUTOMATED (LEVEL 2 MODEL-DRIVEN)
Mold level stability	Wider swings, manual correction	Held within a few millimeters automatically
Casting speed	Conservative, fixed, to stay safely within metallurgical length	Dynamically optimized against real-time solidification model
Internal quality	Higher centerline segregation/porosity risk	Reduced via dynamic soft reduction targeting the actual mushy zone
Breakout risk	Relies on operator vigilance and conservative margins	Actively predicted and interrupted by thermocouple pattern recognition
Surface quality	More variable, dependent on manual oscillation/flux practice	Consistent via servo-controlled oscillation and M-EMS

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **CCM** exists to solve one continuous problem: keep a moving liquid-to-solid transition — the **metallurgical length** — within the caster's designed containment and solidification envelope at all times, whether that means fully solid before the straightener or, on machines specifically engineered for it, bending a controlled liquid or mushy core. Casting can't even begin until a **dummy bar** is threaded up to seal the mold's open bottom, since there's nothing else for the withdrawal rolls to grip until real steel has solidified onto it. Once running, steel flows from ladle to **tundish** to a water-cooled **mold**, where a fast **level control loop** holds the meniscus within millimeters while servo-driven oscillation and mold flux keep the fragile new shell from sticking. **Electromagnetic stirring** coils — the same Lorentz-force principle seen elsewhere in this guide series — improve surface and internal quality without any moving parts touching the steel. Further down, zoned **secondary cooling** and precisely positioned **soft reduction** cylinders manage the balance between shell safety, surface cracking, and internal segregation, all continuously recalculated by a dynamic solidification model. Once fully solid, a **Torch Cutting Machine** segments the still-moving strand to length, and a **deburring station** immediately grinds away the resulting burr before each piece moves on. And because a shell failure below the mold — a **breakout** — is the single most dangerous event on the machine, hundreds of mold thermocouples are watched in real time for the telltale sticking signature that lets the system slow or stop the cast before it ever happens.

How This Guide Works

Follow the process in order — from reheated slab to a finished coil

This guide walks through the hot rolling route in the order steel physically travels through a hot strip mill — from a cold or warm slab going into the reheating furnace, through progressively thinner rolling stands, to a finished coil ready for shipping or further processing (cold rolling, pickling, galvanizing, etc.).

WHAT'S ON EVERY STAGE PAGE

1. **A simple diagram** showing the equipment and the material flowing through it.
2. **A "How It Works" walk-through** — numbered, plain-language steps in the order they happen.
3. **An Electrical Equipment table** — every major powered / electrically-controlled item at that stage and what it does.
4. **A parameters panel** — typical temperatures, speeds, and reduction figures.
5. **A safety note** — the main hazard to be aware of at that stage.

THE ROUTE AT A GLANCE

01	Reheating Furnace	Heat slab to rolling temperature
02	Roughing Mill	Coarse thickness reduction
03	Finishing Mill	Final gauge, shape & surface
04	Runout Table	Controlled cooling
05	Down Coiler	Coil the finished strip

WHY HOT ROLLING?

Steel is far easier to deform above its recrystallization temperature (typically >900°C). Hot rolling takes a thick, slow-cast slab or billet and reduces it in a series of passes into a long, thin, coiled product far more efficiently than would be possible cold.

READING THE DIAGRAMS

Vessel / equipment Electrical item Material flow

BEFORE YOU START

This is a training and orientation guide, not an operating or electrical-safety manual. Always follow site lockout/tagout, PPE, and permit-to-work procedures issued by your plant.

1 Reheating Furnace

Stage 1 of 5 | Bringing slabs up to a uniform rolling temperature

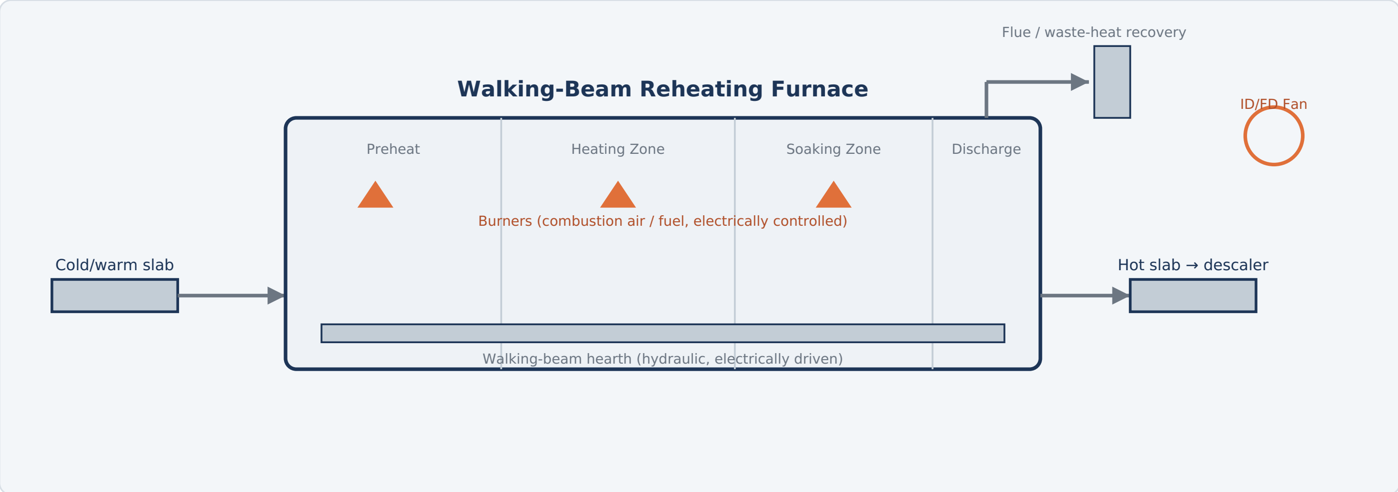


Fig. 1 — Slabs move through preheat, heating, and soaking zones on a walking-beam hearth until they reach a uniform rolling temperature.

HOW IT WORKS

1. **Charge:** Slabs (or billets) are loaded onto the furnace's charging table and pushed or walked into the furnace entrance.
2. **Preheat & heat:** As slabs move through successive zones, fuel burners raise their temperature gradually, avoiding thermal shock and cracking.
3. **Soak:** In the final soaking zone, the slab's temperature is equalized through its full thickness so it deforms uniformly in the mill.
4. **Discharge:** A walking-beam or pusher mechanism moves the slab to the furnace exit, where an extractor pulls it out onto the mill approach table.
5. **Descale:** The hot slab passes through a high-pressure water descaler that blasts off the oxide scale formed in the furnace before it reaches the roughing mill.

TYPICAL PARAMETERS

Discharge temperature	**1150–1280 °C**
Residence time	**2–4 hours**
Hearth type	**Walking beam**
Fuel	**Natural gas / mixed gas**

KEY HAZARD

Extreme radiant heat, natural gas lines under pressure, and moving walking-beam machinery — furnace access requires cooling-down and gas isolation procedures.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Combustion air & ID/FD fan motors	Supply combustion air to burners and manage furnace draft and flue gas extraction.
Burner control system	Electronically regulates fuel/air ratio and firing rate in each zone for precise temperature profiles.
Walking-beam hydraulic drive motors	Electric motors power the hydraulics that lift, advance, and lower the walking beams moving slabs through the furnace.
Charging & extraction machine drives	Motorized pushers/extractors load slabs in and pull them out at the discharge end.
Pyrometers & thermocouples	Electronic temperature sensors feeding the combustion control system.
High-pressure descaler pump motors	Drive the pumps that generate high-pressure water jets to remove furnace scale.
MCC / switchgear	Distributes and protects power to all furnace-area motors and drives.
Level 1/2 automation (PLC/DCS)	Tracks each slab and schedules heating profiles to hit target discharge temperature.

2 Roughing Mill

Stage 2 of 5 | Breaking the slab down into a long transfer bar

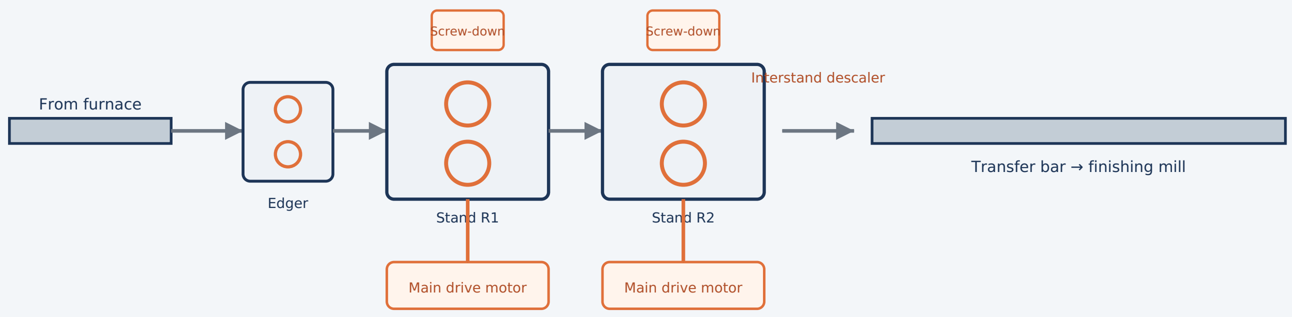


Fig. 2 — The roughing mill uses a series of reversing/tandem stands (with edgers) to reduce the slab's thickness in repeated passes, producing a long transfer bar.

HOW IT WORKS

1. **Descale:** The hot slab passes through high-pressure water sprays to remove surface scale before rolling.
2. **Edge:** Vertical edger rolls square up the slab's width, keeping edges straight for later stands.
3. **Rough down:** The slab passes through one or more horizontal roughing stands (often reversing — back and forth) that squeeze it thinner and longer with each pass.
4. **Repeat passes:** Depending on mill design, the slab may pass through the same stand several times, or through 2–5 stands in sequence, progressively reducing thickness.
5. **Hand off:** The slab emerges as a much longer, thinner "transfer bar," ready for the finishing mill.

TYPICAL PARAMETERS

Entry thickness	**~200–250 mm**
Exit thickness	**~25–50 mm**
Rolling temperature	**~1000–1150 °C**
Passes	**3–9 (mill dependent)**

KEY HAZARD

High-force pinch points at roll bites, hot moving metal, and reversing stand motion — strict guarding and clear zones are maintained around the mill line.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Main mill drive motors	Large AC or DC motors that rotate the work rolls in each roughing stand.
Screw-down / roll gap positioning motors	Electrically driven screws (or hydraulic actuators) that set the gap between rolls for the target reduction.
Edger stand drive motors	Power the vertical edging rolls that control slab width.
High-pressure descaler pump motors	Drive water pumps producing jets that blast scale off the slab surface.
Roller table drive motors	Move the slab/bar between stands and to the furnace exit.
Hydraulic power pack pumps	Supply pressurized oil for roll balance, bending, and gap-adjustment cylinders.
Automatic Gauge Control (AGC) system	Electronic feedback system adjusting roll gap in real time to hold target thickness.
Level 1/2 automation (PLC/DCS)	Sequences pass schedules, speeds, and reductions for each slab.

3 Finishing Mill

Stage 3 of 5 | Rolling the transfer bar down to final gauge, shape, and surface

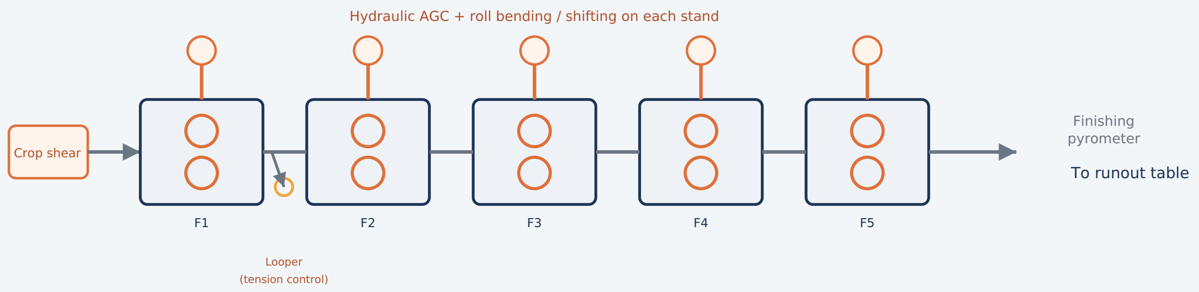


Fig. 3 — A tandem group of 5–7 close-coupled stands (F1–F7) rolls the transfer bar continuously down to final gauge, with hydraulic gauge and shape control on every stand.

HOW IT WORKS

1. **Crop & descale:** A crop shear trims the leading/trailing end of the transfer bar, and a final descaler removes any scale formed since roughing.
2. **Thread the mill:** The bar enters the first finishing stand (F1) and is threaded continuously through all stands at once — unlike the roughing mill, there's no reversing.
3. **Progressive reduction:** Each stand reduces thickness a little further while speed increases stand-to-stand to keep mass flow constant, since the strip gets thinner and faster as it goes.
4. **Control gauge & shape:** Hydraulic Automatic Gauge Control (AGC), roll bending, and roll shifting on each stand continuously correct thickness, flatness, and profile in real time.
5. **Exit at final gauge:** The strip leaves the last stand (often F5–F7) at its target thickness — commonly 1.2–25 mm — and full rolling speed.

TYPICAL PARAMETERS

Entry thickness	**~25–50 mm**
Exit thickness	**~1.2–25 mm**
Exit speed	**Up to ~20 m/s**
Number of stands	**5–7 (typical)**
Finishing temperature	**~850–950 °C**

KEY HAZARD

Very high strip speeds and stored tension between stands; a strip break ("cobble") can whip loose material — the mill line is a restricted, guarded area during operation.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Main drive motors (per stand)	Individually speed-controlled AC/DC motors driving each finishing stand's work rolls, synchronized for constant mass flow.
Hydraulic AGC servo valves & cylinders	Electrically controlled hydraulics that adjust roll gap within milliseconds to hold gauge tolerance.
Work roll bending & shifting drives	Electric/hydraulic actuators that correct strip flatness and crown profile.
Interstand looper motors	Small motorized arms that control tension and loop height of strip between stands.
Crop shear drive motor	Powers the shear that trims the bar's leading and trailing ends.
Roll cooling & lubrication pump motors	Circulate water/oil to cool rolls and reduce friction during rolling.
Strip gauge, profile & temperature sensors	X-ray/laser gauges and pyrometers feeding real-time process control.
Level 2 automation (mill setup model)	Calculates roll gaps, speeds, and tensions for each stand before and during rolling.

Runout Table (Laminar Cooling)

4

Stage 4 of 5 | Cooling the strip in a controlled way to set its final properties

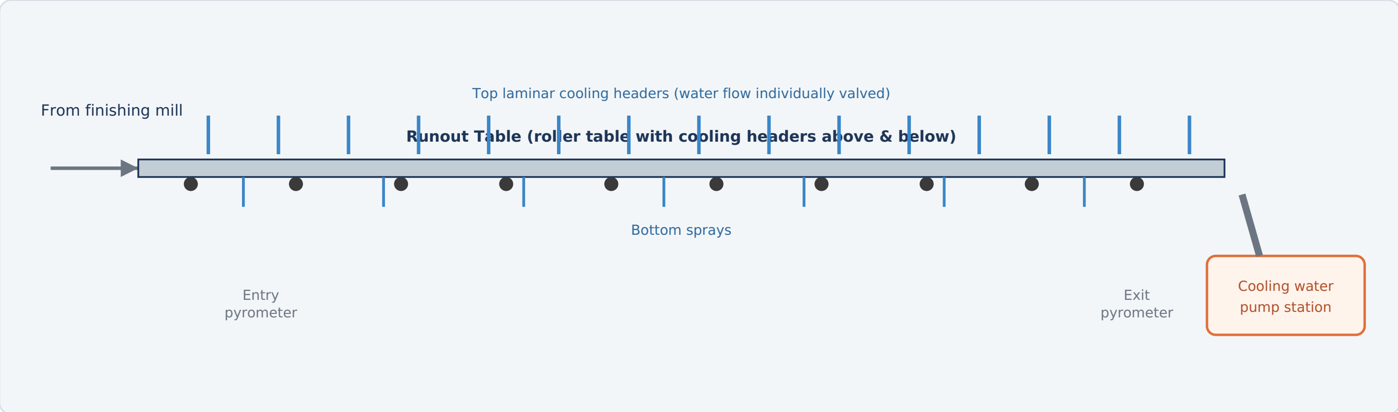


Fig. 4 — Rows of individually-controlled water headers cool the strip at a precise, programmed rate as it travels along the runout table to the coiler.

HOW IT WORKS

1. **Enter hot:** The strip leaves the finishing mill at ~850–950°C and travels along the roller table toward the coiler.
2. **Programmed cooling:** Rows of overhead (and sometimes underside) water headers switch on and off in sequence, following a cooling curve calculated for the steel grade being produced.
3. **Control cooling rate:** The rate and pattern of cooling determines the steel's final microstructure (grain size, phase balance) and therefore its mechanical properties — strength, ductility, and hardness.
4. **Monitor temperature:** Pyrometers at the entry and exit of the runout table measure actual strip temperature and feed back into the header control system to correct water flow in real time.
5. **Hand off to coiler:** By the time the strip reaches the coiler, it has cooled to the target coiling temperature.

TYPICAL PARAMETERS

Entry temperature	**~850–950 °C**
Coiling temperature	**~500–700 °C**
Cooling rate	**Grade-dependent, controlled**
Table length	**Often 100+ m**

KEY HAZARD

Large volumes of high-pressure water near hot moving steel create steam and slip hazards; electrical enclosures near the table are rated for the wet environment.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Cooling water pump motors	Large motors supplying high-volume water flow to the header banks.
Header solenoid/control valves	Electrically actuated valves that switch individual water header rows on and off per the cooling model.
Roller table drive motors	Move the strip along the runout table at controlled, synchronized speed.
Entry & exit pyrometers	Non-contact infrared sensors measuring strip temperature for closed-loop cooling control.
Strip tracking sensors (photoeyes)	Detect the strip's position to trigger the correct header sequence as it passes.
Cooling model computer / Level 2 automation	Calculates which headers to fire and for how long to hit the target coiling temperature and cooling rate.

5 Down Coiler

Stage 5 of 5 | Winding the finished strip into a shippable coil

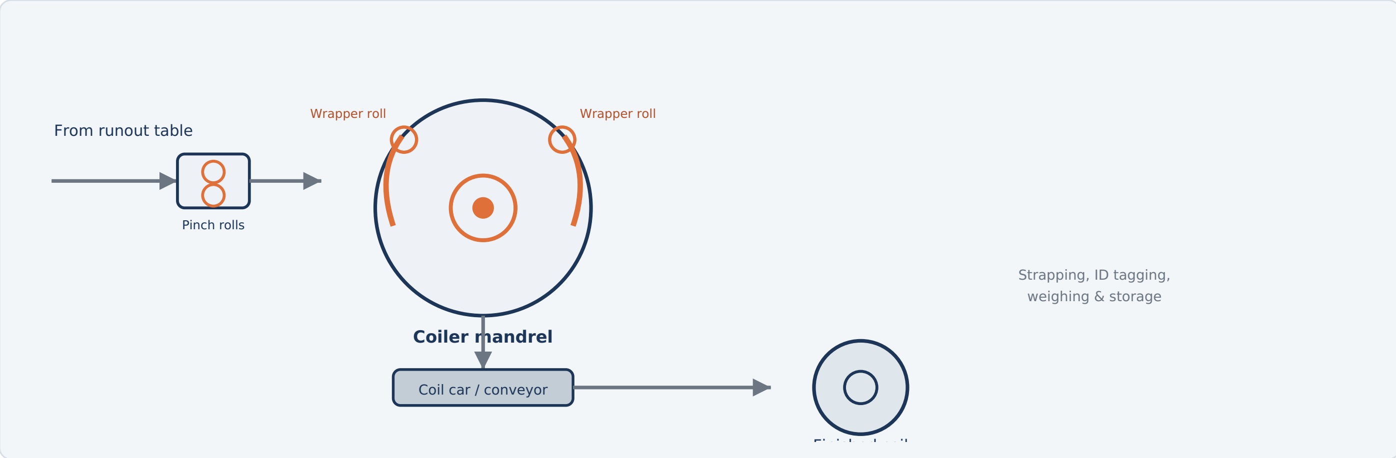


Fig. 5 — Pinch rolls feed the strip onto a motorized mandrel; wrapper rolls curl the leading end around it as the coil builds, then the finished coil is stripped off and moved to storage.

HOW IT WORKS

1. **Feed in:** Pinch rolls grip the strip and guide it onto the coiler entry, keeping consistent tension.
2. **Wrap onto the mandrel:** Wrapper rolls curl the strip's leading edge tightly around the expandable coiler mandrel to start the wind.
3. **Build the coil:** The mandrel rotates, winding the strip under controlled tension as more material feeds in, building up a tight, uniform coil.
4. **Cut to length:** Once the full strip has been wound (or at a scheduled coil weight), a shear cuts the strip and the trailing end is secured.
5. **Strip & transfer:** The mandrel contracts, releasing the finished coil onto a coil car, which moves it to strapping, weighing, ID tagging, and storage — ready for shipping or further cold processing.

TYPICAL PARAMETERS

Coiling temperature	**~500–700 °C**
Coil weight	**~15–35 t (typical)**
Coil OD	**~1.8–2.2 m**
Output	**Hot-rolled coil**

KEY HAZARD

Rotating mandrel and heavy coil handling; hot coils retain significant heat and sharp/springy strip ends — strict exclusion zones apply during winding and coil transfer.

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Coiler mandrel drive motor	Rotates the mandrel under controlled torque/tension to wind the coil evenly.
Mandrel hydraulic expansion system	Electrically controlled hydraulics that expand the mandrel to grip the coil and contract it for release.
Pinch roll drive motors	Feed the strip into the coiler at line speed with controlled tension.
Wrapper roll drive & positioning motors	Curl the strip's leading end around the mandrel and retract once the coil is self-supporting.
Coiler shear drive	Cuts the strip at the end of each coil.
Coil car / conveyor drive motors	Transfer finished coils from the coiler to strapping and storage.
Coil weighing & ID marking system	Electronic scales and marking equipment for tracking and traceability.
Tension control / Level 2 automation	Coordinates mandrel speed and pinch roll speed to maintain correct winding tension throughout the coil build.

Process Summary

The full hot rolling route, and every stage's electrical equipment at a glance

One continuous, high-speed line from hot slab to finished coil

REHEAT FURNACE → ROUGHING MILL → FINISHING MILL → RUNOUT TABLE → COILER

REHEAT FURNACE	ROUGHING MILL	FINISHING MILL	RUNOUT TABLE	COILER
Heat to temp.	Coarse reduction	Final gauge	Controlled cooling	Coil the strip

ELECTRICAL EQUIPMENT — ALL STAGES

STAGE	MAJOR ELECTRICAL EQUIPMENT	PRIMARY ROLE
Reheating Furnace	Combustion air/ID fans, burner controls, walking-beam hydraulic drives, charging/extraction drives, descaler pumps	Bring the slab to a uniform rolling temperature
Roughing Mill	Main drive motors, screw-down/AGC motors, edger drives, descaler pumps, roller table drives, hydraulic power packs	Coarse thickness reduction of the slab
Finishing Mill	Per-stand main drive motors, hydraulic AGC servo valves, roll bending/shifting drives, looper motors, crop shear drive	Roll down to final gauge, shape, and surface finish
Runout Table	Cooling water pump motors, header control valves, roller table drives, pyrometers, cooling model computer	Controlled cooling to set final microstructure
Down Coiler	Mandrel drive motor, mandrel hydraulic expansion system, pinch/wrapper roll drives, coiler shear, coil car drives	Wind the finished strip into a shippable coil

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A cold or warm slab enters the **reheating furnace** and is brought to a uniform ~1200°C before being descaled and pushed through the **roughing mill**, where reversing or tandem stands (with edgers) reduce it from roughly 200–250 mm down to a long transfer bar. The transfer bar then threads continuously through the **finishing mill** — a tandem group of 5–7 stands under hydraulic gauge and shape control — emerging at final thickness and full rolling speed. On the **runout table**, precisely sequenced water headers cool the strip at a programmed rate to set its mechanical properties, before the **coiler** winds it into a tight, tensioned coil, cuts it to length, and transfers it to strapping and storage — ready for shipping or further cold processing.

PART 11 — REHEATING FURNACE

Why Reheat? The Metallurgy of Hot Working

Steel has to be hot enough to flow — and uniformly hot, or it won't flow evenly

Cold steel is strong and brittle: deforming it by the amount a rolling mill needs — cutting a 250 mm slab down toward a few millimeters over several stages — would take enormous force and would crack the material long before it reached final gauge. Heating steel above roughly 1000–1100°C drops its flow stress dramatically and lets it deform plastically instead of fracturing, which is exactly why every hot strip mill starts with a furnace. But getting the surface hot is the easy part; the **reheating furnace's real job is getting the entire cross-section to a uniform temperature**, because a slab with a hot surface and a cold core rolls unevenly, overloads the mill, and can crack internally under the rolling forces.

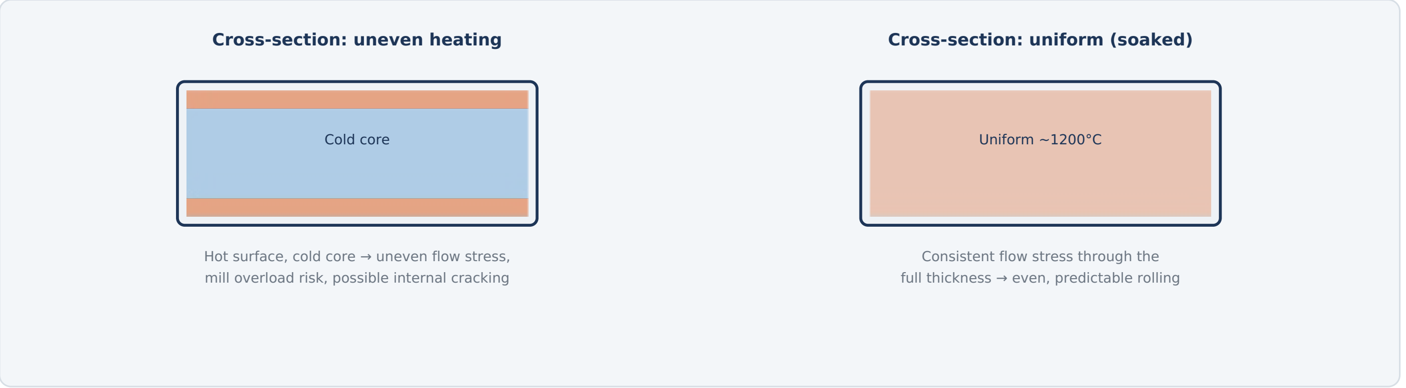


Fig. 1 — Surface temperature is easy to achieve quickly; the furnace's soaking zone exists specifically to let heat conduct all the way to the core before the slab ever reaches the mill.

WHY WALKING BEAM OVER OLDER PUSHER FURNACES

DESIGN	HOW IT WORKS	TRADE-OFF
Pusher furnace	Slabs are pushed from behind, sliding along fixed skid rails the whole way through the furnace.	Simpler, but skid rails contact and locally cool the slab underside, leaving "skid marks" that can affect rolling and surface quality.
Walking-beam furnace	Moving beams lift each slab clear off its rails, advance it forward, and set it back down — repeated in a continuous cycle.	No sustained skid contact, more uniform heating, precise individual slab spacing and residence time control — the modern standard.

TYPICAL PARAMETERS

Discharge temperature	**1150–1280 °C**
Residence time	**2–4 hours**
Hearth type	**Walking beam**

WHY THIS MATTERS FOR EVERYTHING THAT FOLLOWS

Every electrical system in this guide — burner control, walking-beam drives, and the slab-pacing model — exists to serve one goal: deliver each slab to the mill at the right temperature, uniformly, exactly when the mill is ready for it.

1 The Reheating Process

From charging to discharge — three zones, one continuous walking-beam cycle

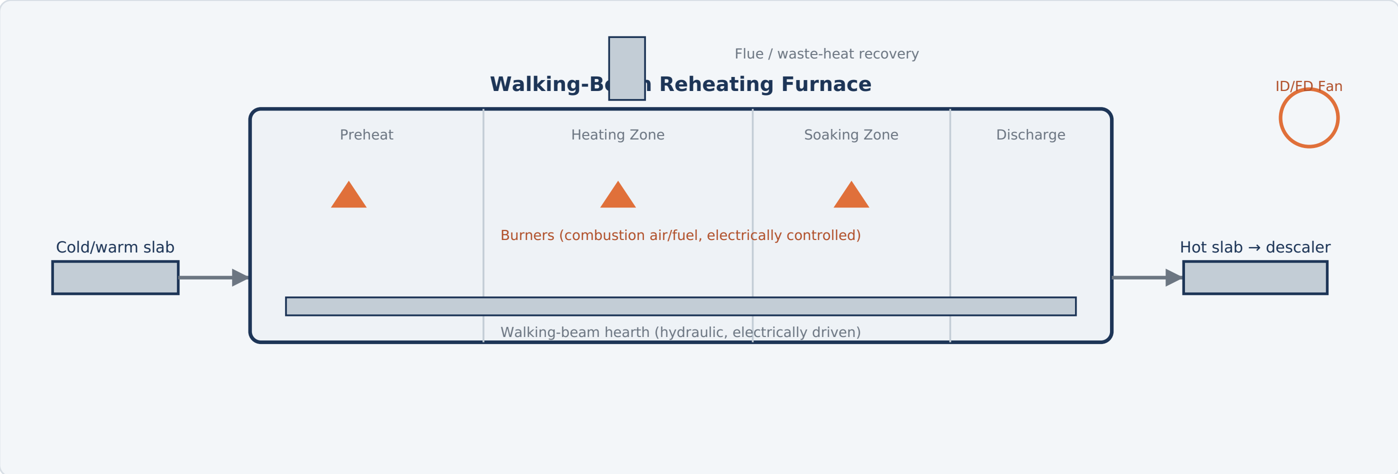


Fig. 2 — Slabs move through preheat, heating, and soaking zones on a walking-beam hearth until they reach a uniform rolling temperature, then discharge through a descaler toward the roughing mill.

HOW IT WORKS

1. **Charge:** Slabs are loaded onto the furnace's charging table and pushed or walked into the entrance, spaced according to the furnace pacing schedule.
2. **Preheat:** In the first zone, slabs absorb heat gradually from the hot flue gases moving toward the stack — a gentle warm-up that also serves as waste-heat recovery.
3. **Heat:** Burners in the main heating zone(s) deliver the bulk of the energy, raising the slab surface toward target temperature.
4. **Soak:** In the final zone, temperature is held steady long enough for heat to conduct all the way to the core, equalizing the through-thickness temperature.
5. **Discharge:** A walking-beam or extractor mechanism moves the slab to the furnace exit and onto the mill approach table.
6. **Descale:** A high-pressure water descaler blasts off the oxide scale formed during heating before the slab reaches the roughing mill.

TYPICAL PARAMETERS

Discharge temperature	**1150–1280 °C**
Residence time	**2–4 hours**
Fuel	**Natural gas / mixed gas**
Furnace zones	**3–5 typical**

2 Combustion & Zone Control

Dozens of burners, each individually metered, working together to hold a precise temperature profile

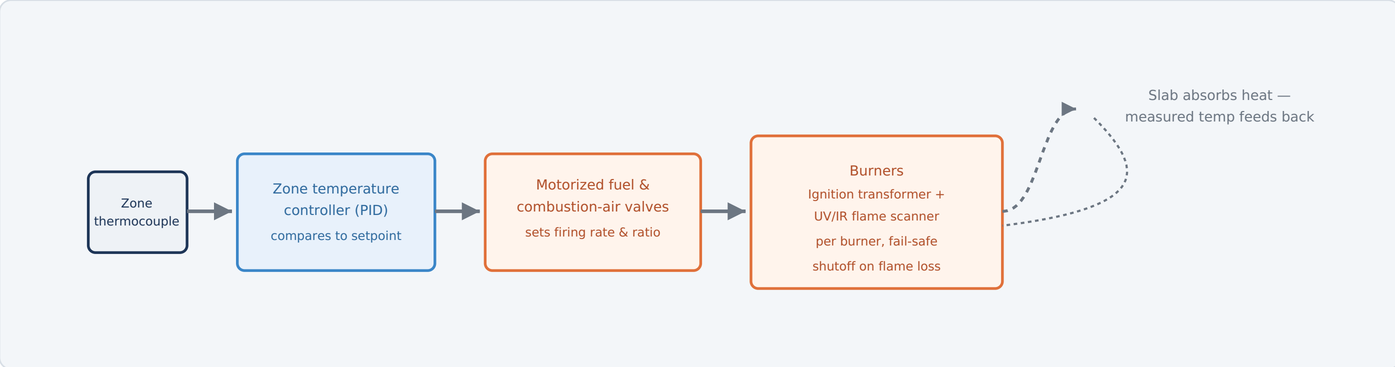


Fig. 3 — Each zone runs its own closed loop: a thermocouple measures actual temperature, a controller compares it to setpoint, and motorized valves adjust firing rate and fuel/air ratio to correct any drift.

TWO BURNER TECHNOLOGIES

TYPE	HOW IT WORKS
Conventional (recuperative) burners	Burn continuously; a separate heat exchanger (recuperator) preheats incoming combustion air using heat recovered from the exhaust flue gas.
Regenerative burners	Burners are paired and fire alternately — while one fires, its partner's ceramic core absorbs heat from the outgoing exhaust; roles swap every 20–60 seconds. This recovers dramatically more waste heat than recuperative designs, cutting specific fuel consumption significantly, at the cost of more complex electrically-sequenced flow-switching valves.

WHY FUEL/AIR RATIO CONTROL MATTERS BEYOND EFFICIENCY

Excess oxygen in the furnace atmosphere doesn't just waste fuel — it accelerates oxidation of the slab surface, producing more scale. Since scale represents pure yield loss (metal that never reaches the customer as product), tight ratio control at every burner is as much a yield issue as an energy one.

ELECTRICAL EQUIPMENT

Zone thermocouples	**Multiple points per zone**
Burner ignition transformers	**Per-burner**
Flame scanners	**UV/IR, per-burner**
Motorized fuel/air valves	**4–20 mA control**
Regenerative switching valves	**Electrically sequenced**

KEY POINT

Furnace pressure is also actively controlled (slightly positive) via ID/FD fan speed and damper position — too negative and cold air infiltrates, wasting energy and cooling slabs unevenly near the furnace shell.

3 Electrical Equipment in Detail

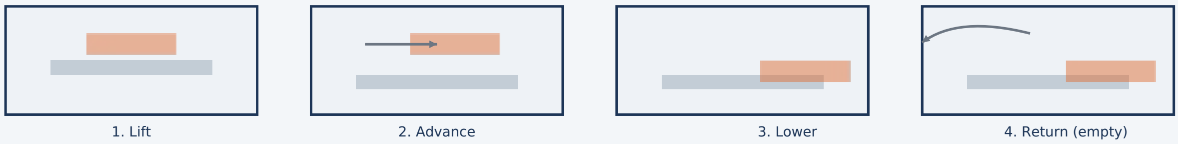
From combustion air fans to the slab-tracking automation that paces the whole furnace

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
COMBUSTION SYSTEM		
Combustion air / ID / FD fan motors	Supply combustion air to burners and manage furnace draft, flue gas extraction, and internal pressure.	Multiple fans, VFD-controlled for turndown
Burner ignition & flame safety system	Electric ignition transformers spark each pilot; flame scanners confirm the flame and trigger fail-safe shutoff within about a second of flame loss.	Per-burner protection relay
Motorized fuel & combustion-air valves	Modulate firing rate and fuel/air ratio at each zone or burner group per the temperature control loop.	4–20 mA actuated
Regenerative burner switching valves	Electrically sequence which burner of a pair is firing versus recovering heat, on a short cycle.	Where regenerative burners are installed
WALKING-BEAM & MATERIAL HANDLING		
Walking-beam hydraulic drive motors	Electric motors power the hydraulics executing the beam's lift-advance-lower-return cycle.	Typically 3 independent hydraulic axes
Charging machine drive	Motorized pusher/loader that moves slabs from the approach table into the furnace entrance.	Position-controlled
Extraction machine drive	Motorized mechanism that pulls finished slabs out at the discharge end.	Synchronized with beam cycle
High-pressure descaler pump motors	Drive the pumps generating high-pressure water jets that remove furnace scale from the discharged slab.	Located at furnace exit
INSTRUMENTATION & CONTROL		
Zone thermocouples	Multiple measurement points per zone feeding each zone's temperature control loop.	Type K/N, roof and/or sidewall mounted
Slab discharge pyrometer	Non-contact infrared sensor confirming actual slab surface temperature at furnace exit.	Continuous monitoring
Waste-heat recovery fan motors	Circulate flue gas through recuperators or support regenerative burner operation.	Where fitted
MCC / switchgear	Distributes and protects power to all furnace-area motors and drives.	Low/medium voltage distribution
Level 1/2 automation (slab tracking & pacing model)	Tracks each slab's position and thermal history individually, and paces the walking-beam cycle to deliver every slab at target temperature exactly when the mill needs it.	Coordinates all systems above

4 The Walking Beam & Slab Pacing Model

A furnace full of individually-tracked slabs, each on its own heating schedule

The walking-beam cycle (repeated continuously)



Every cycle advances every slab in the furnace by one fixed step —
cycle time sets the residence time (and therefore heating time) for all of them at once

Fig. 4 — Three independent hydraulic axes execute a repeating lift-advance-lower-return cycle; because every slab in the furnace moves together each cycle, the cycle time is the single knob that sets throughput for the whole furnace.

TRACKING EVERY SLAB INDIVIDUALLY

Even though the walking beam moves every slab together, different slabs can need different total heating times — a thicker slab, a different grade, or one that's being hot-charged directly from the caster all need a different thermal history. The Level 2 automation system tracks each slab's identity, dimensions, grade, and entry temperature individually as it moves through the furnace, continuously predicting its current core temperature using a heat-conduction model, and confirms it's on track to reach target temperature by the time it reaches the discharge end.

PACING TO THE MILL, NOT JUST THE FURNACE

The walking-beam cycle time isn't set in isolation — it's paced to match how fast the downstream roughing mill can actually accept slabs. Too fast a cycle and slabs queue up outside the furnace or leave under-heated; too slow and the mill sits idle waiting. This is the same "pacing to the next station's schedule" problem seen at the LMF pacing itself to the caster in the melt shop guides — here it's the furnace pacing itself to the mill.

WHAT THE MODEL TRACKS

Slab identity & dimensions	**Per-slab record**
Predicted core temperature	**Continuous heat-conduction model**
Target discharge temp.	**Grade-specific**
Mill readiness	**Signal from roughing mill**

5 Throughput vs. Temperature Uniformity

The furnace's central balancing act — and why you can't simply speed up the cycle

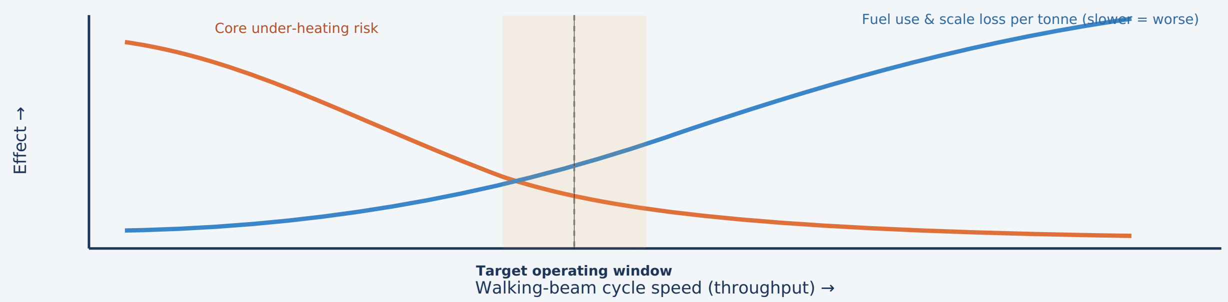


Fig. 5 — Cycle too fast and heat doesn't reach the slab core in time; cycle too slow and specific fuel consumption and scale losses climb from longer steady-state exposure. The pacing model continuously targets the window between them.

WHAT HAPPENS AT EACH EXTREME

EXTREME	CONSEQUENCE
Cycle too fast	Insufficient time for heat to conduct to the slab core; slabs leave with a hot surface but a cold interior, risking mill overload, uneven rolling, and possible internal cracking under rolling force.
Cycle too slow	Slabs spend longer than necessary at high temperature, increasing scale formation (yield loss) and steady-state fuel consumption per tonne produced — and reduces overall mill throughput.

HOW THE BALANCE IS ACTUALLY HELD

Rather than a fixed cycle time, the pacing model continuously recalculates the minimum safe residence time for the specific slabs currently in the furnace (thickest, coldest-charged, or hardest-to-heat grade sets the pace), and adjusts cycle time to match — trading a little throughput on difficult slabs to avoid compromising temperature uniformity on any of them.

INPUTS TO THE PACING DECISION

Slab thickness & grade	**Sets required soak time**
Charge temperature	**Cold vs. hot-charged**
Mill readiness signal	**Avoids starving or overrunning**
Zone temperature margin	**Headroom to speed up safely**

6 Safety & Protection Systems

Flame safety, gas handling, and the mechanical hazards of a furnace full of moving machinery

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Flame failure interlock	Flame scanner loses signal from a lit burner	Shuts fuel valve to that burner within ~1 second; blocks re-ignition until purged
Furnace pressure control	Internal pressure drifts negative, risking cold-air infiltration, or excessively positive	Adjusts ID/FD fan speed and damper position to hold target pressure
Gas leak detection	Combustible gas detected in the furnace building	Alarms operator; can trigger automatic fuel isolation
Emergency fuel shutoff (ESD)	Manual activation or a critical safety trip	Isolates fuel supply to the entire furnace
Walking-beam hydraulic pressure monitoring	Hydraulic pressure falls outside safe operating range	Halts beam motion before an uncontrolled drop or collision can occur
Walking-beam pinch-point guarding	Access to the beam mechanism while in motion	Physical guarding and interlocked access doors
Charging/extraction machine interlock	Machine motion commanded while personnel access is detected	Blocks motion until the area is clear
Descaler water pressure/flow monitoring	Loss of descaler water pressure	Alarms operator; can hold slab discharge

WHY FLAME SAFETY IS THE ANCHOR OF EVERY OTHER SYSTEM

A reheating furnace burns a continuous, large volume of fuel gas in a confined refractory chamber — flame failure interlocks and gas detection exist to guarantee that unburned fuel can never accumulate to a dangerous concentration, which is why they are hard-wired, per-burner systems rather than a single furnace-wide check.

✓ Summary

What the reheating furnace hands off to the roughing mill, and a recap of the process

FACTOR	POORLY CONTROLLED FURNACE	WELL-CONTROLLED FURNACE
Core temperature	Inconsistent, risk of under-heated cores	Uniform through-thickness temperature at discharge
Scale loss	Higher, from excess time/oxygen at temperature	Minimized via tight ratio and residence-time control
Fuel consumption	Higher specific consumption per tonne	Reduced via zone control and (where fitted) regenerative burners
Mill pacing	Mill starves or backs up waiting on slabs	Slabs arrive exactly when the mill is ready

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

A **reheating furnace** exists to bring a slab from ambient (or hot-charge) temperature to a **uniform** ~1200°C throughout its thickness — the surface heats quickly, but the **soaking zone** is what lets heat conduct all the way to the core. A **walking-beam hearth** carries slabs through preheat, heating, and soaking zones on a repeating lift-advance-lower-return cycle, while independent **zone temperature loops** modulate burner firing rate and fuel/air ratio to hold each zone's setpoint. Because every slab in the furnace moves together each cycle, the **pacing model** has to track each slab's individual thermal history and set cycle time to the slowest-heating slab present — balancing throughput against the risk of an under-heated core, while also matching the pace the downstream roughing mill can actually accept.

Why So Many Passes? The Angle of Bite

A physical limit on how much thickness a single pair of rolls can grip and reduce at once

Going from a 200–250 mm slab to a transfer bar a fraction of that thickness looks like it should be one big squeeze — but a pair of rolls can only bite into the workpiece and pull it through if friction between the roll surface and the metal is enough to overcome the roll's tendency to simply push the metal away. That limit is governed by the **angle of bite**: the contact angle between roll and workpiece can't exceed the point where the friction coefficient can still grip it. Try to take too large a reduction in one pass and the rolls simply can't grab the slab at all. That single constraint is why a roughing mill takes 5–9 separate passes to do what looks like it should be one operation.

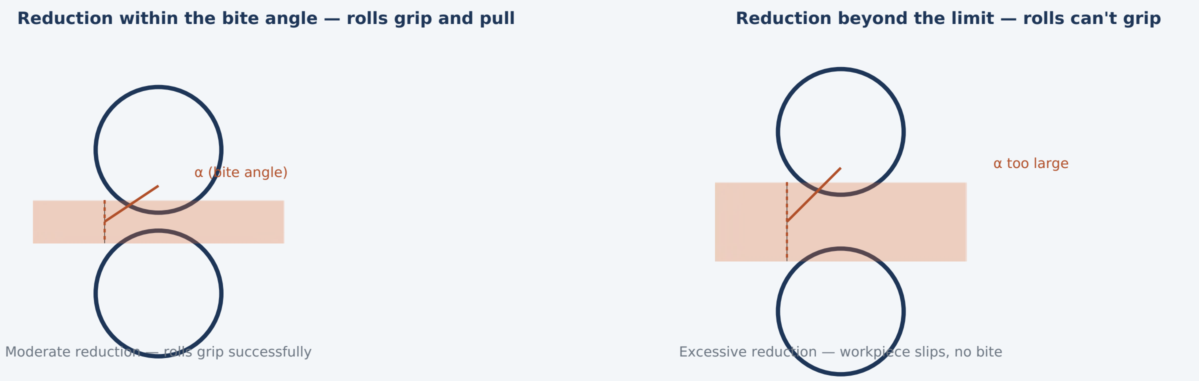


Fig. 1 — Friction can only grip the workpiece up to a maximum bite angle; beyond it the rolls slip rather than pull the metal through, forcing multiple smaller-reduction passes instead of one large one.

TWO WAYS TO ARRANGE THE PASSES

CONFIGURATION	HOW IT WORKS
Reversing stand(s)	The slab passes back and forth through the same one or two stands repeatedly, direction reversing each pass, until it reaches transfer bar thickness.
Tandem roughing line	Several stands arranged in a row, each taking one reduction as the bar passes through once — no reversing, but more total stands and floor space.

TYPICAL PARAMETERS

Entry thickness	**~200–250 mm**
Exit thickness	**~25–50 mm**
Passes	**5–9**
Rolling temperature	**~1000–1150 °C**

1 The Roughing Process

Descale, edge, and reduce — repeated until the bar is ready for finishing

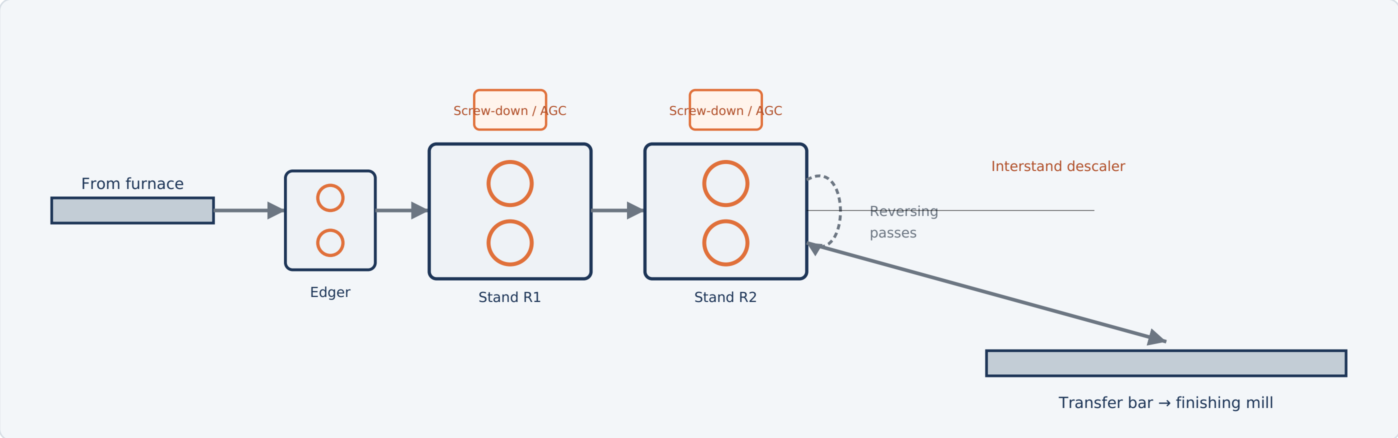


Fig. 2 — The slab descales, passes through an edger to control width, then reduces in thickness through one or more stands, reversing repeatedly, until it emerges as a long transfer bar.

HOW IT WORKS

1. **Descale:** High-pressure water sprays remove furnace scale before the slab enters the first stand.
2. **Edge:** Vertical edger rolls square up the slab's width, keeping edges straight and controlling final bar width.
3. **Rough down:** The slab passes through one or more horizontal roughing stands — often reversing — squeezing it thinner and longer with each pass.
4. **Repeat:** Depending on mill design, the slab passes through the same stand several times or through 2–5 stands in sequence, with an interstand descaler between passes to keep the surface clean.
5. **Hand off:** The slab emerges as a much longer, thinner transfer bar, ready for the finishing mill.

TYPICAL PARAMETERS

Entry thickness	**~200–250 mm**
Exit thickness	**~25–50 mm**
Rolling temperature	**~1000–1150 °C**
Passes	**5–9**

2 Automatic Gauge Control (AGC)

Holding target thickness on a mill that flexes under enormous force

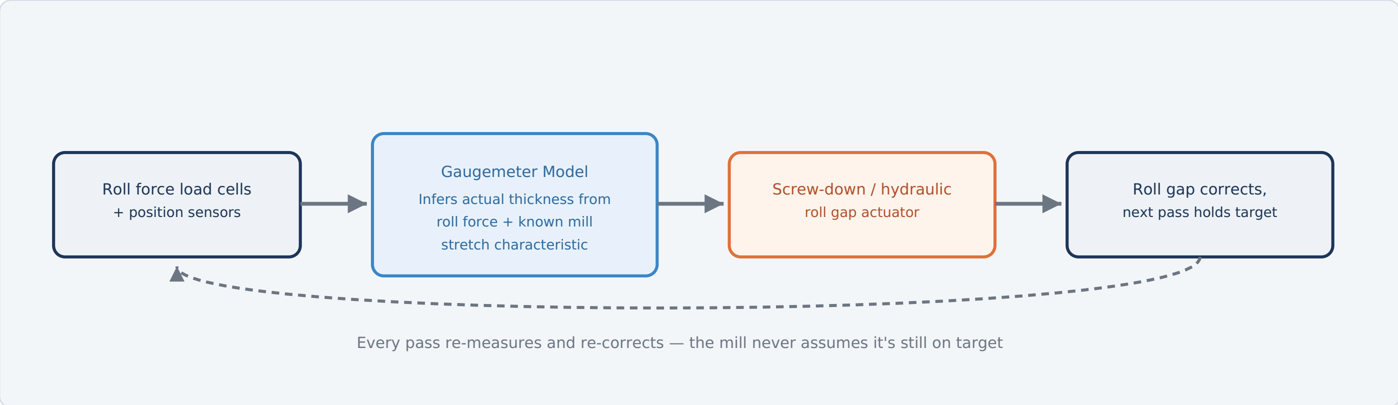


Fig. 3 — Since a direct thickness gauge often isn't practical inside the roughing stand itself, the "gaugemeter" principle infers actual thickness from measured roll force and the mill's known elastic stretch, closing the loop every pass.

WHY THE MILL ITSELF HAS TO BE PART OF THE MODEL

Under the enormous forces of hot roughing, the mill housing and rolls themselves flex elastically — a real, measurable stretch that changes the actual gap between the rolls from what the screw-down position alone would suggest. The gaugemeter principle uses a pre-characterized force-vs-stretch curve for the mill, combined with real-time roll force measurement, to back-calculate actual exit thickness without needing a gauge sensor positioned inside the hot, harsh stand environment itself.

TYPICAL PARAMETERS

AGC response	**Within the pass**
Feedback source	**Roll force + position**
Verification	**X-ray gauge downstream**

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Roll force load cells	Continuously measure force between the rolls during each pass, the key gaugemeter input.
Screw-down / AGC position motors or hydraulic actuators	Physically adjust roll gap in response to the gaugemeter's correction signal.
X-ray thickness gauge (downstream)	Provides a direct thickness check after the stand, used to calibrate and verify the gaugemeter model.

3 Electrical Equipment in Detail

From the massive main drive motors to the descaler pumps between passes

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
MAIN DRIVE & GAUGE CONTROL		
Main mill drive motors	Large motors that rotate the work rolls; historically DC for superior low-speed, high-torque, and reversing characteristics, increasingly AC with advanced drives on modern mills.	Very high torque, reversing duty
Screw-down / AGC actuators	Electric motor-driven screws or electro-hydraulic cylinders that set and correct roll gap.	Per-pass correction
Roll force load cells	Feed the gaugemeter model that infers actual exit thickness.	High-capacity, continuous monitoring
X-ray thickness gauge	Direct downstream thickness measurement used for verification and model calibration.	Non-contact
EDGING & DESCALING		
Edger stand drive motors	Power the vertical edging rolls that control slab/bar width.	Independent drive from horizontal stands
High-pressure descaler pump motors	Drive water pumps producing jets that remove scale before and between passes.	Multiple stations along the line
MATERIAL HANDLING		
Roller table drive motors	Move the slab/bar between the furnace, edger, stand(s), and onward to the finishing mill.	Many individually driven sections
Hydraulic power packs	Supply pressurized oil for roll balance, bending, and gap-adjustment cylinders.	Central or stand-local supply
AUTOMATION		
Level 1/2 automation (pass schedule model)	Calculates the number of passes, reduction per pass, and roll gap/speed targets for each slab based on its dimensions and grade.	Sets targets executed by AGC each pass

4 Pass Schedule Optimization

Choosing how to spend a limited number of passes across competing goals

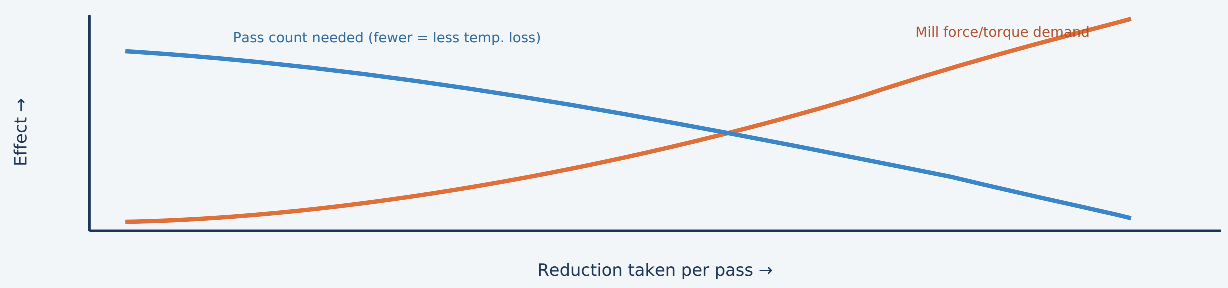


Fig. 4 — Bigger reductions per pass mean fewer total passes (less time, less temperature loss) but demand more force and torque from the mill; the schedule has to respect equipment limits while minimizing pass count.

WHAT THE SCHEDULE HAS TO BALANCE

GOAL	CONSTRAINT IT COMPETES WITH
Minimize number of passes	Fewer passes means less time for the bar to cool between the furnace and the finishing mill — but each pass is capped by the angle of bite and by mill force/torque limits.
Hit target width	Edging passes control width but add their own pass count and time.
Stay within equipment limits	Roll force, motor torque, and mill housing stretch all cap how aggressive any single pass can be.
Manage temperature loss	Every pass extracts heat through contact with the (relatively cool) rolls; too many passes and the bar cools below the temperature the finishing mill needs.

The Level 2 pass-schedule model calculates all of this before the slab even enters the first stand, then AGC executes and fine-tunes it pass by pass based on what the gaugemeter actually measures.

TYPICAL PARAMETERS

Passes	**5–9**
Temperature loss target	**Managed to finishing mill entry spec**
Schedule source	**Level 2 model, per slab**

5 Safety & Protection Systems

High-force pinch points and reversing motion in a mill handling red-hot steel

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Roll force overload trip	Measured roll force exceeds mill design limit	Halts the pass and alarms operator before mechanical damage occurs
Main drive motor protection	Overcurrent or thermal overload on the main drive motor	Trips the motor off-line
Reversing motion interlock	Direction change commanded while the bar/rolls are not in a safe state	Sequences reversal only when safe conditions are confirmed
Roll neck bearing monitoring	Abnormal vibration or temperature at roll bearings	Alarms operator; can reduce speed or halt
Entry/exit guarding	Personnel access near the roll bite while the stand is powered	Physical guards and interlocked access points
Hydraulic pressure loss protection	Loss of pressure in AGC or roll balance hydraulics	Halts affected motion in a controlled manner
Emergency stop (E-stop)	Manual activation anywhere along the line	Removes power from drives per the defined E-stop category

WHY REVERSING OPERATION IS A DISTINCT HAZARD CATEGORY

Unlike a continuous tandem line, a reversing roughing stand changes the direction red-hot steel is moving, repeatedly, in the same physical space — clear zones, directional interlocks, and operator sightlines are all designed specifically around this back-and-forth motion pattern.

✓ Summary

What the roughing mill hands off to the finishing mill, and a recap of the process

FACTOR	BASIC CONTROL	AGC + PASS SCHEDULE OPTIMIZATION
Thickness consistency	Drifts with mill stretch under varying force	Corrected every pass via the gaugemeter model
Pass count	Fixed, conservative	Optimized per slab against force limits and temperature loss
Width control	Coarser	Actively managed via scheduled edging passes
Bar temperature at exit	More variable	Managed to hit finishing mill entry target

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **roughing mill** exists to take a thick, hot slab down toward transfer-bar thickness — but physics caps how much reduction a single pass can take, since rolls can only grip the workpiece within a limited **angle of bite**. That forces **5–9 passes** through one or more reversing stands, with an **edger** controlling width and descalers keeping the surface clean between passes. Because the mill housing itself flexes elastically under the huge forces involved, **automatic gauge control** uses a "gaugemeter" model — measured roll force plus the mill's known stretch characteristic — to infer actual thickness and correct the roll gap every single pass, rather than trusting screw position alone. A **pass-schedule model** decides upfront how to spend those passes, balancing fewer/bigger reductions (faster, less cooling, but higher force) against equipment limits and target width, before AGC takes over to execute and fine-tune the plan in real time.

Why Speed Has to Change at Every Stand

Mass flow continuity — the physics that ties every stand in the finishing mill together

Unlike the roughing mill's reversing passes, the finishing mill threads the strip continuously through 5–7 stands at once — the same piece of steel is inside every stand simultaneously. Since steel is essentially incompressible and the same mass has to pass every point along the line each second, thickness and speed are locked together: as each stand reduces thickness, the strip has to speed up proportionally to keep the same mass flowing through per second. By the last stand, the strip can be moving many times faster than it entered the first one. This single physical fact — **mass flow continuity** — is why every stand's speed has to be precisely coordinated with every other stand's, all the time.

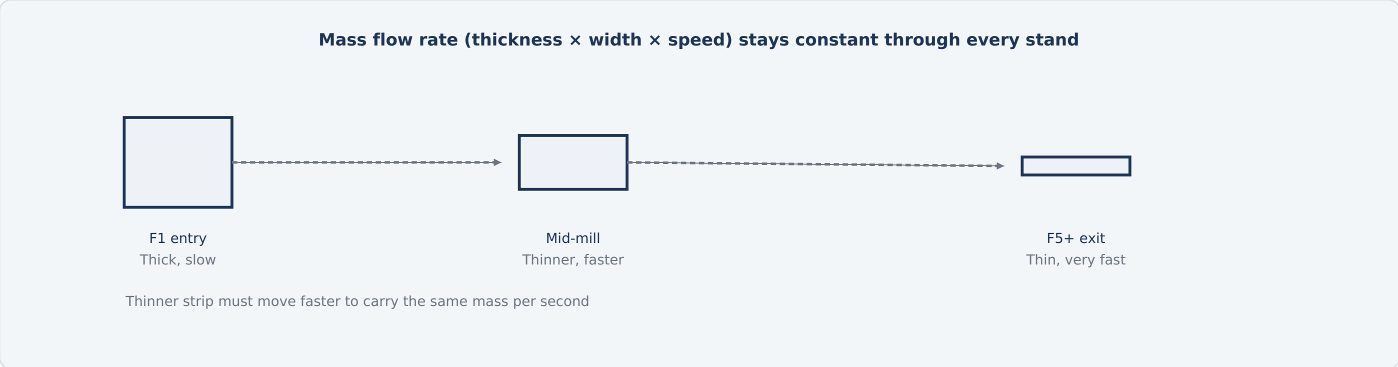


Fig. 1 — As thickness drops through the mill, speed rises to match, so the same tonnage per hour is always flowing past every point on the line simultaneously.

WHY THIS MAKES THE FINISHING MILL HARDER TO CONTROL THAN THE ROUGHING MILL

A reversing roughing stand only has to control itself. A finishing mill's stands are all mechanically coupled through the strip at every instant — a speed error at one stand doesn't just affect that stand's output gauge, it immediately changes tension to its neighbors, which can cascade into a shape defect or, in the worst case, a strip break. Every electrical system on this line ultimately exists to manage that coupling.

TYPICAL PARAMETERS

Entry thickness	**~25–50 mm**
Exit thickness	**~1.2–25 mm**
Exit speed	**Up to ~20 m/s**
Number of stands	**5–7**

1 The Finishing Process

One continuous thread through every stand — no stopping, no reversing

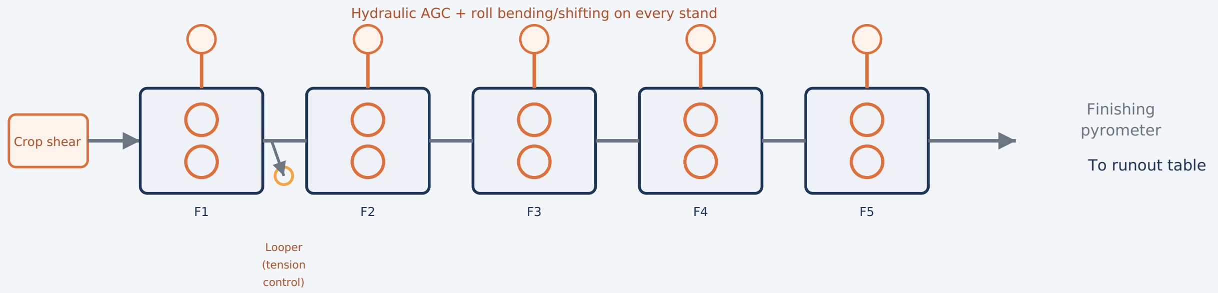


Fig. 2 — The transfer bar threads through all stands at once; each reduces thickness a little further while speed increases to match, with loopers between stands managing tension.

HOW IT WORKS

1. **Crop & descale:** A crop shear trims the bar's leading/trailing end, and a final descaler removes scale before entry.
2. **Thread the mill:** The bar enters stand F1 and is threaded continuously through every stand at once.
3. **Progressive reduction:** Each stand reduces thickness further while speed rises to maintain constant mass flow.
4. **Control gauge & shape:** Hydraulic AGC, roll bending, and roll shifting on each stand correct thickness, flatness, and profile in real time.
5. **Exit at final gauge:** The strip leaves the last stand at target thickness and full rolling speed, headed for the runout table.

TYPICAL PARAMETERS

Entry thickness	**~25–50 mm**
Exit thickness	**~1.2–25 mm**
Finishing temperature	**~850–950 °C**

2 Hydraulic Automatic Gauge Control

Correcting thickness within milliseconds, on a strip already moving many meters per second

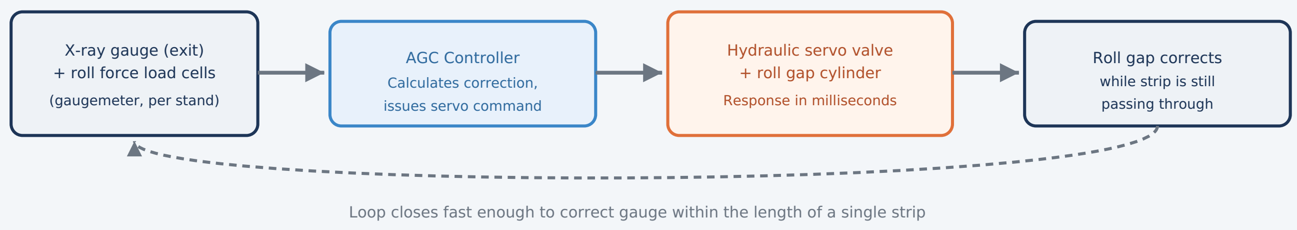


Fig. 3 — Hydraulic AGC responds far faster than the old electro-mechanical screw-down systems it replaced, correcting roll gap within milliseconds — essential at exit speeds where the strip may move tens of meters per second.

WHY HYDRAULIC, NOT ELECTRO-MECHANICAL, AT THE FINISHING MILL

A screw-down motor system, adequate for the roughing mill's slower reversing passes, simply can't react fast enough at finishing-mill speeds. Hydraulic cylinders controlled by high-response servo valves can adjust roll gap within milliseconds — fast enough to correct a gauge deviation before more than a short length of strip is affected, holding tolerances far tighter than mechanical actuation ever could.

TYPICAL PARAMETERS

AGC response time	**Milliseconds**
Actuator type	**Hydraulic servo cylinder**
Feedback	**Gaugemeter + exit X-ray**

TWO FEEDBACK SOURCES, WORKING TOGETHER

SOURCE	ROLE
Exit X-ray gauge	Direct, accurate thickness measurement — but physically located after the last stand, so it can't correct that stand's own output in real time.
Per-stand gaugemeter (roll force)	Infers thickness at each individual stand instantly from measured force, enabling every stand to self-correct even though the direct gauge is elsewhere.

3 Electrical Equipment in Detail

From per-stand drive motors to the loopers holding tension between them

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
MAIN DRIVES & GAUGE CONTROL		
Main drive motors (per stand)	Individually speed-controlled motors driving each stand's work rolls, precisely synchronized so mass flow stays continuous across every stand.	Highest precision speed control on the mill
Hydraulic AGC servo valves & cylinders	Adjust roll gap within milliseconds to hold gauge tolerance at each stand.	Per-stand, high-response
Roll force load cells (gaugemeter)	Feed each stand's own real-time thickness inference.	Continuous monitoring
X-ray / laser gauge (exit)	Direct final gauge measurement, used for calibration and quality confirmation.	Downstream of last stand
SHAPE & PROFILE CONTROL		
Work roll bending actuators	Hydraulic cylinders that flex the rolls slightly to correct strip flatness and crown.	Per-stand
Work roll shifting drives	Axially shift the rolls to even out wear across the roll face and manage edge profile.	Per-stand
Shapemeter	Roll-mounted sensor array measuring flatness across strip width in real time.	Typically at or near exit
TENSION & MATERIAL HANDLING		
Interstand looper motors	Small motorized arms that measure and control tension/loop height of strip between stands.	One per interstand gap
Crop shear drive motor	Powers the shear trimming the bar's leading and trailing ends before threading.	Entry end
Roll cooling & lubrication pump motors	Circulate water/oil to cool rolls and reduce friction during rolling.	Per-stand or shared header
Finishing pyrometer	Measures strip temperature at mill exit, a key input to the runout table's cooling model.	Non-contact
AUTOMATION		
Level 2 automation (mill setup model)	Calculates roll gaps, speeds, and tensions for every stand before threading, then Level 1 executes and fine-tunes in real time.	Coordinates all stands simultaneously

4 Interstand Tension Control — The Balancing Act

Too tight and the strip narrows or breaks; too loose and it buckles — loopers hold the line

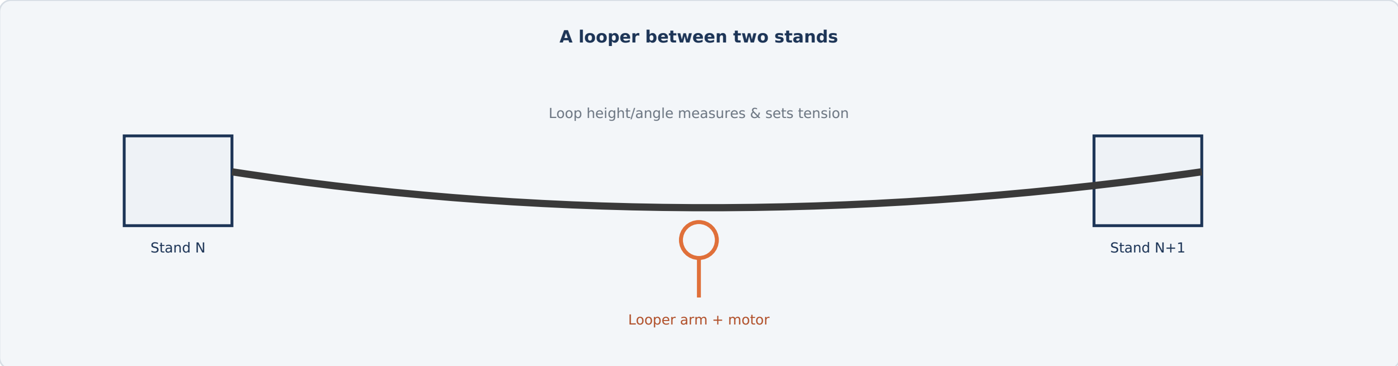


Fig. 4 — The looper rides against the strip's natural sag between stands; its angle indicates tension, and a small motor actively holds that angle — and therefore tension — at the target value set for that gap.

WHAT GOES WRONG AT EACH EXTREME

CONDITION	CONSEQUENCE
Too much tension	The strip narrows (necks) under the pulling force, and in severe cases can pull apart entirely — a strip break.
Too little tension	The strip buckles or wanders off-center between stands, risking a fold or a "cobble" (a tangled pile-up) as it enters the next stand misaligned.

WHY IT'S A WHOLE-MILL COORDINATION PROBLEM

Because every stand's speed is coupled to every other stand's through the strip itself, an error anywhere cascades everywhere. The Level 2 model sets a coordinated speed cascade for all stands before threading even begins, then loopers and each stand's own speed regulator make continuous small corrections to hold every interstand tension at target — simultaneously, across the whole mill, for as long as strip is running.

ELECTRICAL CONTROL EQUIPMENT

Looper motors	**Per interstand gap**
Tension setpoints	**Level 2, per gap**
Stand speed regulators	**Continuous coordination**

KEY POINT

A "cobble" — a strip tangle inside the mill — is one of the most costly and hazardous failure modes in a hot strip mill; tension control exists first and foremost to prevent it.

5 Safety & Protection Systems

Extreme speed, stored tension, and the specific risks of a cobble

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Cobble detection	Sudden tension loss or looper position anomaly consistent with a strip tangle	Rapidly decelerates affected stands and can trip the line to prevent further pile-up
Strip break detection	Tension sensor or looper indicates a sudden loss of strip continuity	Alarms operator; halts affected drives
Hydraulic AGC pressure monitoring	Loss of hydraulic pressure to AGC cylinders	Halts the affected stand in a controlled manner
Stand speed deviation trip	A stand's actual speed deviates from its coordinated setpoint beyond tolerance	Alarms and can slow the whole line to prevent tension cascade
Roll cooling flow monitoring	Loss of roll cooling water flow	Alarms operator; can reduce speed to protect rolls
Entry/exit guarding	Personnel access near a stand while powered	Physical guards and interlocked access points
Emergency stop (E-stop)	Manual activation anywhere along the line	Removes power from drives per the defined E-stop category

WHY SPEED IS THE DOMINANT HAZARD HERE

At exit speeds that can exceed 20 m/s, a strip break or cobble develops and escalates far faster than an operator can react manually — nearly every protection system on this page exists to detect and respond automatically, in a fraction of a second, rather than relying on human intervention.

✓ Summary

What the finishing mill hands off to the runout table, and a recap of the process

FACTOR	BASIC CONTROL	HYDRAULIC AGC + TENSION CONTROL
Gauge tolerance	Wider, slower to correct	Held within tight limits, corrected in milliseconds
Interstand tension	Prone to drift and cascade errors	Actively held by loopers at every gap
Cobble risk	Higher, relies on manual response	Detected and interrupted automatically
Shape/flatness	Less consistent	Actively corrected via roll bending/shifting

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **finishing mill** threads the transfer bar continuously through **5–7 stands** at once, and because steel is essentially incompressible, **mass flow continuity** forces strip speed to rise at every stand as thickness drops — coupling every stand's control to every other stand's in real time. **Hydraulic AGC**, responding within milliseconds using both a per-stand gaugemeter (roll force) and the final exit X-ray gauge, holds tight thickness tolerance despite that speed constantly changing. Between each pair of stands, a **looper** measures and actively holds interstand tension — too much and the strip narrows or breaks, too little and it buckles into a costly, dangerous **cobble**. All of it is set up in advance by a **Level 2 mill model** that calculates a coordinated speed cascade for the whole line before the strip is ever threaded, then continuously fine-tuned as it runs.

Why Cooling Rate Is a Metallurgical Choice

The runout table doesn't just cool the strip — it decides what microstructure it freezes into

When hot steel leaves the finishing mill, it's in a high-temperature phase called austenite. As it cools, that austenite transforms into different room-temperature microstructures — soft ferrite and pearlite, tougher bainite, or hard martensite — and **which one it becomes depends strongly on how fast it cools through the transformation range**, acting together with the steel's chemistry, prior austenite grain size, and finish-rolling temperature. This is a close cousin of the temperature-vs-time control problem seen in the annealing guide's grain structure control, but running in the opposite direction: instead of controlling heating to recrystallize a grain structure, the runout table controls cooling — alongside the alloy design and rolling history already built into the strip — to help select which transformation product the steel ends up as.

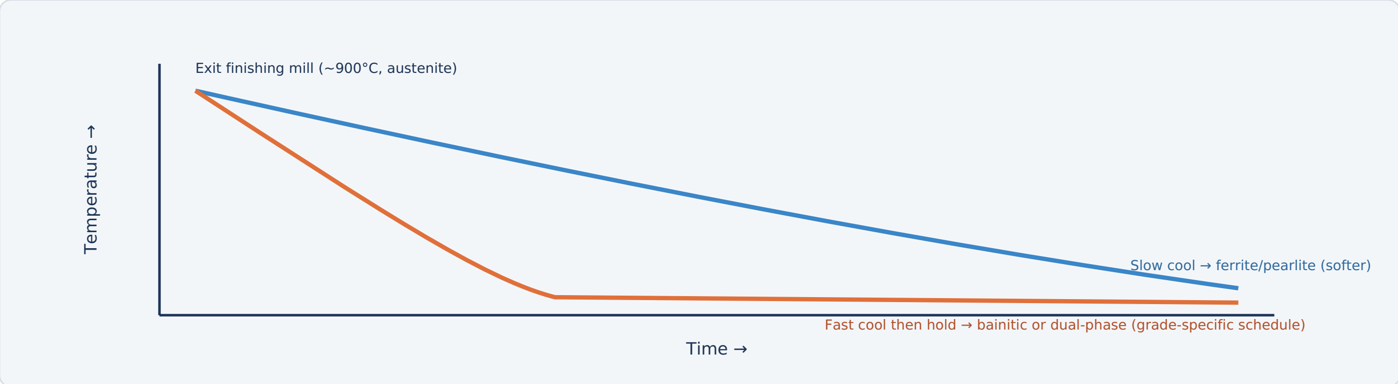


Fig. 1 — Two different cooling paths from the same starting temperature produce two different final microstructures; the diagram is conceptual — bainitic and dual-phase (ferrite + martensite) products are metallurgically distinct and each depends on its own carefully designed chemistry and cooling schedule, not one shared recipe.

WHY THIS BECAME MORE DEMANDING OVER TIME

Standard commodity grades tolerate a fairly simple, roughly uniform cooling curve. Advanced high-strength automotive grades (dual-phase, TRIP, and similar steels) need a precisely shaped multi-stage curve — often a rapid quench followed by a controlled hold — to land the microstructure they're designed around. The runout table's electrical control systems exist specifically to make that level of precision repeatable, strip after strip.

TYPICAL PARAMETERS

Entry temperature	**~850–950 °C**
Coiling temperature	**~500–700 °C**
Table length	**Often 100+ m**

1 The Cooling Process

A precisely timed sequence of water headers along a long roller table

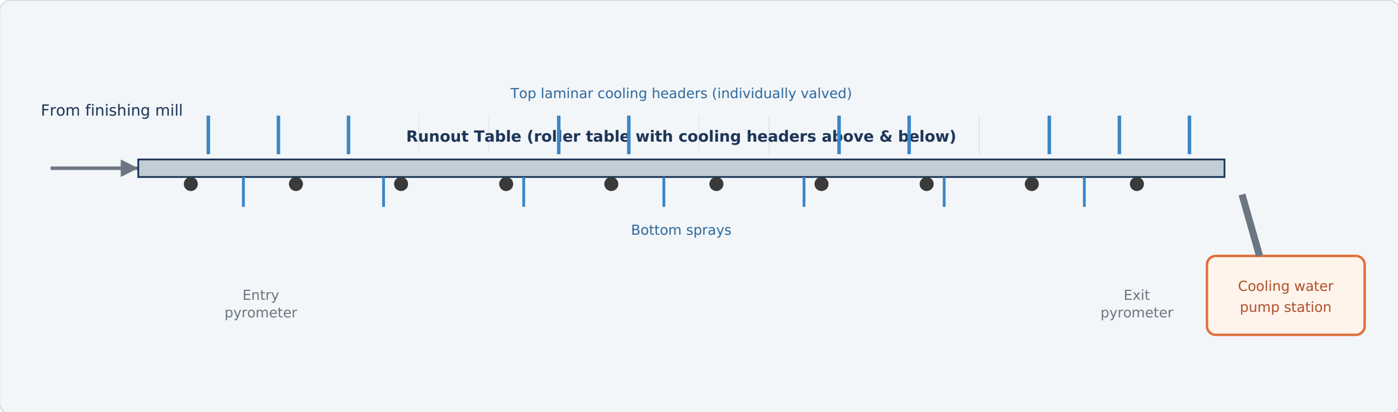


Fig. 2 — Rows of individually-controlled water headers cool the strip at a precise, programmed rate as it travels along the runout table to the coiler.

HOW IT WORKS

1. **Enter hot:** The strip leaves the finishing mill at ~850–950°C and travels along the roller table toward the coiler.
2. **Programmed cooling:** Header rows switch on and off in sequence, following a cooling curve calculated for the steel grade being produced.
3. **Control transformation:** The rate and pattern of cooling sets the steel's final microstructure and therefore its mechanical properties.
4. **Monitor temperature:** Entry and exit pyrometers measure actual strip temperature and feed the header control system in real time.
5. **Hand off to coiler:** By the time the strip reaches the coiler, it has cooled to the target coiling temperature.

TYPICAL PARAMETERS

Entry temperature	**~850–950 °C**
Coiling temperature	**~500–700 °C**
Cooling rate	**Grade-dependent, controlled**

2 The Cooling Model & Header Sequencing

Hundreds of valves, fired in a traveling wave timed to the strip's own motion

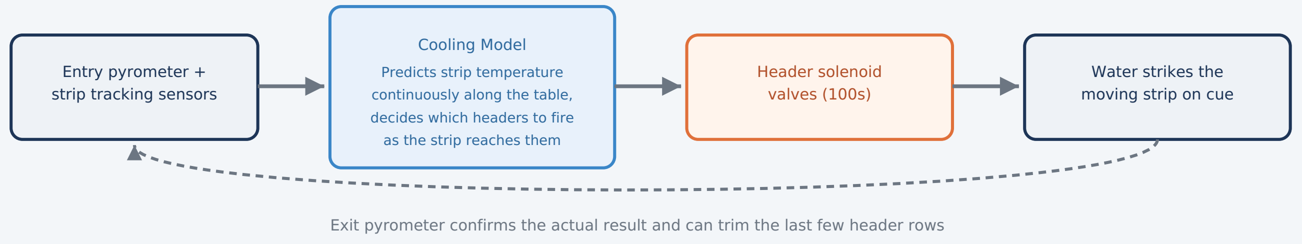


Fig. 3 — As the strip physically moves under each header row, the cooling model fires exactly the headers needed at that moment — effectively painting a moving "cooling recipe" onto the strip as it travels.

WHY TIMING, NOT JUST HEADER COUNT, IS THE HARD PART

It isn't enough to know how much total cooling a grade needs — the model has to know exactly which headers to fire at exactly which moment as a specific point on the strip passes beneath them, continuously, for the entire length of a coil that might be over a kilometer long once unrolled. Strip speed, table position, and predicted temperature all feed into deciding which header solenoids are open at any given instant.

ELECTRICAL EQUIPMENT

Header solenoid valves	**Individually addressable**
Cooling water pump motors	**High flow rate**
Entry/exit pyrometers	**Non-contact infrared**
Strip tracking sensors	**Photoeyes / encoders**

A PARALLEL TO FUME SYSTEM DAMPER SEQUENCING

This "traveling sequence tied to a moving process state" is structurally the same idea as the Fume Extraction System's damper sequencing tied to furnace operating state — except here the sequence is tied to strip position along the table rather than furnace mode, and repeats continuously rather than switching between a handful of discrete states.

3 Electrical Equipment in Detail

From the pump station supplying every header to the roller drives carrying the strip beneath them

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
COOLING WATER SYSTEM		
Cooling water pump motors	Supply high-volume water flow to the header banks at the pressure the cooling model demands.	Multiple pumps, zone-matched pressure
Header solenoid/control valves	Electrically actuated valves switching individual header rows on and off per the cooling model's sequence.	Large numbers, individually addressable
Water treatment & recirculation system	Filters and conditions cooling water for reuse, supporting continuous high-volume operation.	Shared plant utility
INSTRUMENTATION		
Entry pyrometer	Measures strip temperature as it leaves the finishing mill and enters the runout table.	Non-contact infrared
Exit pyrometer	Confirms actual coiling temperature, closing the loop on the cooling model's prediction.	Non-contact infrared
Intermediate pyrometers	On advanced lines, check progress partway along the table for multi-stage cooling curves.	Where fitted
Strip tracking sensors	Detect the strip's leading/trailing edge and position to trigger the correct header sequence as it passes.	Photoeyes or shaft encoders
MATERIAL HANDLING & AUTOMATION		
Roller table drive motors	Move the strip along the runout table at controlled, synchronized speed.	Many individually driven sections
Cooling model computer / Level 2 automation	Calculates which headers to fire and for how long to hit the target coiling temperature and cooling rate for each grade.	Real-time, tied to strip position

4

Cooling Uniformity — The Balancing Act

Getting the average cooling rate right isn't enough if it isn't even across the strip

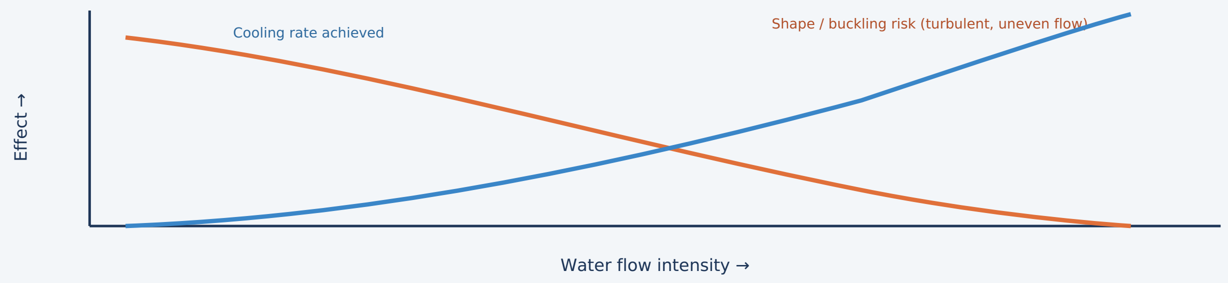


Fig. 4 — Faster cooling is often desirable metallurgically, but excessive or uneven flow — especially near the edges, which naturally cool faster than the center — risks shape defects and property variation across the strip.

WHY THE EDGES ARE A SPECIAL PROBLEM

A strip's edges lose heat in two dimensions instead of one, so they naturally cool faster than the center under identical header flow — left uncorrected, this produces a strip with different mechanical properties (and sometimes different flatness) at the edges than in the middle. Many runout tables use edge masking — reducing or blocking spray near the strip edges — specifically to even out through-width cooling.

TYPICAL PARAMETERS

Edge masking	**Common on wide strip**
Cooling zones	**~5–15 along table**
Speed compensation	**Real-time, per header**

WHAT THE MODEL HAS TO BALANCE

FACTOR	TRADE-OFF
Cooling rate	Faster achieves advanced-grade properties, but raises turbulence and shape risk.
Through-width uniformity	Edge masking corrects natural edge over-cooling, but adds valve complexity.
Along-length consistency	Line speed variation (e.g., during threading or tail-out) changes exposure time under each header, requiring the model to compensate in real time.

5 Safety & Protection Systems

High-volume water, flash steam, and electrical equipment in a genuinely wet environment

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Header flow/pressure monitoring	A header bank fails to reach commanded flow	Alarms operator; cooling model can compensate with adjacent headers
Water pump motor protection	Overcurrent or thermal overload	Trips the affected pump off-line
Strip tracking loss detection	Tracking sensors lose the strip's position	Alarms operator; can default to a safe, conservative header sequence
Roller table drive protection	Overcurrent or stall on a table section motor	Trips the affected section
Electrical enclosure rating & monitoring	Continuous exposure to water spray and flash steam	NEMA/IP-rated enclosures for all electrical equipment in the wet zone
Emergency water shutoff	Manual activation or critical safety trip	Isolates water supply to the header banks
Emergency stop (E-stop)	Manual activation anywhere along the table	Removes power from drives per the defined E-stop category

WHY THIS AREA DEMANDS ITS OWN ELECTRICAL STANDARD

Unlike most of the mill, the runout table is a deliberately wet environment by design — every motor, sensor, and junction box here has to be rated for continuous water spray and flash steam exposure, not just splash protection, which is why runout table electrical specifications are treated as a distinct category from the rest of the line.

Summary

What the runout table hands off to the coiler, and a recap of the process

FACTOR	BASIC COOLING CONTROL	FULL COOLING MODEL + SEQUENCING
Microstructure consistency	Coarser, grade-limited	Precisely tailored per grade, including multi-stage curves
Through-width uniformity	Edge over-cooling uncorrected	Actively corrected via edge masking
Coiling temperature accuracy	Wider variation	Held tightly via closed-loop pyrometer feedback
Achievable grades	Standard commodity grades only	Advanced high-strength / dual-phase grades possible

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **runout table** takes strip fresh off the finishing mill at ~900°C and cools it to coiling temperature — and the **cooling rate path**, working together with the steel's chemistry and rolling history, is what steers the final microstructure toward soft ferrite/pearlite or toward stronger, distinct bainitic or dual-phase structures. Rows of individually valved **laminar cooling headers** fire in a traveling sequence, timed by a **cooling model** that continuously tracks the strip's position and predicted temperature, effectively painting a grade-specific cooling recipe onto the moving strip. Because strip edges naturally cool faster than the center, **edge masking** and careful uniformity management keep properties consistent across the width — all while every electrical component in the area has to survive a continuous bath of water spray and flash steam by design.

Why Coiling Tension Is a Moving Target

The same tension force means something completely different at the start of a coil than at the end

Winding strip onto a mandrel under too much tension crushes and distorts the inner wraps as the coil builds up — like over-tightening a spring until it deforms. Too little tension, and the coil winds loosely, telescopes (wraps slide sideways relative to each other), and can unravel entirely. Holding tension constant doesn't hold everything else constant, though: **mandrel torque scales directly with coil radius** (torque equals tension times radius), so the drive works harder as the coil grows even at an unchanged tension setpoint. More important for coil quality, each new wrap presses inward on every wrap already beneath it, and that radial pressure compounds through the pack as the coil builds — a cumulative effect, not a simple case of "the same tension causing different stress." Left unmanaged, that accumulating pressure can crush or plastically deform the innermost wraps. The coiler's electrical control exists specifically to keep it in check.

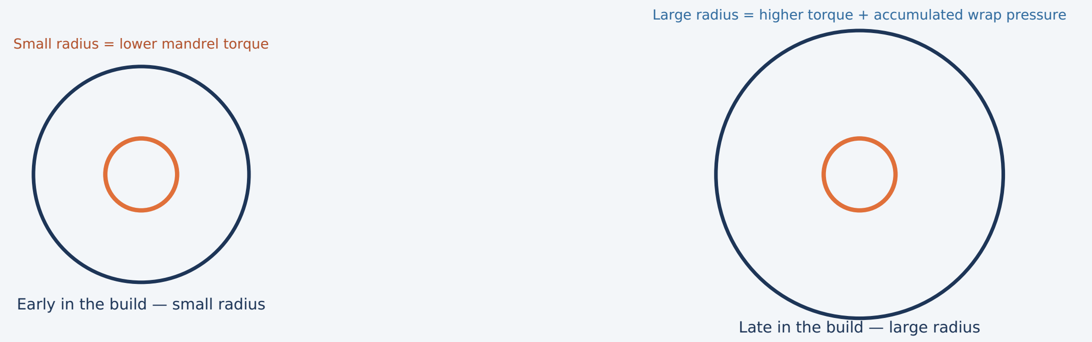


Fig. 1 — Torque scales directly with radius at a given tension; separately, each new outer wrap adds to the radial pressure already built up on the wraps beneath it — together, these are why tension setpoint has to change through the build, not stay constant.

WHAT GOES WRONG AT EACH EXTREME

CONDITION	CONSEQUENCE
Too much tension	Inner wraps crush under accumulated pressure from the outer wraps — visible as "telescoping" damage or coil set defects.
Too little tension	Loose winding lets wraps slide sideways relative to each other, producing a coil that can unravel or ships and handles poorly.

TYPICAL PARAMETERS

Coiling temperature	**~500–700 °C**
Coil weight	**~15–35 t typical**
Coil OD	**~1.8–2.2 m**

1 The Coiling Process

From pinch rolls to a finished, strapped coil ready for shipment

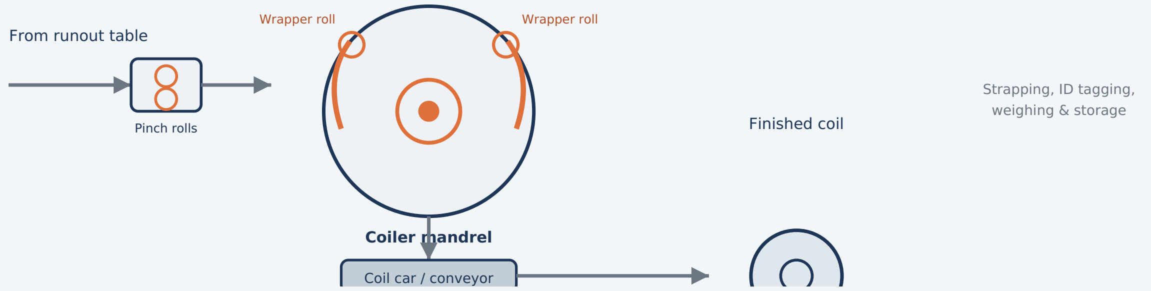


Fig. 2 — Pinch rolls feed the strip onto a motorized mandrel; wrapper rolls curl the leading end around it as the coil builds, then the finished coil is stripped off and moved to storage.

HOW IT WORKS

1. **Feed in:** Pinch rolls grip the strip and guide it onto the coiler entry, keeping consistent tension.
2. **Wrap onto the mandrel:** Wrapper rolls curl the strip's leading edge tightly around the expandable coiler mandrel to start the wind.
3. **Build the coil:** The mandrel rotates, winding the strip under a tension setpoint that changes through the build, as more material feeds in.
4. **Cut to length:** Once the full strip has wound (or at a scheduled coil weight), a shear cuts the strip and the trailing end is secured.
5. **Strip & transfer:** The mandrel contracts, releasing the finished coil onto a coil car, which moves it to strapping, weighing, ID tagging, and storage.

TYPICAL PARAMETERS

Coiling temperature	**~500–700 °C**
Coil weight	**~15–35 t typical**
Coil OD	**~1.8–2.2 m**

2 Tension Taper Control

Continuously reducing the tension setpoint as the coil grows, without ever stopping the wind

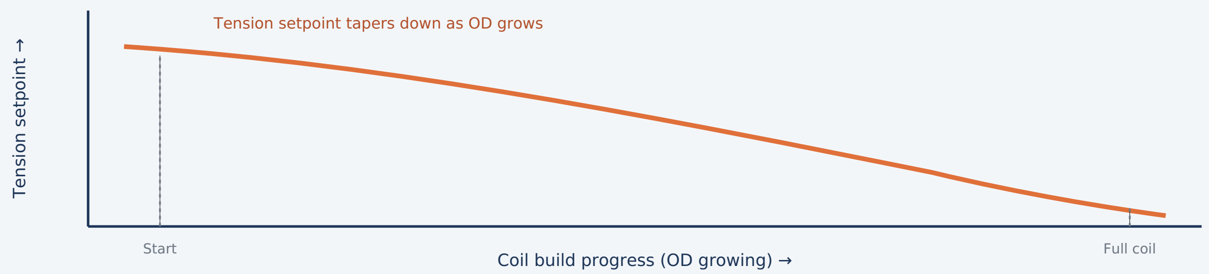


Fig. 3 — Rather than a fixed setpoint, tension is scheduled to fall as coil diameter grows, holding both mandrel torque and accumulated interlayer pressure within a safe band throughout the build.

HOW THE CONTROL LOOP ACTUALLY WORKS

Tension is set by the difference between the mandrel's rotational torque and the pinch rolls' feed speed — the same fundamental principle as bridle-roll tension control seen elsewhere in a steel mill. What's specific to coiling is that the target itself keeps changing: as coil diameter grows (tracked in real time via wrap counting or a diameter sensor), the tension control model reduces the tension setpoint following a pre-calculated taper curve. That keeps mandrel torque from climbing unchecked as radius increases, and limits how much radial pressure accumulates on the inner wraps from everything wound on top of them.

TYPICAL PARAMETERS

Tension control	**Torque vs. speed differential**
Taper schedule	**Function of coil OD**
OD tracking	**Wrap count or diameter sensor**

ELECTRICAL EQUIPMENT

EQUIPMENT	FUNCTION
Mandrel drive motor	Torque-controlled motor rotating the mandrel, the primary actuator for tension.
Pinch roll drive motors	Feed strip in at line speed; the speed differential with the mandrel sets actual tension.
Coil diameter/wrap sensor	Tracks coil OD growth in real time, feeding the tension taper schedule.

3 Electrical Equipment in Detail

From the mandrel's hydraulic expansion system to the ID marking that tracks every coil

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
MANDREL & WRAPPING		
Coiler mandrel drive motor	Rotates the mandrel under controlled torque to wind the coil evenly and hold the tension taper schedule.	Torque-controlled
Mandrel hydraulic expansion system	Electrically controlled hydraulics that expand the mandrel to grip the coil ID and contract it for release.	High force, precision control
Wrapper roll drive & positioning motors	Curl the strip's leading end around the mandrel and retract once the coil is self-supporting.	Position-controlled
FEED & CUTTING		
Pinch roll drive motors	Feed strip into the coiler at line speed with controlled tension differential against the mandrel.	Speed-synchronized to line
Coiler shear drive	Cuts the strip once each coil is complete, releasing it from the incoming strip; on semi-endless/endless lines, a high-speed flying shear instead divides a continuous strip into coil lengths on the fly.	Timed to coil weight/length
HANDLING & AUTOMATION		
Coil car / conveyor drive motors	Transfer finished coils from the coiler to strapping and storage.	Heavy-duty, precision stopping
Coil diameter/wrap sensor	Feeds the tension taper schedule in real time.	Continuous monitoring
Coil weighing & ID marking system	Electronic scales and marking equipment for tracking and traceability.	Per-coil record
Tension control / Level 2 automation	Coordinates mandrel torque and pinch roll speed to maintain correct, tapered winding tension throughout the coil build.	Continuous, real-time

4 Multi-Coiler Switching in Conventional Operation

Alternating between discrete strips — distinct from the flying-shear cut used in endless or semi-endless rolling

Two discrete strips, no cut needed at the coiler — just a mandrel handoff

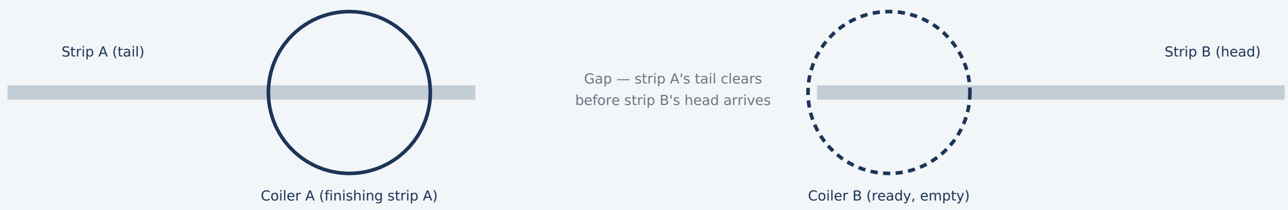


Fig. 4 — Each slab already produces a discrete strip with its own head and tail; the second coiler exists so the next strip's head has a ready, empty mandrel waiting, without needing to cut anything at the coiler itself.

WHY NO SHEAR IS NEEDED AT THE COILER

In conventional batch hot strip mill operation, each slab produces one finite transfer bar and one finished coil — the strip already has a defined head and tail by the time it reaches the coiler. Multiple downcoilers alternate *availability* between successive strips: the tail of one strip clears the coiling area before the head of the next strip arrives and gets directed to whichever mandrel is ready. There's no continuous strip joining one batch strip to the next, so there's nothing that needs to be cut by a flying shear at the changeover — the switch is a handoff between separate pieces of strip, not a cut through a moving one.

WHERE A FLYING SHEAR ACTUALLY BELONGS: SEMI-ENDLESS AND ENDLESS ROLLING

A high-speed "flying" shear immediately upstream of the coilers is characteristic of semi-endless or endless rolling — configurations where successive transfer bars are welded together earlier in the line so a genuinely continuous or jumbo-length strip has to be divided into coil-length pieces on the fly. That's a distinct process choice, typically for thin-gauge or difficult-to-roll grades, requiring additional welding and shearing equipment that a conventional batch mill doesn't have.

ELECTRICAL EQUIPMENT

Coiler selection logic	**Level 1/2 automation**
Second/third mandrel drive	**Standing ready, pre-staged**
Flying shear drive	**Semi-endless/endless lines only**

5 Safety & Protection Systems

A rotating mandrel, heavy coil handling, and coils that are still very hot

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Mandrel torque overload trip	Measured winding torque exceeds safe limit	Alarms operator; can reduce speed or halt winding
Tension loss detection	Sudden drop in tension feedback, suggesting a strip break	Halts the coiler and upstream drives in a controlled sequence
Mandrel hydraulic pressure monitoring	Loss of pressure in the expansion/contraction system	Blocks release sequence until pressure is confirmed safe
Flying shear timing interlock (semi-endless/endless lines)	Shear commanded outside its safe speed-tracking window	Blocks the cut until timing conditions are met
Coil car position interlock	Car not correctly positioned for coil transfer	Blocks mandrel release until alignment is confirmed
Rotating mandrel guarding	Personnel access near the mandrel while rotating	Physical guarding and interlocked access points
Emergency stop (E-stop)	Manual activation anywhere along the coiler area	Removes power from drives per the defined E-stop category

WHY HOT COIL HANDLING IS A DISTINCT HAZARD

Coils leave the mandrel still hot enough to cause serious burns and can retain significant heat for hours afterward; coil car and storage procedures account for this separately from the coiler's own electrical and mechanical interlocks.

Summary

The end of the hot strip mill line, and a recap of the process

FACTOR	FIXED TENSION CONTROL	TENSION TAPER (CONVENTIONAL COILING)
Coil quality	Risk of core crush or loose winding	Torque and interlayer pressure held within a safe band
Coil changeover	Single coiler, brief pause between strips	Alternating coilers reduce downtime between successive strips
Product consistency	More variable coil-to-coil	Uniform, well-formed coils at full line speed

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **down coiler** is where the finished strip becomes a shippable product: pinch rolls feed it onto a **mandrel**, **wrapper rolls** curl the leading edge to start the wind, and the mandrel then winds the rest under torque-controlled tension. Because mandrel torque scales directly with coil radius, and because each new wrap adds to the radial pressure already accumulated on the wraps beneath it, a fixed tension setpoint would eventually over-stress the core — so a **tension taper** schedule continuously reduces the setpoint as diameter grows, tracked in real time by a diameter or wrap-count sensor. In conventional batch hot strip mill operation, each slab already produces a discrete strip with its own head and tail; most mills use **2–3 coilers** so the next strip's head always has a ready, empty mandrel waiting, with the tail of one strip clearing before the head of the next is directed to it — no cut is needed at the coiler itself. A high-speed **flying shear** immediately upstream of the coilers belongs to a different configuration entirely: semi-endless or endless rolling, where successive transfer bars are welded together upstream into one genuinely continuous strip that has to be divided into coil lengths on the fly.

PART 16 — AIR COMPRESSOR PLANT

Why Compressed Air? Two Grades, One Utility

The invisible utility running thousands of valves, dampers, and cylinders across every part of this book

Nearly every process guide in this book has quietly depended on compressed air without naming it directly — every pneumatically actuated damper, control valve, and cylinder mentioned across the melt shop and hot strip mill ultimately traces back to this single utility plant. Compressing air stores usable energy in it; released through a valve into a cylinder or actuator, that stored pressure does mechanical work. Pneumatic actuation is simple, robust, and tolerates harsh, hot, dirty industrial environments far better than many electric alternatives — which is exactly why it's everywhere in a steel mill. It can also be *engineered* for a defined failure position: spring-return actuators drive to a selected fail-open or fail-closed state purely from stored spring force the moment air pressure is lost, no controller required. That specific, deliberately designed behavior — not some inherent property of "pneumatic" as a category — is why so many critical shutoff and safety valves in this book are spring-return pneumatic rather than electric.

Fig. 1 — The same compressor plant typically supplies two distinct grades: ultra-clean instrument air that process control depends on, and a lower-spec plant air supply for less sensitive uses.

WHY THE DISTINCTION MATTERS

Moisture, oil carryover, or particulate in instrument air can freeze in outdoor lines, corrode precision valve internals, or foul sensitive instrumentation — a contamination event here can cause process upsets across multiple areas simultaneously, since a huge number of independent control loops throughout the mill share the same instrument air header.

TYPICAL PARAMETERS

Instrument air dew point	**≤-40 °C pressure dew point**
Typical supply pressure	**~6–8 bar (85–115 psi)**
Oil content (instrument air)	**Effectively zero**

1 The Compressed Air Process

Intake, compress, cool, dry, store, and distribute — a continuous cycle sized to the mill's peak demand

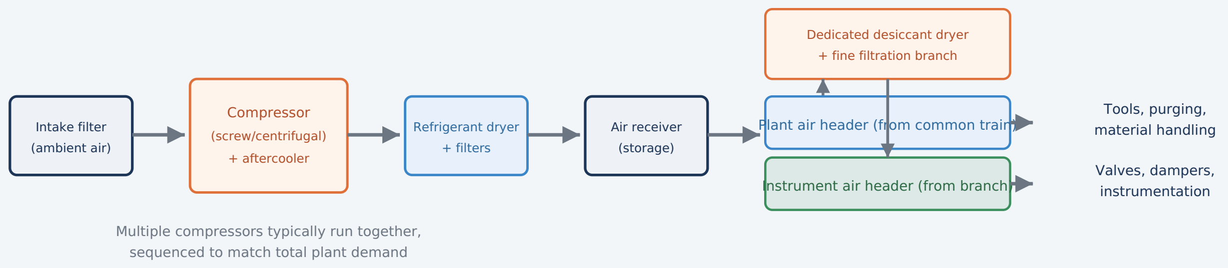


Fig. 2 — Air is filtered, compressed, cooled, and dried/filtered to plant-air quality in a common train; the plant-air header draws directly from that train, while a dedicated desiccant and fine-filtration branch dries a portion further to instrument-air quality — drying only where the tighter spec is actually needed.

HOW IT WORKS

1. **Intake:** Ambient air is drawn in through an inlet filter that removes coarse dust before it ever reaches the compressor.
2. **Compress:** A compressor (screw, centrifugal, or reciprocating) raises the air to distribution pressure, generating significant heat of compression in the process.
3. **Cool:** An aftercooler removes that heat, condensing much of the moisture the compression process concentrated.
4. **Dry & filter (common train):** A refrigerant dryer and filters remove enough moisture and particulate for general plant-air use, condensing much of what the aftercooler didn't already catch.
5. **Store:** An air receiver buffers supply against sudden demand spikes, smoothing out compressor loading.
6. **Dry further for instrument air:** A dedicated desiccant dryer and fine coalescing filter treat a branch of that same air further, down to the much lower dew point instrument air specifically needs.
7. **Distribute:** The plant-air header draws from the common train; the instrument-air header draws from the additional desiccant branch — each header only as dry as its own use actually requires.

TYPICAL PARAMETERS

Distribution pressure	**~6–8 bar**
Instrument air dew point	**≤-40 °C**
Compressor types	**Screw, centrifugal**

2 Compressor Types & Capacity Control

Matching output to a demand curve that never sits still

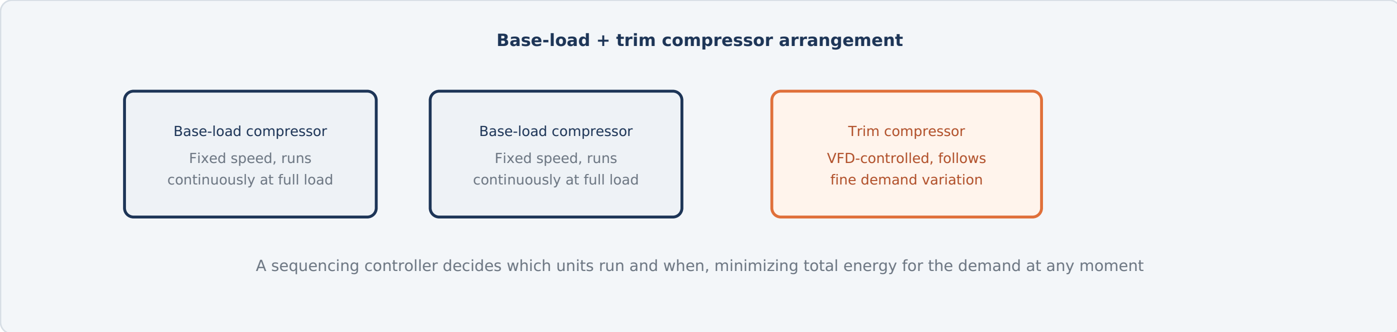


Fig. 3 — Rather than one compressor throttling up and down, plants commonly run fixed-speed units at their efficient full-load point and let a single VFD-controlled trim unit absorb the moment-to-moment demand swings.

THREE COMPRESSOR TECHNOLOGIES

TYPE	CHARACTERISTICS
Reciprocating	Piston-based; simple and robust at smaller capacities, more maintenance-intensive at large scale.
Rotary screw	Two meshing helical rotors compress air continuously; the workhorse of most industrial plant air systems, good efficiency across a wide range.
Centrifugal	High-speed impeller compresses air dynamically; efficient at very large, continuous capacities, less suited to frequent load cycling.

ELECTRICAL EQUIPMENT

Compressor motors	**Fixed-speed and VFD**
Sequencing controller	**Coordinates multiple units**
Pressure transmitters	**Header pressure feedback**

WHY FIXED-SPEED UNITS STILL MAKE SENSE

Running a compressor at a fixed, efficient full-load point and only trimming the last portion of demand with a variable-speed unit is often more energy-efficient overall than running several variable-speed units all partially loaded — full-load operation is typically where a fixed-speed compressor's efficiency curve peaks.

3 Electrical Equipment in Detail

From the compressor motors themselves to the drain traps quietly protecting every downstream valve

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
COMPRESSION		
Compressor motors	Drive the screw, centrifugal, or reciprocating compression element; sized to the plant's base and peak air demand.	Mix of fixed-speed and VFD units
Aftercooler fan/pump motors	Remove the heat of compression before the air reaches drying equipment.	Air-cooled or water-cooled design
Inlet filtration & intake louvers	Remove coarse particulate before air enters the compressor.	Passive, periodically serviced
DRYING & FILTRATION		
Refrigerant dryer compressor	Drives the refrigeration cycle that condenses moisture out of compressed air.	Where refrigerant-type drying is used
Desiccant dryer heaters & valves	Regenerate desiccant material and switch between drying towers for continuous operation.	Where very low dew points are required
Coalescing & particulate filters	Remove oil carryover and fine particulate, especially ahead of the instrument air header.	Staged filtration
Automatic condensate drain valves	Remove accumulated condensate from receivers and drying equipment without manual intervention or air loss.	Zero-loss or timed solenoid type
STORAGE & DISTRIBUTION		
Air receiver instrumentation	Pressure and level monitoring on the storage vessel(s) buffering supply against demand spikes.	Pressure-rated vessel, code-inspected
Distribution header pressure transmitters	Continuously monitor pressure at multiple points across the plant-wide piping network.	Feeds sequencing control
Sequencing controller (Level 1/2 automation)	Decides which compressors run, and at what output, to meet current demand at minimum energy cost.	Coordinates all compressor units

4 Air Quality — Drying & Filtration

The cheapest-looking utility in the plant is also one of the easiest to quietly contaminate

What has to come out, and why

Moisture

Freezes in outdoor lines, corrodes valve internals, washes out lubrication

Oil Carryover

From lubricated compressors; contaminates instruments, can affect sensitive processes

Particulate

Erodes valve seats and seals, causes sticking or premature failure

Fig. 4 — Three contaminant types, each with its own failure mode downstream; the drying and filtration train exists specifically to remove all three before air reaches sensitive instrumentation.

TWO DRYING TECHNOLOGIES, APPLIED WHERE EACH IS NEEDED

TYPE	HOW IT WORKS
Refrigerant dryer	Cools air to condense out moisture, achieving a moderate dew point suitable for most plant air needs — lower energy cost than desiccant drying; typically sized for the full common train.
Desiccant dryer	Passes air through a moisture-absorbing material, achieving the very low dew points instrument air needs, especially outdoor piping in cold climates; requires periodic regeneration of the desiccant. Sizing it only for the instrument-air branch — rather than the whole plant's air volume — keeps energy cost down while still meeting the tighter spec where it matters.

TYPICAL PARAMETERS

Instrument air dew point	**≤-40 °C**
Plant air dew point	**Less strict, often +3 to +5 °C**
Filtration stages	**Coarse → coalescing**

WHY FILTRATION IS STAGED

Coarse particulate filtration happens early, protecting downstream equipment; fine coalescing filters — which remove oil aerosol down to very small droplet sizes — are typically positioned specifically ahead of the instrument air header, where contamination has the widest-reaching consequences across the plant's control systems.

5 Pressure vs. Energy — The Balancing Act

Compressed air is often called the most expensive utility in an industrial plant — here's why

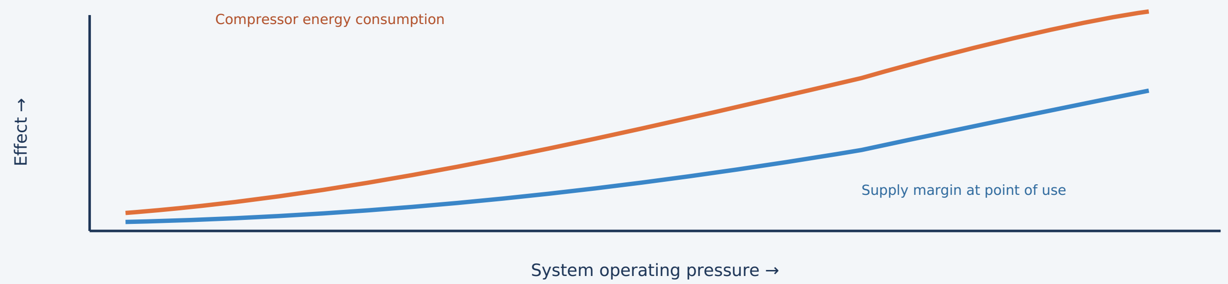


Fig. 5 — Every additional bar of system pressure costs real compressor energy; running higher than necessary "just in case" is one of the most common and avoidable sources of wasted energy in an industrial plant.

WHY LEAKS MATTER MORE HERE THAN ALMOST ANYWHERE ELSE

A compressed air leak wastes energy continuously, 24 hours a day, whether or not the plant is producing anything — unlike most other utility losses, leak losses don't scale down with reduced production. Aging plant-wide piping networks with hundreds of fittings, quick-connects, and valve packings routinely lose a significant fraction of total compressor output to leaks that individually seem too small to matter.

TYPICAL PARAMETERS

Typical leak loss (unmanaged)	**20–30% of output**
Distribution pressure	**~6-8 bar**

THE STANDARD EFFICIENCY TOOLKIT

PRACTICE	WHY IT HELPS
Pressure/flow optimization	Running the distribution header at the lowest pressure that still satisfies every point of use.
Leak detection programs	Systematic ultrasonic leak surveys find losses invisible to the ear in a noisy plant environment.
Base-load/trim sequencing	Keeps compressors running near their efficient full-load point rather than partially loaded.

6 Safety & Protection Systems

Pressure vessels, high-speed rotating machinery, and stored energy that has to go somewhere if released suddenly

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Air receiver relief valve	Vessel pressure exceeds safe rated limit	Automatically vents excess pressure to atmosphere
Pressure vessel inspection program	Scheduled code-required inspection interval	Vessel taken offline for certified inspection
Compressor high-temperature trip	Discharge temperature exceeds safe operating limit	Shuts down the affected compressor
Low oil pressure/level trip (lubricated units)	Lubrication system fault detected	Shuts down the compressor before bearing damage occurs
Condensate drain monitoring	Drain valve fails to operate, risking moisture carryover	Alarms operator; can be backed up by a redundant drain
Emergency stop (E-stop)	Manual activation at the compressor house	Removes power from the affected compressor

WHY STORED AIR ENERGY IS TREATED WITH REAL CAUTION

A large air receiver holds a genuinely significant amount of stored energy under pressure; uncontrolled release — from a failed fitting or an improperly isolated line during maintenance — can cause serious injury, which is why lockout/tagout procedures for compressed air systems explicitly require confirmed depressurization, not just power isolation.

FAIL-SAFE IS A DESIGN CHOICE, NOT A PROPERTY OF "PNEUMATIC"

A spring-return actuator drives to its designed fail-open or fail-closed position on loss of air, because a physical spring — not the air itself — provides that force. A double-acting actuator has no such tendency: on air loss it can simply stay wherever it was, or move unpredictably, unless it's specifically fitted with an accumulator, trip valve, or lock-up device to give it a defined failure action. Anyone specifying or reviewing a safety-critical valve needs to know which type it is — the two are not interchangeable assumptions.

✓ Summary

The utility every other chapter in this book quietly depends on, and a recap

FACTOR	POORLY MANAGED SYSTEM	WELL-MANAGED SYSTEM
Energy cost	High — leaks and excess pressure waste continuously	Minimized via leak programs and pressure optimization
Instrument reliability	At risk from moisture/oil contamination	Protected by proper drying and filtration
Compressor efficiency	Multiple units partially loaded	Base-load units at full efficiency, trim unit absorbs variation

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **air compressor plant** supplies two distinct grades of compressed air — ultra-clean, dry **instrument air** for process control valves and instrumentation, and lower-spec **plant air** for tools and material handling — both produced by compressing, cooling, drying, and filtering ambient air before buffering it in a storage **receiver**. Multiple compressors typically run in a **base-load/trim** arrangement, with fixed-speed units held at their efficient full-load point and a single variable-speed unit absorbing moment-to-moment demand swings. Because compressed air is one of the most energy-expensive utilities in an industrial plant, and because **leaks** waste that energy continuously regardless of production level, pressure optimization and systematic leak management are treated as ongoing operational disciplines rather than one-time projects — the payoff is felt as reduced electrical demand everywhere else in the mill this utility quietly supports.

PART 17 — OXYGEN PLANT

Why a Dedicated Oxygen Plant? The Physics of Air Separation

One utility, feeding nearly every high-temperature process covered elsewhere in this book

The BOF's oxygen lance, the EAF's oxygen injection, the LMF's argon stirring, and countless purging and inerting duties across the plant all draw from the same source: a centralized **Air Separation Unit (ASU)** that separates ordinary atmospheric air into its component gases. This isn't incidental infrastructure — building one large, efficient plant to supply oxygen, nitrogen, and argon to every consumer is dramatically more economical than each process area sourcing its own gas independently. The separation itself relies on a simple physical fact: **nitrogen, oxygen, and argon each boil at a different temperature**, so cooling air down to the point where it liquefies and then carefully controlling that temperature lets the plant separate them by fractional distillation — precisely the same principle used to refine crude oil, just at cryogenic temperatures instead of high ones.

Three gases, three boiling points, one separation

Nitrogen (N_2)
Boils at -196°C
Lightest, separates first

Argon (Ar)
Boils at -186°C
Separates between N_2/O_2

Oxygen (O_2)
Boils at -183°C
Heaviest, separates last

Fig. 1 — Cooling air to cryogenic temperatures and carefully distilling it separates all three gases by their distinct boiling points — the same air that's 78% nitrogen and 21% oxygen becomes three separate industrial products.

WHY ONE PLANT SERVES THE WHOLE MILL

CONSUMER	WHAT IT USES
BOF (Part 5)	High-volume oxygen for the supersonic decarburization blow.
EAF (Parts 3–4)	Oxygen for refining, scrap cutting, and burner assistance.
LMF & CCM (Parts 7, 9)	Argon for bath and mold stirring.
Throughout the plant	Nitrogen for inerting, purging, and blanketing (including DRI hot-charge conveyors, Part 2).

TYPICAL PARAMETERS

Separation temperature	**~-185 °C**
Oxygen purity (industrial)	**~95–99.5%**
Products	**O_2, N_2, Ar**

1 The Air Separation Process

Compress, purify, cool, distill, and store — a continuous cryogenic cycle running around the clock

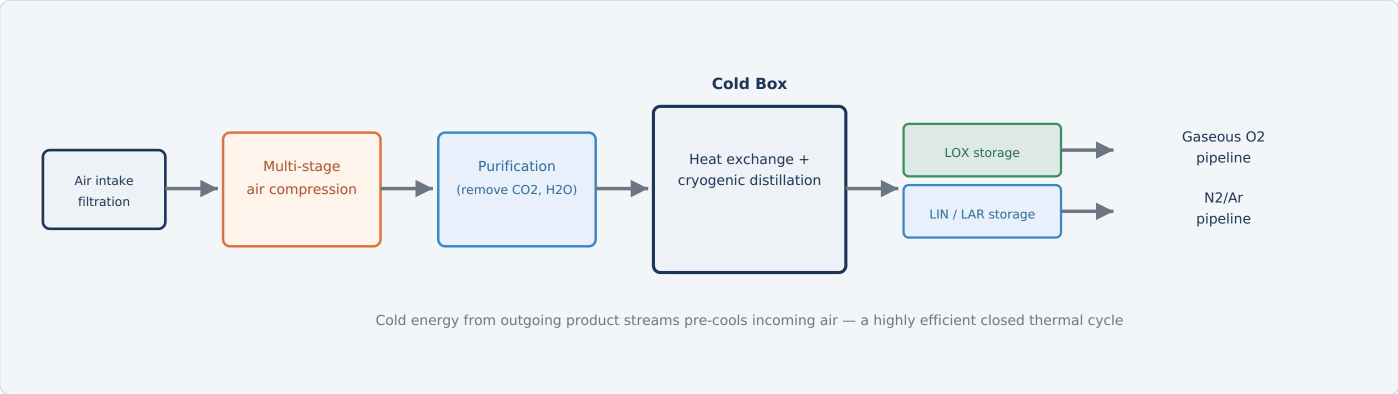


Fig. 2 — Air is compressed, purified of anything that would freeze downstream, then cooled and distilled in the cold box; separated gases either vaporize directly into distribution pipelines or accumulate as liquid backup storage.

HOW IT WORKS

1. **Intake & compress:** Ambient air is filtered and compressed in multiple stages toward the pressure the separation process needs.
2. **Purify:** Water vapor and carbon dioxide are scrubbed out — both would freeze solid at cryogenic temperatures and plug the cold box.
3. **Cool:** Heat exchangers use the cold energy of outgoing product streams to progressively cool incoming air toward liquefaction.
4. **Distill:** Cryogenic distillation columns separate liquid air into oxygen, nitrogen, and argon fractions by their different boiling points.
5. **Store or distribute:** Products either vaporize directly into plant gas pipelines or accumulate as liquid in insulated storage tanks as backup supply.

TYPICAL PARAMETERS

Separation temperature	**~-185 °C**
Operation	**Continuous, 24/7**
Products	**GOX/LOX, GAN/LIN, LAR**

2 Cryogenic Distillation — The Cold Box

The heart of the plant: an elegant thermal cycle that recycles its own cold energy

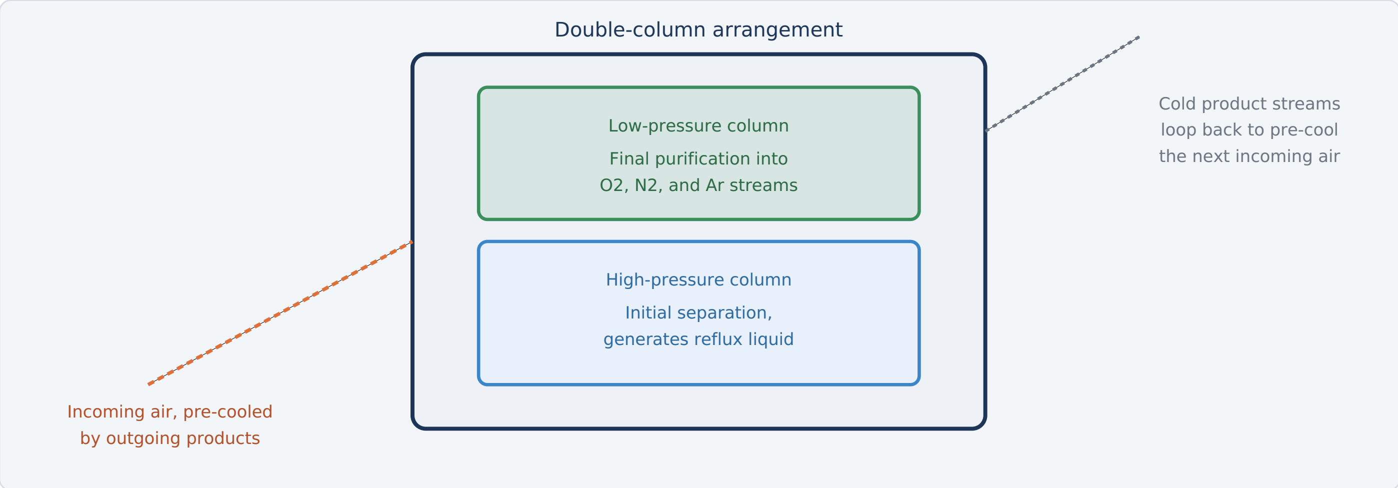


Fig. 3 — A double-column design separates air in two stages; critically, the cold outgoing product streams are routed back through heat exchangers to pre-cool incoming air, recovering cooling energy the plant would otherwise have to generate fresh.

WHY IMPURITIES HAVE TO BE SCRUBBED OUT FIRST

Even small amounts of water vapor or carbon dioxide will freeze solid the moment they reach cryogenic temperatures inside the cold box — and a frozen deposit inside a heat exchanger passage or distillation tray isn't a minor inconvenience, it can plug flow paths and force an unplanned shutdown. The purification stage exists specifically to remove these before air ever reaches the cold section, protecting equipment that's extremely difficult and costly to access once it's built into the insulated cold box structure.

WHY THE HEAT EXCHANGE NETWORK MATTERS SO MUCH

Cooling air from ambient temperature down to roughly -185°C takes real energy — recovering as much of that cold energy as possible from the products already at cryogenic temperature, rather than paying for it twice, is what makes continuous air separation practical at industrial scale.

ELECTRICAL & CONTROL EQUIPMENT

Purification system valves	**Automated switching/regeneration**
Cold box instrumentation	**Cryogenic temp/level sensors**
Column level control	**Continuous, closed-loop**

KEY POINT

Because the cold box can't easily be opened for inspection during operation, purification failures are treated as a serious operational risk, not a minor maintenance item.

3 Electrical Equipment in Detail

From the plant's single largest motor load to the purity analyzers guarding every product stream

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
AIR COMPRESSION & PURIFICATION		
Main air compressor	Compresses incoming air to the pressure the separation process requires; often the single largest motor load at the plant.	Very large, continuous duty
Booster compressors	Provide additional pressure for specific product streams or liquefaction duty.	Where higher-pressure product is needed
Purification system valves & heaters	Cycle adsorbent beds that remove CO2 and moisture, with periodic regeneration.	Automated switching sequence
COLD BOX & DISTILLATION		
Cold box instrumentation	Cryogenic-rated temperature, pressure, and level sensors throughout the distillation columns.	Specialized cryogenic sensors
Column level & reflux control valves	Maintain correct liquid levels and reflux ratios for stable separation.	Continuous closed-loop control
Purity analyzers	Continuously verify oxygen, nitrogen, and argon product purity before release to distribution.	Per product stream
LIQUID STORAGE & DISTRIBUTION		
Cryogenic storage tank instrumentation	Monitor pressure and liquid level in vacuum-insulated LOX/LIN/LAR tanks.	Cryogenic-rated sensors
Cryogenic transfer pumps	Move liquid product between storage and vaporization or tanker-loading systems.	Specialized low-temperature pump design
Ambient air vaporizers	Convert liquid product back to gas using ambient heat, typically without added energy input.	Passive finned-tube design
Distribution pipeline pressure control	Regulates gas pressure delivered to each consuming process area.	Multiple zones across the plant
Level 1/2 automation	Coordinates compression, purification cycling, and column control to maintain stable, continuous separation.	Runs the plant around the clock

4 Liquid Storage & Vaporization — The Accumulation Function

Why a plant that runs continuously still needs a buffer of stored product

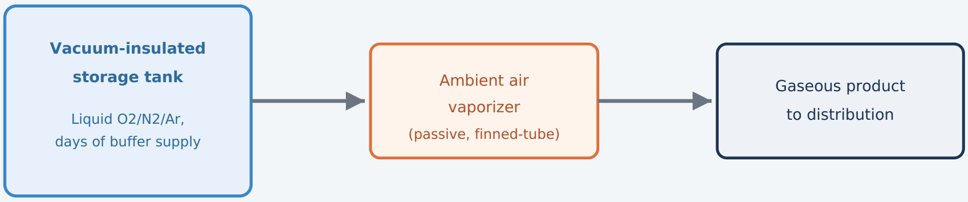


Fig. 4 — Liquid storage tanks hold days' worth of buffer supply; ambient air vaporizers convert liquid back to usable gas on demand, using surrounding heat rather than added energy.

WHY CONTINUOUS PRODUCTION STILL NEEDS A BUFFER

Cryogenic ASUs run most efficiently — and most reliably — as a genuinely continuous process; stopping and restarting a cold box is slow and energy-intensive, similar in spirit to why a blast furnace or a continuous annealing line resists being cycled on and off. But a BOF mid-blow or an EAF mid-heat cannot tolerate an interrupted oxygen supply. Liquid storage resolves this mismatch: the ASU keeps running steadily regardless of moment-to-moment plant demand, with liquid inventory absorbing the difference and providing a genuine buffer if the ASU itself trips or goes down for maintenance.

VAPORIZATION WITHOUT ADDED ENERGY

Ambient air vaporizers use nothing more than the temperature difference between liquid product and the surrounding air to convert liquid back to gas — finned-tube heat exchangers exposed to outdoor air provide enough heat transfer surface to vaporize product on demand without a dedicated heat source, though electric or steam-heated vaporizers exist as a backup for high-demand or cold-climate conditions.

TYPICAL PARAMETERS

Liquid storage temperature	**~-183 to -196 °C**
Storage buffer	**Typically days of supply**
Vaporization method	**Ambient air (primary)**

KEY POINT

Liquid storage also supports tanker delivery to other sites and provides a market-facing product beyond what the plant itself consumes.

5 Continuous Production vs. Fluctuating Demand

Conventional ASUs favor steady operation; modern designs can also follow load — storage and flexibility are complementary

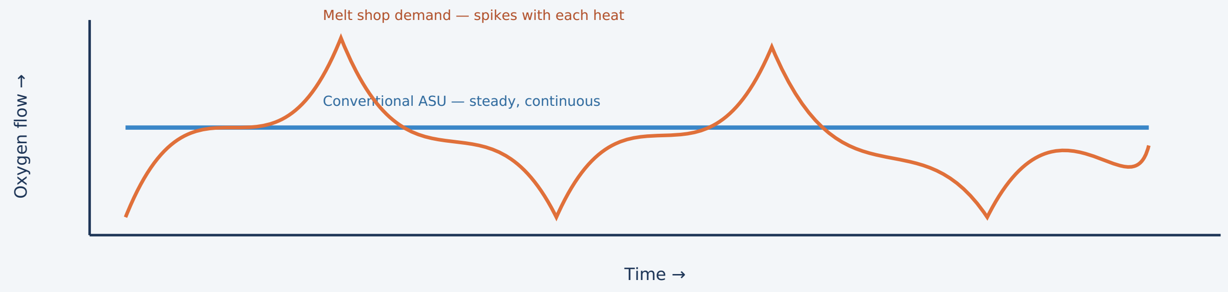


Fig. 5 — A conventional ASU's flat, continuous output and the melt shop's spiky, batch-driven demand curve rarely match at any given instant; liquid storage buffers the difference, and on modern ASU designs the production curve itself can also be engineered to track demand more closely.

WHY STORAGE HAS TRADITIONALLY DONE THE BUFFERING

A BOF blow or an EAF oxygen injection happens in short, intense bursts tied to each heat's schedule, while a conventionally designed cryogenic process favors steady, continuous operation — rapid load swings are harder on a cold box built only for steady-state running. The traditional solution is **decoupling**: liquid storage absorbs the difference between steady ASU output and spiky plant demand, rather than asking the ASU itself to track demand directly.

MODERN ASUS CAN ALSO BE ENGINEERED TO FOLLOW LOAD

This is no longer an either/or. Current-generation ASU designs — sized and controlled specifically for demand-side management — can follow load and cycle output deliberately, with components qualified for thousands of pressure and load changes; some are even operated to track variable power availability rather than running flat-out regardless of grid conditions. The permissible ramp rate and turndown are design-specific, so storage and production flexibility work as complementary strategies on a modern plant, not substitutes for each other.

TYPICAL PARAMETERS

ASU operation	**Steady-state (conventional) or load-following (modern)**
Demand pattern	**Batch-driven, spiky**
Buffer mechanism	**Liquid inventory management**

6 Safety & Protection Systems

Cryogenic cold, oxygen-enriched atmospheres, and a counterintuitive nitrogen hazard

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Oxygen cleanliness enforcement	Standard procedure for all oxygen-contact equipment	Strict no oil/grease policy — same severity as covered in the BOF guide
Cryogenic tank pressure relief	Tank pressure rises from normal boil-off beyond safe limit	Automatically vents to atmosphere via relief devices
Purification system monitoring	CO_2/moisture breakthrough detected ahead of the cold box	Alarms operator; can force early bed regeneration
Oxygen-enriched atmosphere detection	Elevated oxygen concentration detected in a work area	Alarms and restricts access — enriched atmospheres dramatically increase fire risk
Nitrogen/oxygen deficiency monitoring	Reduced oxygen concentration detected in confined or enclosed spaces	Alarms and blocks entry — nitrogen displaces breathable oxygen without warning
Cryogenic PPE requirements	Any work near liquid product or cold piping	Mandatory insulated gloves and face protection against cryogenic burns
Emergency stop (E-stop)	Manual activation anywhere on the platform	Removes power from drives per the defined E-stop category

WHY NITROGEN — THE "SAFE, INERT" GAS — IS STILL A SERIOUS HAZARD HERE

Nitrogen is non-toxic and non-flammable, which makes it easy to underestimate: in a confined space, it silently displaces breathable oxygen with no warning odor or color, and asphyxiation can occur before anyone realizes the atmosphere has changed. This plant produces bulk nitrogen alongside oxygen, and confined-space procedures account for both hazards with equal seriousness.

Summary

The utility behind every oxygen lance and argon stir in this book, and a recap

FACTOR	WITHOUT LIQUID STORAGE	WITH LIQUID STORAGE (ACCUMULATION)
Supply reliability	Vulnerable to any ASU trip or outage	Buffered by days of liquid inventory
ASU operation	Would have to chase spiky demand directly on a conventional design	Can run steady (or load-follow, on modern designs) as intended
Melt shop exposure	A single ASU fault could halt production	Isolated from short-term ASU disruptions

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **oxygen plant** — an air separation unit — supplies oxygen, nitrogen, and argon to nearly every high-temperature process covered elsewhere in this book, by exploiting the simple fact that each gas boils at a different cryogenic temperature. Ambient air is compressed, scrubbed of anything that would freeze solid downstream, then cooled and fed into a **double-column cold box** that recycles cold energy from its own outgoing product streams to keep the whole cycle efficient. A conventionally designed cryogenic plant favors steady operation while the melt shop's actual oxygen demand spikes with every heat, so **liquid storage** — the accumulation function — traditionally buffers that mismatch and protects against any single ASU outage; modern ASU designs can also be engineered for deliberate load-following, making storage and production flexibility complementary strategies rather than either/or. The same plant that keeps a BOF blowing and an EAF refining also produces bulk nitrogen, whose quiet, odorless ability to displace breathable air makes it just as serious a safety consideration as the oxygen it's made alongside.

PART 18 — WATER TREATMENT PLANT

Why Treat Water? Two Jobs, One Plant

Preparing clean water going in, and cleaning contaminated water coming back out

A steel mill's water treatment plant actually runs two largely independent operations under one roof. **Makeup water treatment** takes raw water from a river, lake, or municipal source and prepares it for use across the mill's various cooling and process circuits — removing suspended solids and, for the most demanding applications, dissolved minerals that would scale inside heat exchangers and boilers. **Wastewater treatment** does the opposite job: it takes water that's already been used — carrying mill scale, oil, or in the pickling line's case, dissolved metal and residual acid — and cleans it enough to either discharge within environmental permit limits or recycle back into the plant.

Fig. 1 — Two distinct treatment jobs, running as parallel operations, connect this plant to nearly every water-consuming process covered elsewhere in this book.

WHY QUALITY REQUIREMENTS DIFFER SO MUCH BY END USE

Not every water circuit needs the same purity. General cooling duties tolerate ordinary clarified water, while boiler feedwater or certain sensitive closed-loop circuits need minerals removed almost entirely — pushing hard water through those systems would scale heat exchanger surfaces and steadily reduce their efficiency.

TYPICAL PARAMETERS

Raw water source	**River, lake, or municipal**
Suspended solids removal	**Clarification + filtration**
Discharge	**Permit-limited**

1 The Water Treatment Process

Two parallel trains, each purpose-built for the water quality it needs to deliver

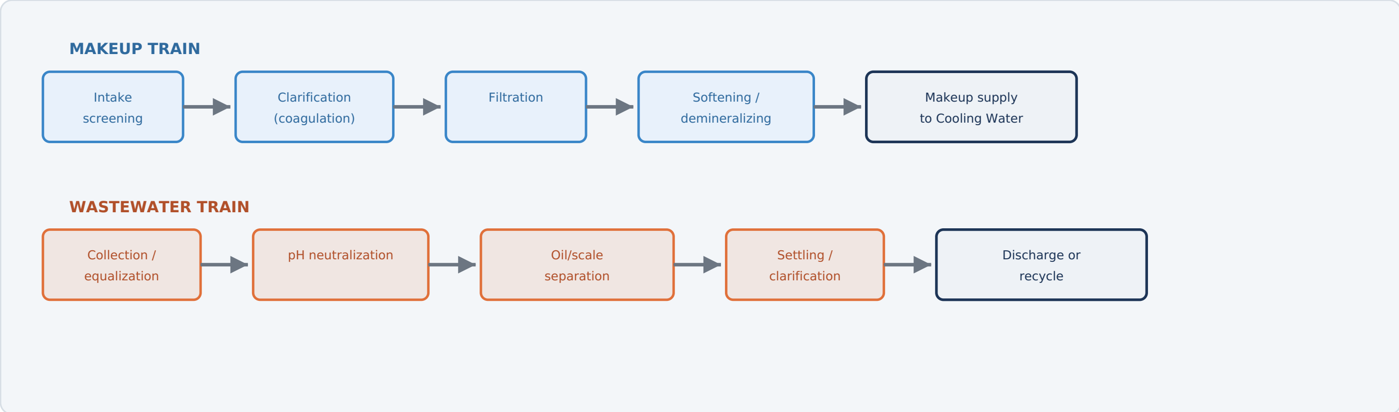


Fig. 2 — The makeup train prepares fresh water for use; the wastewater train cleans used water enough to leave the plant or return to it — running independently but sharing the same overall water balance.

HOW IT WORKS

1. **Intake & screen:** Raw water is drawn in and screened to remove debris before any chemical treatment begins.
2. **Clarify:** Coagulant chemicals help suspended solids clump together and settle out.
3. **Filter:** Remaining fine particulate is removed by filtration.
4. **Soften/demineralize (where needed):** Dissolved minerals are stripped for the most sensitive applications.
5. **Collect & treat wastewater:** Used process water is collected, neutralized, separated of oil and scale, and settled before discharge or recycling.

TYPICAL PARAMETERS

Raw water source	**River, lake, or municipal**
Discharge limits	**Permit-defined**
Recycled fraction	**Plant-dependent, often substantial**

2 Makeup Water Treatment — Clarification to Demineralization

Not every circuit needs the same purity — the treatment train branches by end use

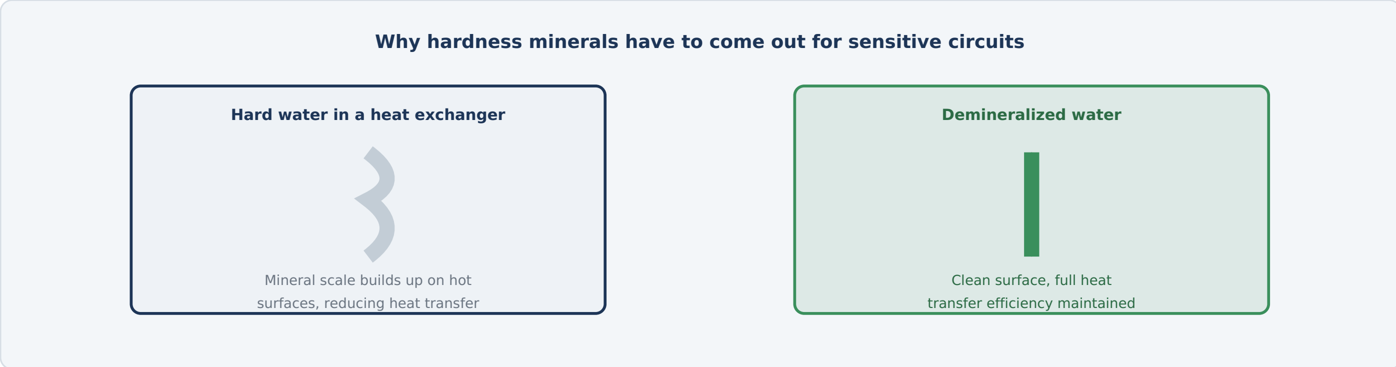


Fig. 3 — Scale buildup from dissolved minerals steadily degrades heat exchanger performance; demineralized water avoids this entirely for the circuits that can't tolerate it.

THE CLARIFICATION STEP IN DETAIL

Coagulant chemicals (commonly alum or a synthetic polymer) neutralize the electrical charge that keeps fine suspended particles apart, letting them clump into larger flocs that settle out under gravity in a clarifier basin — dramatically faster than the particles would settle on their own. Filtration afterward polishes out whatever fine material remains.

ELECTRICAL EQUIPMENT

Chemical dosing pumps	**Coagulant metering**
Clarifier rake drive	**Slow continuous rotation**
RO/ion exchange system	**Pumps + automated valves**

SOFTENING VS. DEMINERALIZATION — NOT THE SAME PROCESS

METHOD	HOW IT WORKS
Softening (Na-cycle)	Sodium-form cation exchange resin swaps out dissolved calcium and magnesium (hardness) for sodium — good enough for scale prevention, but the water still carries dissolved salts, just not "hard" ones.
Demineralization (H-form + OH-form)	True demineralization needs *both* halves: H-form cation resin removes cations while OH-form anion resin removes anions — as separate beds or combined in a single mixed bed — leaving water essentially free of dissolved ionic salts, not just hardness.
Membrane (RO)	Reverse osmosis forces water through a semi-permeable membrane, rejecting dissolved minerals as concentrate — an alternative route to demineralized-grade water without resin regeneration chemicals.

Softening alone is often enough for general cooling makeup; boiler feedwater and other sensitive applications need the full cation-plus-anion demineralization step, not softening by itself.

3 Electrical Equipment in Detail

From the intake pumps to the sludge dewatering equipment closing the loop on solids removed from both trains

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
MAKEUP WATER TREATMENT		
Raw water intake pump motors	Draw water from the source (river, lake, or municipal connection) into the treatment plant.	Sized to plant peak demand
Chemical dosing pump motors	Meter coagulant and other treatment chemicals into the raw water stream.	Precision metering pumps
Clarifier rake drive motor	Slowly rotates a rake mechanism that moves settled solids to a collection point.	Low speed, continuous duty
Filtration system backwash pumps	Periodically reverse-flush filter media to remove accumulated solids.	Automated backwash cycle
Ion exchange / RO system pumps & valves	Drive membrane-based demineralization or resin-based softening/demineralization (cation and, where fitted, anion exchange) for sensitive water circuits.	Automated regeneration sequencing
WASTEWATER TREATMENT		
Wastewater collection/lift station pumps	Gather process wastewater from across the plant and move it to the treatment train.	Multiple collection points
pH dosing pumps	Add lime or caustic (for acidic streams) or acid (for alkaline streams) to neutralize pH before further treatment.	Automated pH-feedback control
Oil skimmer drive motors	Remove floating oil from wastewater surfaces ahead of further treatment.	Continuous or scheduled operation
Sludge dewatering equipment	Belt press or centrifuge motors that reduce collected sludge volume for disposal or recovery.	Scale sludge often recovered/recycled
INSTRUMENTATION & AUTOMATION		
pH, conductivity & turbidity analyzers	Continuously monitor water quality at multiple points across both treatment trains.	Feeds automated dosing control
Level 1/2 automation	Coordinates chemical dosing, pump staging, and filter backwash cycles across both trains.	Runs continuously

4 Wastewater Treatment — Scale, Oil & Acid

Three different contaminants, three different treatment steps, one combined effluent stream

Fig. 4 — Different process areas contribute different contaminants; the wastewater train has to address all three, sometimes via dedicated pre-treatment before streams combine.

SCALE RECOVERY — TURNING WASTE INTO RAW MATERIAL

Iron oxide mill scale washed off during casting and rolling settles readily in scale pits given enough residence time, and because it's largely iron oxide, many mills recover and recycle it as a raw material input rather than treating it purely as waste to dispose of.

WHY ACID STREAMS NEED DEDICATED PRE-TREATMENT

Pickling line rinse water carries both residual acid and dissolved iron chloride — simply diluting it with other wastewater isn't sufficient. Automated pH-feedback dosing neutralizes the acid first (typically with lime or caustic), precipitating dissolved metals out as settleable solids before the stream joins general wastewater treatment.

TYPICAL PARAMETERS

Scale pit residence time	**Sized for settling**
pH neutralization target	**Discharge permit range**
Oil removal method	**Skimming / API separator**

5 Recycling vs. Fresh Water — The Balancing Act

Recycling saves intake and discharge costs — but recycled water gets dirtier with every pass

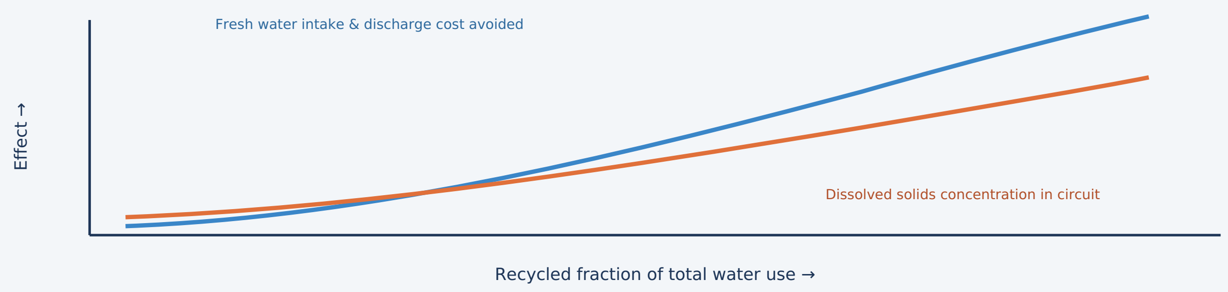


Fig. 5 — Recycling water avoids fresh intake and discharge costs, but dissolved solids concentrate with every cycle unless periodically diluted — the balance is managed through controlled blowdown.

WHY RECYCLED WATER CAN'T JUST CIRCULATE FOREVER

Every time water evaporates from an open circuit (as it does continuously in cooling towers) or is otherwise consumed, the dissolved minerals it carried stay behind, concentrating in whatever water remains. Left unchecked, that buildup would eventually cause scaling and corrosion throughout the system — which is exactly why a controlled **blowdown** (periodic discharge of a portion of circulating water) is a deliberate part of the design, balanced against fresh makeup water addition from this plant.

THE WATER BALANCE AS A GENUINE OPTIMIZATION

More recycling reduces both fresh water costs and wastewater treatment/discharge volume, but requires more careful chemistry management and, sometimes, additional treatment steps to make recycled water suitable for its next use — the right balance point depends on local water costs, discharge regulations, and the water quality each internal use actually requires.

TYPICAL PARAMETERS

Blowdown rate	**Set by cycles of concentration target**
Recycled fraction	**Plant- and permit-dependent**

6 Safety & Protection Systems

Corrosive treatment chemicals, confined tanks, and the specific hazards of concentrated wastewater

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Chemical containment & spill detection	Leak or overflow detected at a dosing or storage tank	Alarms operator; isolates the affected area
pH monitoring interlock	Treated discharge falls outside permitted pH range	Holds discharge and redirects for further treatment
Confined space entry procedures	Entry into a clarifier, tank, or pit required	Requires isolation, ventilation, and atmosphere testing before entry
Chemical dosing pump interlocks	Dosing rate deviates from setpoint	Alarms operator; can halt dosing automatically
Discharge quality monitoring	Continuous analyzers detect an out-of-spec parameter	Alarms operator; can automatically divert discharge
Emergency stop (E-stop)	Manual activation anywhere in the treatment area	Removes power from drives per the defined E-stop category

WHY CHEMICAL HANDLING GETS THE SAME SERIOUSNESS AS ACID ON THE PICKLING LINE

Coagulants, pH-adjustment chemicals, and concentrated wastewater streams involve genuinely corrosive and hazardous materials — the same PPE, containment, and handling discipline applied to the pickling line's hydrochloric acid applies here to the treatment chemicals this plant uses every day.

Summary

The plant behind every cooling and process water circuit in this book, and a recap

FACTOR	LOW RECYCLING	MANAGED RECYCLING
Fresh water cost	Higher intake volume required	Reduced via internal reuse
Discharge volume	Higher, more permit exposure	Reduced, easier compliance
Water chemistry management	Simpler	Requires active blowdown control

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **water treatment plant** runs two parallel jobs: a **makeup water train** that clarifies, filters, and — for the most sensitive circuits — demineralizes raw water before it feeds the Cooling Water System, and a **wastewater train** that neutralizes, separates, and settles used process water carrying mill scale, oil, and (from the pickling line) residual acid, before it's discharged or recycled. Because dissolved minerals concentrate every time water evaporates or is consumed in a circuit, **recycling** water back into the plant — while it saves real fresh-water and discharge cost — has to be balanced against controlled **blowdown** to keep water chemistry within safe operating limits. The same corrosive-chemical handling discipline that governs the pickling line's hydrochloric acid applies here too, since this plant works with genuinely hazardous treatment chemicals every day to keep the rest of the mill supplied with water it can actually use.

PART 19 — COOLING WATER SYSTEM

Why Active Cooling? Heat That Has Nowhere Else to Go

Nearly every hot process covered in this book rejects heat somewhere — this system carries it away

From the EAF's water-cooled panels to the continuous caster's mold and secondary cooling sprays, from hot mill roll cooling to the DC EAF's rectifier and transformer cooling, almost every major system in this book rejects substantial heat that has to go somewhere — left in place, that heat would rapidly overwhelm equipment ratings and cause failure. The cooling water system is the shared infrastructure that carries heat away from all of it and rejects it to the atmosphere, continuously, for as long as the mill runs.

A shared utility behind equipment across every part of this book

Fig. 1 — Every one of these systems, covered in its own chapter elsewhere in this book, ultimately depends on the cooling water system to carry away the heat it generates or absorbs.

EVAPORATIVE COOLING — A WITHDRAWAL/CONSUMPTION TRADE-OFF, NOT A FREE EFFICIENCY WIN

Rather than simply discharging warm water, cooling towers reject heat primarily through evaporation: a small fraction of circulating water evaporates as it cascades through the tower, carrying a disproportionate amount of heat away with it. This dramatically reduces how much water has to be **withdrawn** from the source compared with once-through cooling, which pulls a large flow straight through equipment once and returns most of it, warmed, to the source. But withdrawal isn't the same as **consumption**: a wet tower actually consumes more water overall, permanently losing it to evaporation and periodic blowdown, while a once-through system returns nearly all the water it withdraws. Which arrangement is more "water-efficient" genuinely depends on which of the two metrics matters for the site — reduced withdrawal favors the tower, minimized consumptive loss favors once-through where it's permitted.

WITHDRAWAL VS. CONSUMPTION, DEFINED

Withdrawal is water taken from the source, whether or not it comes back. **Consumption** is water that doesn't return — lost to evaporation, blowdown, or other means. A wet tower has low withdrawal but real consumption; a once-through system has high withdrawal but low consumption. Neither number alone tells the whole story.

TYPICAL PARAMETERS

Supply water temperature	**~25–32 °C typical**
Return water temperature	**~35–45 °C typical**
Circuit types	**Closed-loop and open (tower)**

1 Where Cooling Water Is Used, Plant-Wide

Every major heat-rejection duty covered elsewhere in this book, in one place

Each process guide in this book covers its own cooling-related equipment in its own electrical equipment table. This page pulls those threads together into a single reference — showing not just *where* cooling water is used, but which duties are direct (open, contacting the product or process gas) and which are indirect (closed, protecting equipment without touching anything the water shouldn't).

AREA / EQUIPMENT	COOLING DUTY	PURPOSE	CIRCUIT
MELT SHOP			
AC/DC EAF vessel	Roof & sidewall water-cooled panels	Protect the shell from arc radiant heat	Closed, indirect
DC EAF rectifier & transformer	Rectifier heat sinks, transformer oil coolers	Remove heat from high-current power electronics	Closed, indirect
BOF	Oxygen lance body	Prevents the lance melting in the furnace during the blow	Closed, indirect
Fume Extraction System	Off-gas cooling (radiant/spray cooler)	Cools hot furnace gas before it reaches baghouse filter bags	Mixed — some direct-contact stages
LMF	Electrode holders & roof	Same arc-radiant protection duty as EAF	Closed, indirect
VTD	Intercondenser spray pumps	Condense motive steam/process vapor between ejector stages	Direct, open
CCM — mold (primary)	Copper mold cooling channels	Highest heat-flux duty on the caster; freezes the initial shell	Closed, indirect
CCM — spray chamber (secondary)	Zoned nozzle banks on the strand	Continues heat extraction until the core is fully solid	Direct, open
CCM — machine cooling	Roll bearings, segment frames, hydraulics	Protects caster hardware from the strand's radiant heat	Closed, indirect
HOT STRIP MILL			
Reheat Furnace	Skid pipe cooling	Keeps the slab-support rails from melting inside the furnace	Closed, indirect
Roughing & Finishing Mills	Work roll cooling sprays	Controls roll surface temperature and lubrication, protects roll life	Direct, open
Runout Table	Laminar cooling headers (Part 14)	Sets the strip's phase-transformation microstructure	Direct, open
AUXILIARY PLANTS & ELECTRICAL			
Air Compressor Plant	Aftercooler / intercooler	Removes heat of compression before drying	Closed, indirect
Oxygen Plant (ASU)	Main air compressor cooling	Removes heat from the plant's largest continuous compressor load	Closed, indirect
Throughout the plant	Hydraulic oil coolers	Keeps hydraulic fluid within operating temperature on tilting, lifting, and positioning systems	Closed, indirect
Throughout the plant	SVC/STATCOM & large transformers	Removes heat from reactive-power and power-conversion equipment	Closed, indirect

READING THE "CIRCUIT" COLUMN

Direct/open duties are where water actually contacts the strand, gas, or another process stream — these need makeup water and typically drain to the plant's scale pits or wastewater system (Part 18). **Closed/indirect** duties never touch the product or process directly; they protect equipment and reject their heat to the same tower infrastructure via a heat exchanger, as covered on the next page.

2 The Cooling Water Cycle

Pump, supply, absorb heat, return, and reject — continuously, for as long as the mill runs

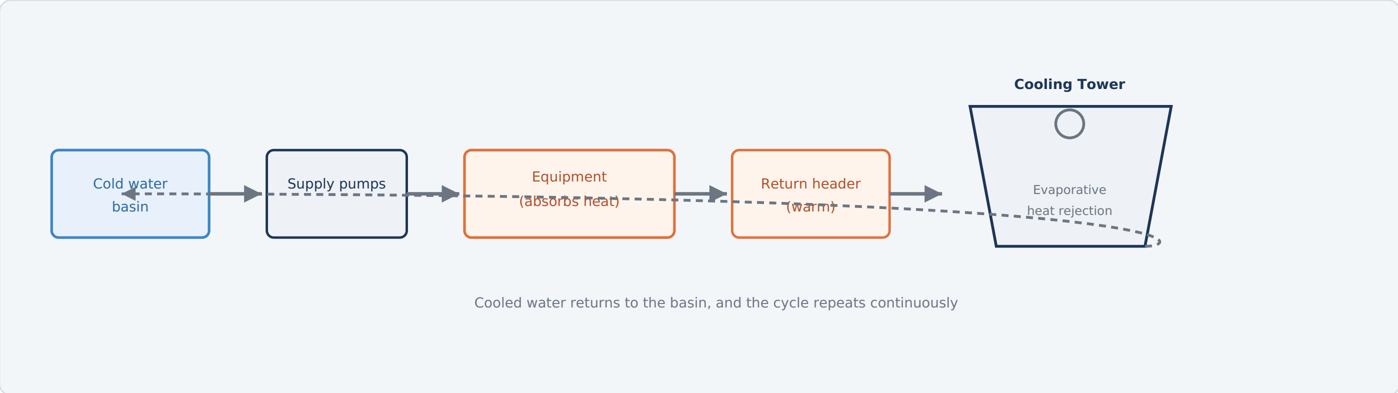


Fig. 2 — Water pumped from a cold basin absorbs heat at equipment across the mill, returns through a warm header to the cooling tower, and cycles back to the basin — a continuous loop running as long as the plant operates.

HOW IT WORKS

1. **Draw from the basin:** Supply pumps pull cooled water from the tower's collection basin.
2. **Distribute to equipment:** Water flows via supply headers to every cooling duty across the mill.
3. **Absorb heat:** As water passes through or around hot equipment, it picks up heat, raising its temperature.
4. **Return:** Warmed water flows back via return headers toward the cooling tower.
5. **Reject heat:** The tower sprays warm water through its fill while air draws heat and a small fraction of the water away, cooling the remainder before it collects in the basin again.

TYPICAL PARAMETERS

Supply temperature	**~25–32 °C**
Return temperature	**~35–45 °C**
Operation	**Continuous**

3 Cooling Towers & Evaporative Heat Rejection

Why the achievable water temperature depends on humidity, not just air temperature

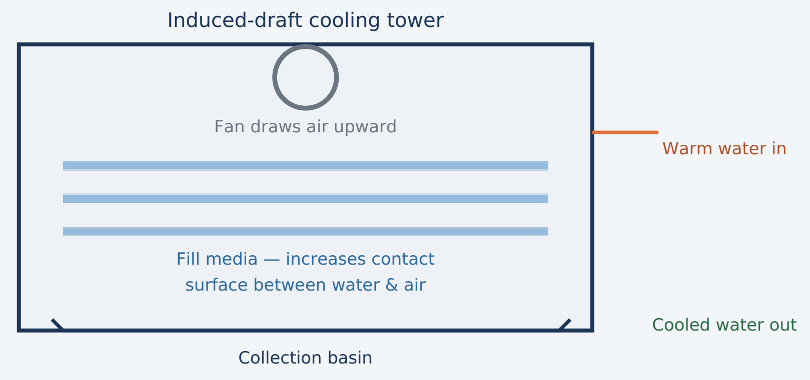


Fig. 3 — Fill media maximizes contact time between falling water and air drawn through by the fan; evaporation of a small fraction of the water carries heat away far more effectively than air contact alone.

WHY WET-BULB TEMPERATURE IS THE REAL LIMIT

Because the cooling effect comes from evaporation, the coldest water a tower can practically achieve is limited by the **wet-bulb temperature** of the ambient air — a measure that accounts for humidity, not just air temperature. On a hot, dry day a tower can achieve surprisingly cold water; on a hot, humid day, achievable cooling is much more limited even at the same air temperature, since humid air already carries more moisture and accepts less through evaporation.

INDUCED VS. FORCED DRAFT

ARRANGEMENT	HOW IT WORKS
Induced draft	Fan mounted at the top draws air up through the falling water — the more common industrial arrangement.
Forced draft	Fan mounted at the base pushes air through — quieter at the top, but more prone to air recirculation issues.

ELECTRICAL EQUIPMENT

Tower fan motors	**Induced or forced draft**
Fill media	**Passive, maximizes contact**
Drift eliminators	**Limit water droplet carryover**

4 Electrical Equipment in Detail

From the plant's largest continuous pump load to the chemical dosing that keeps the water itself from becoming a problem

EQUIPMENT	FUNCTION & DETAIL	TYPICAL RATING / NOTES
CIRCULATION		
Circulation pump motors	Move water from the basin to every cooling duty across the mill; often among the largest continuous electrical loads in the plant given the enormous flow rates involved.	Multiple large pumps, often VFD-equipped
Distribution header valves	Balance and control flow to each cooling zone or equipment area.	Motorized or manual, zone-specific
HEAT REJECTION		
Cooling tower fan motors	Draw or force air through the tower fill to drive evaporative heat rejection.	Induced or forced draft, often VFD-controlled
Makeup water control valves	Replace water lost to evaporation and blowdown, sourced from the Water Treatment Plant.	Level-controlled
Blowdown control valves	Discharge a controlled portion of circulating water to limit dissolved solids buildup.	Conductivity-controlled
WATER QUALITY & INSTRUMENTATION		
Chemical dosing pumps	Add corrosion inhibitor, biocide, and pH-adjustment chemicals to protect equipment and control biological growth.	Automated, feedback-controlled
Water temperature & conductivity sensors	Monitor supply/return temperature and dissolved solids concentration throughout the system.	Continuous monitoring
Heat exchangers (closed-loop circuits)	Transfer heat from closed, chemically conditioned loops to the open tower water without direct contact.	Isolates sensitive equipment from open water
Level 1/2 automation	Coordinates pump and fan staging against actual heat load, and manages blowdown/makeup balance.	Runs continuously

5 Closed-Loop vs. Open Systems

Not every piece of equipment can tolerate the same water quality

Closed-Loop (Indirect)

Treated, chemically conditioned water, isolated via heat exchanger

Used for: CCM mold, sensitive electrical/electronic cooling

Open (Tower) System

Larger volume, direct exposure to atmosphere via the tower

Used for: general area cooling, less sensitive furnace panels

Fig. 4 — Sensitive equipment gets isolated, chemically controlled closed-loop cooling; less sensitive duties run on the larger, simpler open tower circuit directly.

WHY ISOLATE SENSITIVE EQUIPMENT AT ALL

Open tower water is directly exposed to atmosphere — it picks up airborne dust, biological contamination, and, through evaporation, steadily concentrating dissolved solids. That's perfectly fine for robust, less sensitive cooling duties, but equipment like a continuous caster mold or precision electrical/electronic cooling needs consistently clean, chemically stable water without those variables. A closed loop, isolated from the open tower circuit by a heat exchanger, delivers that stability while still ultimately rejecting its heat to the same tower infrastructure.

THE TRADE-OFF

Closed loops cost more — they need their own pumps, heat exchangers, and chemical treatment — but protect equipment that would otherwise suffer fouling, scaling, or corrosion from open water quality. The decision of which duties get a closed loop versus open tower water is a deliberate equipment protection choice, not an arbitrary design detail.

TYPICAL PARAMETERS

Closed-loop chemistry	**Corrosion inhibitor, biocide, pH control**
Open system exposure	**Atmosphere, dust, biological**

6 Cooling Capacity vs. Energy — The Balancing Act

Running every pump and fan at maximum guarantees cold water — and wastes enormous energy doing it

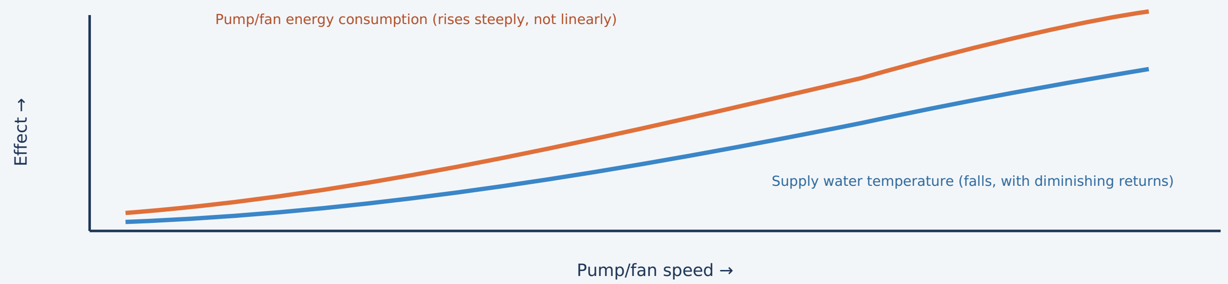


Fig. 5 — Pump and fan power both rise steeply with speed while the resulting temperature benefit flattens out — matching capacity to actual heat load, rather than running flat-out constantly, is where the real savings live.

WHY MATCHING LOAD BEATS RUNNING AT MAXIMUM

Pump and fan power both scale steeply with speed — roughly with the cube of speed for fans — so a modest reduction in flow or airflow during periods of lower heat load (reduced production, cooler ambient conditions) saves disproportionately more energy than the temperature benefit lost. Variable speed pumps and fans, controlled against actual return water temperature rather than run at a fixed maximum, are the standard modern approach.

WHY THIS CONNECTS BACK TO PRODUCTION SCHEDULE

Heat load isn't constant — it rises and falls with what's actually running upstream at any given moment (an EAF mid-heat vs. between heats, a caster running vs. idle). Level 2 automation coordinates cooling capacity against that real-time demand rather than sizing everything for worst-case peak load continuously.

TYPICAL PARAMETERS

Fan/pump control	**VFD, return-temperature feedback**
Heat load variation	**Tracks production schedule**

7 Safety & Protection Systems

Biological hazards in open water, high-flow rotating equipment, and warm-water burn risk

PROTECTION	TRIGGER CONDITION	PROTECTIVE ACTION
Biocide dosing & water quality monitoring	Scheduled treatment program for open tower water	Controls biological growth, including Legionella risk management
Low-flow/low-level pump protection	Insufficient water available at pump suction	Trips the affected pump before dry-running damage occurs
Cooling loss alarm (critical equipment)	Flow or temperature deviation detected on equipment-critical closed loops	Alarms operator; can force a controlled process slowdown to protect equipment
Confined space entry procedures	Entry into a tower basin or sump required	Requires isolation, ventilation, and atmosphere testing before entry
Fan/pump motor overload protection	Overcurrent or thermal overload detected	Trips the affected motor off-line
Emergency stop (E-stop)	Manual activation at the pump house or tower	Removes power from drives per the defined E-stop category

WHY LEGIONELLA MANAGEMENT IS A NAMED, ONGOING PROGRAM

Open cooling towers create exactly the warm, wet, aerosol-generating conditions that can support Legionella bacteria growth if water treatment lapses — this is a genuine public health concern taken seriously industry-wide, with biocide dosing and regular water testing treated as a standing operational requirement, not optional housekeeping.

✓ Summary

The shared infrastructure behind nearly every hot process in this book, and a recap

FACTOR	FIXED-SPEED, MAX CAPACITY	LOAD-MATCHED, VFD-CONTROLLED
Energy consumption	High, constant regardless of actual load	Scales down with reduced heat load
Equipment protection	Adequate but wasteful	Maintained while reducing energy
Response to demand swings	None — runs flat-out always	Tracks real-time return temperature

QUICK RECAP — THE PROCESS IN ONE PARAGRAPH

The **cooling water system** is the shared infrastructure carrying heat away from nearly every hot process covered elsewhere in this book — furnace panels, caster molds, mill rolls, and electrical equipment cooling all ultimately depend on it. Water pumped from a cold basin absorbs heat at equipment across the mill, then returns to a **cooling tower**, where evaporation — limited by ambient **wet-bulb temperature**, not just air temperature — rejects that heat back to atmosphere. Sensitive equipment gets isolated, chemically conditioned **closed-loop** cooling via a heat exchanger, while less sensitive duties run directly on the larger **open tower** circuit, which needs active **blowdown** and biocide treatment to manage concentrating dissolved solids and biological growth. Because pump and fan energy both rise steeply with speed while the cooling benefit flattens out, matching circulation capacity to actual, real-time heat load — rather than running everything at maximum constantly — is where this system's biggest efficiency gains live.